



LAGUNA UNIVERSITY

EDUCOMPOSE: KNOWLEDGE GRAPH-ENHANCED NLP FOR TEACHER-ASSISTED ESSAY EVALUATION

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APPROVAL SHEET



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ABSTRACT



ACKNOWLEDGEMENT



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CHAPTER I

INTRODUCTION

Essay writing remains one of the most critical forms of academic assessment in senior high school and higher education, as it measures not only students' writing proficiency but also their critical thinking, comprehension, organization, and communication skills (Simon Fraser University, 2024). Consequently, educators are expected to provide feedback that is detailed, constructive, and personalized to facilitate student improvement (Oducado et al., 2022). However, this expectation has become increasingly difficult to meet due to systemic challenges such as large class sizes, heavy teaching loads, and limited time for individualized assessment (Adams et al., 2023). In many cases, teachers are forced to review numerous essays within short periods, creating a persistent tension between the need for high-quality, meaningful assessment and the practical constraints of manual classroom instruction (OECD, 2024).

This challenge is further pronounced by the global shift toward formative assessment practices, which emphasize continuous improvement over mere numerical grading (UNESCO, 2021). Formative assessment requires teachers to deliver ongoing, specific, and actionable feedback that guides students in revising and strengthening their logical arguments (William, 2021). While pedagogically effective, this approach is labor-intensive and difficult to sustain at scale. In the Philippine educational context, where public and private school teachers often manage 40 to 50 students per class, the burden of manual essay evaluation contributes significantly to instructional fatigue and a potential reduction in feedback quality (DepEd, 2021; Oducado et al., 2022).

At present, many available educational technology tools focus primarily on surface-level writing issues. Grammar checkers such as Grammarly are effective for mechanical corrections, while plagiarism detection systems like Turnitin support academic integrity (University of Melbourne, 2023). However, these tools often fail to provide a deeper diagnostic view of student performance, as they lack the capacity for deep conceptual and structural analysis (Kakkad et al., 2023). Existing solutions generally do not



explain how well an essay demonstrates logical coherence or the strength of its argumentative structure—elements that are vital for informing future instruction (Gao et al., 2022). In this sense, while current tools are helpful, they remain incomplete for educators who require a comprehensive understanding of a student's conceptual development.

EduCompose addresses this gap by offering a teacher-centered diagnostic system that enhances essay evaluation without replacing human judgment. The system utilizes a Hybrid AI architecture, integrating Optical Character Recognition (OCR) for document digitizing, Knowledge Graph analysis for semantic mapping, and Large Language Models (LLM) for nuanced content feedback. By applying Toulmin's Model of Argumentation, EduCompose examines student essays across multiple dimensions—grammar, readability, coherence, and structural logic. The system then presents these findings as diagnostic evidence, visualizing conceptual connections and identifying logical gaps that teachers can use to guide their instructional decisions.

This study is grounded in the belief that educational technology should augment, rather than substitute, the expertise of professional educators (Ateneo de Zamboanga University, 2025). By extending the diagnostic capacity of teachers through Knowledge Graph-enhanced NLP, EduCompose provides a more structured and evidence-based approach to writing instruction. The system is designed to respond to the practical realities of Philippine classroom assessment, ensuring that the evaluation process remains both efficient and pedagogically meaningful.

Project Context

The global landscape of educational technology has seen a rapid proliferation of AI-driven solutions designed to enhance academic writing. At the most general level, tools such as automated grammar checkers and plagiarism detection systems have become standard in higher education, primarily supporting student autonomy in basic error correction and integrity maintenance. However, while these technologies assist in surface-level improvements, they offer limited diagnostic value for educators, often leaving teachers with the task of performing the deep analytical work required to understand a student's underlying writing weaknesses.

In the practical classroom setting, this creates a significant operational gap.



Teachers are tasked with evaluating high volumes of student essays—a process that is notoriously time-consuming and labor-intensive. While Automated Essay Scoring (AES) systems have attempted to mitigate this by providing numerical grades based on linguistic features, they frequently generate generalized results that lack actionable diagnostic insights. For teachers, a numerical score is insufficient; they require a detailed breakdown of a student's logical flow, evidence usage, and conceptual organization to inform their instructional interventions.

This challenge is further intensified within the Philippine educational landscape. In local institutions, educators often navigate large class sizes and heavy administrative loads, making it nearly impossible to provide timely and consistent feedback for every submission. This systemic pressure often results in "feedback delay," where students receive evaluations too late to apply them to their learning process. Consequently, there is an urgent need for a system that does not just "grade" essays, but "diagnoses" them in a way that aligns with the teacher's existing workflow.

Specifically, EduCompose is developed as a response to these multi-layered challenges. It serves as a Knowledge Graph-Enhanced diagnostic tool that shifts the focus from simple automation to teacher-assisted evaluation. By utilizing Hybrid NLP architectures, the system provides a specialized context where handwritten and digital essays are transformed into machine-readable data, visualized through semantic networks, and analyzed using the Toulmin Model of Argumentation. In this context, EduCompose functions as a "Force Multiplier" for educators—reducing the analytical workload of manual checking while empowering them with the diagnostic evidence needed to deliver high-quality, formative feedback.



Research Problem and Research Objectives

Statement of the Problem

The main problem addressed is the challenge English language teachers at Laguna University face in managing heavy workloads and large class sizes when evaluating student essays. Manual essay checking is time-consuming and often results in fatigue and inconsistency. Without a system that can standardize the evaluation process and provide diagnostic support, teachers are limited in their ability to provide the detailed, personalized feedback necessary for student writing development.

Specifically, the study aimed to address the following:

1. What are the problems encountered by English language teachers of Laguna University in the current manual process of essay evaluation?
2. What system can be developed that provides the following features:
 - 2.1. extraction of multidimensional diagnostic metrics such as grammar, readability, coherence, and argument structure;
 - 2.2. generation of semantic network visualizers such as Knowledge Graphs;
 - 2.3. plagiarism and similarity checking;
 - 2.4. extraction of handwritten text using Optical Character Recognition (OCR);
3. What is the technical performance of the developed AI models in terms of:
 - 3.1. classification accuracy for argument components (F1-Score);
 - 3.2. precision and reliability of grammar and style analysis;
 - 3.3. semantic similarity and quality of generated feedback (ROUGE and BERTScore)?
4. What is the overall performance of the developed EduCompose if evaluated using ISO 25010:2023 in terms of:
 - 4.1. Functional Suitability;
 - 4.2. Performance Efficiency;
 - 4.3. Compatibility;
 - 4.4. Interaction Capability;
 - 4.5. Reliability;
 - 4.6. Security;



4.7. Maintainability;

4.8. Flexibility;

4.9. Safety;

Objectives of the Study

The main objective of the study is to design and develop EduCompose, a teacher-assisted diagnostic system that helps English language teachers evaluate student essays efficiently using NLP and semantic network analysis.

Specifically, the study aims to:

1. To identify the problems encountered by English language teachers of Laguna University in the current manual process of essay evaluation.
2. To develop a system that can provide the following:
 - 2.1. extraction of multidimensional diagnostic metrics such as grammar, readability, coherence, and argument structure;
 - 2.2. generation of semantic network visualizers such as Knowledge Graphs;
 - 2.3. plagiarism and similarity checking;
 - 2.4. extraction of handwritten text using Optical Character Recognition (OCR);
3. To evaluate the technical performance of the integrated AI models in terms of:
 - 3.1. classification accuracy for argument components (F1-Score);
 - 3.2. precision and reliability of grammar and style analysis;
 - 3.3. semantic similarity and quality of generated feedback (ROUGE and BERTScore).
4. To evaluate the overall performance of the developed EduCompose using the software quality standard ISO 25010:2023 in terms of:
 - 4.1. Functional Suitability;
 - 4.2. Performance Efficiency;
 - 4.3. Compatibility;
 - 4.4. Interaction Capability;
 - 4.5. Reliability;
 - 4.6. Security;
 - 4.7. Maintainability;



4.8. Flexibility;

4.9. Safety;

Scope and Limitations

The primary scope of this research focuses on the development and pilot testing of EduCompose, a web-based teacher-assisted diagnostic system strictly tailored to the pedagogical workflows of the English language faculty at Laguna University. The system is a specialized web application designed to evaluate digital text-based student essays submitted within the current academic year. Its functional capabilities encompass the AI-driven extraction of multidimensional diagnostic metrics, the generation of semantic Knowledge Graphs to map conceptual structures, and academic integrity verification through the integration of the Copyscape API.

In terms of user accessibility, the system is categorized into three distinct roles: Administrators, Teachers, and Students. The Administrator possesses the privilege to provision faculty accounts and manage the overall system health. Teachers operate within a dedicated portal where they manage the entire classroom environment, including the capability to add courses, enroll students, and create specific writing activities. The teacher-end integrates trained AI models, enabling educators to view automated diagnostic insights, interactive Knowledge Graph visualizers, and detailed similarity reports generated from every student submission.

Students are provided with their own independent portal where they can submit their essays through direct text input or by uploading Portable Document Format (PDF) files. This student interface allows for an autonomous submission process where the integrated AI processes the documents in real-time. Once the analysis is complete, students can use their portal to independently view the diagnostic results and the Knowledge Graph representations of their own essays, fostering a self-regulated learning environment.

Regarding the technical feasibility of the project, several limitations are established concerning data input and extraction. When processing uploaded PDFs, the system's text extraction module may occasionally encounter minor formatting artifacts, such as irregular spacing, depending on the file's original encoding. Furthermore, regarding Optical



Character Recognition (OCR), the system's capability to extract text from scanned PDFs is optimized for documents consisting of standard printed typography. For handwritten submissions, the system only accepts image-based inputs when they are embedded within a PDF file, rather than as raw standalone image files.

Algorithmically, the Natural Language Processing (NLP) models utilized for deep text analysis specifically SentenceTransformers and Distilled Bidirectional Encoder Representations from Transformers (DistilBERT) are primarily fine-tuned versions of pre-trained models. This selection was made due to practical constraints regarding high-performance computational resources and the development timeline required for training custom models from scratch. These models are selected based on their balance of computational efficiency and diagnostic accuracy suited for the school's infrastructure.

Finally, the study is subject to several institutional and financial limitations. The Copyscape API is employed as the primary engine for plagiarism checking, serving as a practical alternative to proprietary systems like Turnitin due to existing financial constraints. Furthermore, while the architecture of EduCompose is designed with scalability in mind, the current deployment is strictly aligned with the specific English curricula and grading standards of Laguna University. Any future adoption by other academic institutions would require necessary adjustments to the system's logic to align with their specific pedagogical standards.

Significance of the Study

The researchers developed a web-based teacher-assisted diagnostic system known as EduCompose that can help in the evaluation of students currently taking up English language courses, especially managing heavy teaching workloads. This system enables the automated extraction of multidimensional diagnostic metrics, generation of semantic Knowledge Graphs, and checking of document similarity and plagiarism of submitted digital essays. This system also provides a structured approach and systematic framework in dealing with writing assessments by having a platform that can be easily accessed and viewed through dedicated teacher and student portals.

The result of this study will benefit the following:

Educational Institutions. Maximize resource allocation and support faculty wellbeing by providing a scalable, cost-effective evaluation tool that combats teacher



burnout without requiring expensive commercial enterprise software.

English Language Teachers. Assist the educators in mitigating workload cognitive fatigue by automating the mechanical aspects of text analysis, allowing them to maximize their time providing data-informed pedagogical interventions rather than manual error-checking.

Students. Convenience in having an accessible portal where essay submissions are done and immediate, evidence-based visual feedback (Knowledge Graphs) can be viewed to understand specific areas for writing improvement.

Researchers. This study provided the proponents with the invaluable opportunity to apply their theoretical knowledge in data science, software engineering, and natural language processing to build a functional, real-world application. It significantly enhanced their technical proficiencies in developing hybrid AI architectures and deepened their understanding of how technology can be leveraged to solve pressing educational challenges.

Future researchers. This may serve as a reference for future researchers that are into developing and improving human-in-the-loop AI systems and integrating semantic networks into educational diagnostic tools.

Definition of Terms

The following words are defined to better understand their usage in this research. This is necessary for future researchers in fostering better understanding to the contents of this study.

AES (Automated Essay Scoring) is a field of educational technology that uses computer programs to grade and provide feedback on written assignments in EduCompose.

API (Application Programming Interface) refers to a set of rules and protocols that allows different software applications to communicate and exchange data. In this study, APIs are used to connect EduCompose with external services and tools needed for essay analysis, feedback generation, and plagiarism detection.

BERTScore An automatic evaluation metric for text generation that leverages pre-trained contextual embeddings to measure the semantic similarity between the AI's feedback and a teacher's gold standard.



Copyscape API is an external service integrated into EduCompose used for checking document similarity, academic integrity, and plagiarism.

DistilBERT (Distilled Bidirectional Encoder Representations from Transformers) Is a pre-trained Natural Language Processing model utilized by EduCompose for argument component classification and sentence-level categorization.

EduCompose A web-based, teacher-assisted diagnostic platform developed for the essay evaluation of Laguna University, utilizing Hybrid AI architectures (Transformers and LLMs) to provide formative feedback and structural analysis.

F1-Score The harmonic mean of precision and recall used by the researchers to measure the accuracy and reliability of the DistilBERT categorization model.

Heuristic Rubric Mapping The automated process of aligning AI-generated diagnostic data with teacher-defined evaluation criteria to calculate a final score.

ISO 25010:2023 is the software quality criteria used in evaluating the system performance of EduCompose in terms of functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety.

Knowledge Graph A semantic network visualizer and analysis framework utilized by EduCompose to map conceptual structures and relational ideas within student essays. It identifies the connectivity between key concepts and detects "conceptual gaps" where logical links between ideas are missing.

LLM (Large Language Models) are advanced artificial intelligence used by EduCompose as intelligence systems designed to process, understand, and generate human-like text by analyzing massive amounts of data.

NLP (Natural Language Processing) is a field of computer science and artificial intelligence utilized by EduCompose that focuses on the computer's interaction with human language in digital text.

OCR (Optical Character Recognition) is a specialized technology used by EduCompose that converts images of typed, handwritten, or printed text into machine-encoded text that a computer can read and edit.

PDF (Portable Document Format). A file format used by EduCompose to present and exchange documents reliably, independent of software or hardware, which serves as one of the primary input modalities for essay submissions in the system.

ROUGE (Recall-Oriented Understudy for Gisting Evaluation) A set of metrics



used by EduCompose to evaluate the quality of AI-generated feedback by comparing it to human-generated reference comments.

SentenceTransformers is a framework used by EduCompose to generate vector embeddings for measuring the semantic coherence and logical flow of essays.

Toulmin's Model of Argumentation The structural framework utilized by EduCompose to analyze and visualize the components of a student's essay, such as claims, data, and warrants.



CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter includes existing foreign and local studies related to the research. The relevant information was gathered from research papers, books, theses, journal articles, and other published materials as references and guides for the study. The content of this chapter will help readers familiarize themselves with the proposed system and understand how it is informed by current research on automated writing evaluation, artificial intelligence, knowledge representation, and teaching practice.

Related Literature and Studies

AI-Powered Learning Analytics and Dashboards

Recent studies in 2025 highlight that AI-powered learning analytics dashboards have become essential for monitoring student progress in real-time. According to Ifenthaler (2025), these visual interfaces allow educators to identify academic patterns and intervene early when students struggle with specific writing dimensions. The latest research emphasizes that dashboards are most effective when they provide "actionable intelligence," translating complex AI data into simple visual charts that teachers can use for immediate instructional adjustments (Survey on Educational Analytics, 2025).

In 2024, researchers explored the role of predictive analytics in writing-intensive courses. A study by Luckin et al. (2024) demonstrated that integrating AI-driven metrics into teacher portals significantly improves the speed of diagnostic assessment. These dashboards serve as a bridge between the raw outputs of Natural Language Processing and the teacher's pedagogical needs, enabling a more data-informed approach to classroom management. Furthermore, the use of historical data tracking within these systems allows for the longitudinal monitoring of a student's argumentative growth over multiple assignments (Bernardo & Mendoza, 2024).

Earlier work within the 2021-2022 period established the foundation for user-centered analytics. Research on digital learning environments indicated that while data extraction is vital, the visualization of that data through interactive graphs and



summary statistics is what truly reduces cognitive load for educators (Ifenthaler & Widmer, 2022). These findings support the development of platforms like EduCompose, which utilize centralized dashboards to consolidate structural, grammatical, and integrity metrics into a single, accessible view for the teacher.

Criterion-Based Assessment and Rubrics

Recent literature continues to show that rubrics are valuable tools for making essay evaluation more consistent, transparent, and aligned with learning outcomes. A 2025 study highlighted the need for rubrics that are specific, measurable, practical, and clearly defined, since these characteristics improve objectivity, feedback quality, and instructional usefulness (Needs Analysis for the Development of an Assessment Rubric for Explanatory Texts, 2025). Another recent study on AI-assisted writing assessment emphasized that traditional rubrics remain important, but they may need refinement when used to evaluate essays supported by AI tools, particularly to maintain fairness, validity, and transparency (Revolutionising Essay Evaluation: A Cutting-Edge Rubric for AI-Assisted Essay Writing, 2024). These findings suggest that criterion-based assessment remains highly relevant in modern writing evaluation because it provides clearer standards for both teachers and students.

Studies published in 2022 also demonstrate that rubrics are effective not only for scoring but also for formative assessment and student development. English et al. (2022) found that rubrics can strengthen teacher practice and support student learning when used as formative tools in classroom settings, although their effectiveness still depends on how well the criteria are designed and applied. Similarly, research on argumentative writing rubrics showed that criterion-based rubrics can achieve strong interrater reliability and validity when the criteria are aligned with specific writing outcomes such as organization, evidence, academic tone, mechanics, and vocabulary (Assessment of Students' Argumentative Writing: A Rubric-Based Approach, 2022). These studies indicate that rubrics are most effective when they are closely connected to the actual skills being assessed.

Taken together, the literature suggests that criterion-based assessment and rubrics remain essential in writing evaluation because they provide structure, improve scoring



consistency, and support more actionable feedback. Their value is especially clear in essay assessment, where teachers need a systematic way to evaluate multiple dimensions of writing beyond surface correctness. This makes rubrics highly relevant to teacher-centered systems that aim to support diagnostic and formative essay evaluation.

Digitization and OCR in Educational Assessment

The integration of Optical Character Recognition (OCR) within educational technologies is a critical factor in ensuring equitable assessment and accessibility. Recent studies highlight that handwritten text recognition is most effective when integrated into a multi-stage pipeline involving text correction and semantic analysis (Pawar et al., 2025). By combining OCR with language-model-based post-correction, researchers can reduce recognition errors to a range suitable for downstream Natural Language Processing (NLP) analysis. This demonstrates that handwritten answer scripts can be processed within the same computational pipelines as typed text, enabling semi-automated assessment and expanding access to advanced feedback systems.

More recent literature also emphasizes the educational importance of OCR integration, particularly in developing educational systems. Research indicates that approximately 35–40% of students still submit handwritten essays due to limited access to digital devices and infrastructure (Chen & Liu, 2023). This issue is especially relevant in the Philippine context, where high teacher-student ratios and limited digital infrastructure increase teacher workload and create demand for automated tools that can process handwritten submissions efficiently (Oducado et al., 2022). These findings suggest that OCR technologies are important for making educational assessment systems more inclusive and accessible for both students and educators.

Earlier studies focused on the technological evolution of OCR systems. Traditional OCR engines such as Tesseract perform well on clean printed text but often struggle with noisy or unstructured handwriting. In contrast, deep learning-based libraries like EasyOCR and PaddleOCR utilize pre-trained neural networks and transformer-based recognizers to achieve higher accuracy on handwritten datasets (Jain et al., 2022). These advancements enabled handwritten image-based submissions to be converted into machine-readable text, allowing them to be integrated into advanced NLP-based essay evaluation and feedback systems.



Academic Integrity: Plagiarism and AI Detection

In the era of Generative AI, academic integrity has faced unprecedented challenges. Recent studies emphasize that educational AI systems must prioritize transparency, accountability, and human oversight to ensure fairness and ethical implementation. Research on the ethical and regulatory challenges of Generative AI in higher education highlighted the importance of preventing bias and ensuring that AI systems do not unfairly disadvantage students based on linguistic or socioeconomic backgrounds (Ethical and Regulatory Challenges of Generative AI in Higher Education, 2025). These concerns reinforce the need for explainable and responsible AI practices in educational assessment systems.

Studies published in 2024 further stressed that academic integrity tools should function as teacher-guided decision supports rather than fully autonomous authorities. Human oversight remains essential to ensure that AI-generated scores and plagiarism indicators are interpreted correctly within the broader pedagogical context (Ethical Considerations for AI Use in Education, 2024). This perspective supports the idea that AI systems should assist educators rather than replace professional judgment in high-stakes evaluation processes.

Research conducted in 2023 focused on the growing limitations of traditional plagiarism detection systems in the age of Large Language Models (LLMs). Khalil and Er (2023) explained that similarity-based plagiarism detectors are increasingly ineffective at identifying AI-generated or AI-assisted writing because modern LLMs can produce text that appears original despite lacking genuine authorship. Similarly, studies on ethical AI for teaching and learning emphasized that educational AI systems should avoid operating as “black boxes” and instead provide explainable results that educators can evaluate alongside other evidence (Ethical AI for Teaching and Learning, 2023). Collectively, these studies suggest that modern academic integrity systems require multidimensional approaches that combine plagiarism detection, AI-generated text analysis, transparency, and human oversight to maintain fairness and academic standards.

Structural Argument Mining and the Toulmin Model

Recent research in 2025 has significantly emphasized the effectiveness of



leveraging Small Large Language Models (sLLMs) for argument mining within educational contexts. According to Favero et al. (2025), these smaller, computationally efficient models are highly capable of performing argument component identification, classification, and assessment specifically for student essays, enabling the generation of more precise diagnostic feedback. This is supported by the latest surveys on Large Language Models in argument mining, which highlight the current transition toward more specialized datasets and the ongoing challenges of real-time feedback in the 2025 academic landscape (Survey, 2025).

In line with these technological advancements, a 2024 study by ACL ARR (2024) demonstrated how generative language models can be harnessed using Toulmin's Theory for "zero-shot" argument explication. This technique allows for the extraction of explicit argument structures and reasoning without the need for extensive task-specific training, making it ideal for diverse academic subjects. Furthermore, a systematic survey on NLP feedback systems framing Toulmin's model as the foundation for modern argumentative writing established that identifying structural components is essential for providing deeper, pedagogically rich feedback (Liu et al., 2023).

Earlier work in this five-year period established the statistical reliability of these structural frameworks. Mirzababaei and Pammer-Schindler (2021) achieved benchmark F1-scores reaching as high as 0.81 when identifying structural "wrongness" in arguments based on Toulmin's schema. Their study reported high success in the precise detection of claims, warrants, and evidence, proving that AI can mirror human-level structural judgment. Additionally, computational coherence measurements evolved during this time to move beyond surface-level grammatical correctness by examining how ideas are semantically connected throughout a text via sentence-level embeddings (McNamara et al., 2021). This progression—from semantic connectivity to high-accuracy structural mapping—enables platforms like EduCompose to deliver a multi-layered analysis that is both technologically advanced and academically rigorous.

Transformer-Based NLP for Text Analysis

Recent advancements in Natural Language Processing (NLP) have demonstrated that transformer-based architectures can detect a wide range of writing issues, from morphological and syntactic errors to complex stylistic nuances, with increasing accuracy



(Alapati & Swathi, 2025). Unlike older, rule-based systems, modern writing analysis utilizes pre-trained models such as DistilBERT to identify deeper patterns of grammatical breakdown. These lightweight transformer models are particularly effective in real-time educational applications because they offer a high balance between predictive accuracy and computational efficiency. Research using the DistilBERT architecture for educational text classification has reported overall accuracy levels as high as 0.812 and an F1-score of 0.801 (Kasneci et al., 2023), proving that distilled models can maintain high diagnostic reliability despite their reduced computational size. This capability allows for near-instantaneous feedback without the high resource demands of larger Large Language Models (LLMs).

Furthermore, the evaluation of text complexity has evolved beyond traditional readability formulas. While indices like Flesch-Kincaid and SMOG provide basic estimates based on word and syllable counts, advanced approaches now utilize linguistic features such as lexical diversity, referential clarity, and conceptual density (Scarborough & Parker, 2023). These metrics correlate more strongly with actual comprehension difficulty and provide a more nuanced understanding of a student's writing proficiency. However, research suggests that readability and grammar analysis should not be treated as standalone solutions, as they primarily focus on surface-level features rather than the development of ideas (Krathwohl & Anderson, 2021).

To address this gap, modern NLP frameworks suggest integrating these text-analysis metrics with structural and conceptual evaluations. By combining transformer-based error detection with knowledge graph-based concept analysis, writing assessment tools can judge not only the readability of a text but also its disciplinary relevance and logical organization. This holistic approach ensures that students receive feedback that addresses both their technical writing skills and their ability to organize complex ideas within a specific academic context.

Teacher-Centered AI: Workload & Burnout

The development of automated writing evaluation systems is increasingly framed within the concept of “hybrid teaching intelligence,” where AI systems support structured analysis while teachers remain the final decision-makers. Recent research highlights that effective implementation of teacher-centered AI requires seamless integration into



classroom workflows through features such as batch processing, customizable rubrics, and intuitive analytics dashboards that support student progress monitoring (Toward Hybrid Teaching Intelligence, 2025). These systems are designed to enhance instructional efficiency while maintaining pedagogical control in the hands of educators.

Earlier research highlights the pressing issue of teacher workload and burnout as a driving force behind the development of automated writing evaluation systems. In the Philippine context, Magtalas and Eduvala (2024) found that public school teachers in Zambales experience high levels of emotional exhaustion due to large class sizes, administrative burdens, and multiple ancillary responsibilities. This workload directly conflicts with the need for frequent, high-quality formative feedback, which is recognized as a high-impact factor in student learning (Hattie & Clarke, 2024). These findings emphasize the necessity of tools that can reduce the analytical burden of assessment while preserving instructional quality.

Modern educational technology research emphasizes an “augmentation philosophy,” where AI is used to enhance rather than replace teacher judgment. Studies show that human-in-the-loop systems achieve better accuracy and fairness when they provide automated first-pass analysis while allowing educators to retain final authority over grading decisions (Holmes et al., 2023). Similarly, Reich (2023) and Holmes et al. (2023) stress that AI in education should function as a supportive tool that strengthens teacher capacity rather than substituting professional expertise. Research also indicates that hybrid grading systems combining AI efficiency and human oversight result in higher student satisfaction and improved assessment quality (Litman et al., 2021).

User Interface (UI) and Experience (UX) in AI-Assisted Pedagogy

The current research landscape in 2025 emphasizes that the success of AI-based educational tools depends heavily on their user experience (UX) design. Favero et al. (2025) argued that for AI to be accepted in the classroom, the interface must be intuitive and require minimal technical training from the teacher. This is particularly important for complex diagnostic tools, where the layout must clearly present AI-generated insights without overwhelming the user. Modern UX principles for education now focus on



"seamless integration," ensuring that the tool fits naturally into the existing grading workflow (sLLM Research, 2025).

By 2023, the focus of educational software design shifted toward "explainable interfaces." Studies conducted by Huot and Neal (2023) emphasized that teachers need to understand why the AI assigned a specific score or identified a certain error. This led to the development of interfaces that feature color-coded highlights and interactive sidebar feedback, similar to the architecture used in modern writing evaluation platforms. These design elements enhance the transparency of the system, fostering trust between the AI and the human evaluator.

Earlier in 2021, research on software quality underscored the importance of Functional Suitability and Interaction Capability as defined by the ISO standards. Krathwohl and Anderson (2021) highlighted that educational software must not only be technically sound but also align with the cognitive levels of the target users. This historical focus on software usability provided the basis for evaluating contemporary systems using the ISO 25010 framework, ensuring that AI-driven tools like EduCompose are both powerful in their analysis and accessible in their operation.

The Evolution of Automated Writing Evaluation (AWE)

The landscape of educational technology has witnessed a significant shift from Automated Essay Scoring (AES), which primarily focuses on assigning grades, to Automated Writing Evaluation (AWE), which prioritizes providing diagnostic feedback (Yang et al., 2024). Recent large-scale investigations into AES have also underscored the challenges of predictive accuracy and demographic fairness, suggesting that while neural models can achieve high performance, they must be treated as support tools rather than autonomous authorities (Schaller et al., 2024). This growing consensus has contributed to the development of AWE systems designed to support self-regulated learning and teacher-led instruction through iterative drafting and revision cycles (Computer Assisted Language Learning, 2023).

Studies published in 2023 further emphasized the transition from surface-level corrections, such as grammar and spelling, toward deeper conceptual analysis in writing



evaluation. Research showed that semantic networks and Knowledge Graph-enhanced systems improve the diagnostic capacity of educational technologies by mapping conceptual relationships within student writing and assessing how effectively students address course concepts (Zhu et al., 2023). These findings demonstrate how educational technologies increasingly focus on understanding conceptual connections rather than simply evaluating mechanical correctness.

Earlier studies established the foundational role of explicit knowledge representation in educational technology. Chicaiza and Valdiviezo-Diaz (2021) found that systems utilizing Knowledge Graphs achieved improved learning outcomes by tracking conceptual understanding over time and providing targeted recommendations. Similarly, Litman et al. (2021) emphasized the importance of maintaining a “Human-in-the-Loop” approach, where AI functions as a decision-support tool while teachers remain the final authority in high-stakes evaluations. Together, these studies suggest that modern AWE platforms aim to balance automated efficiency with human judgment, allowing AI systems to provide multidimensional insights while preserving fairness, transparency, and teacher control in essay evaluation.

Knowledge Graphs and Semantic Concept Mapping

The shift toward concept-based assessment in educational technology is largely driven by the application of Knowledge Graphs (KGs). As defined by Abu-Salih and Alotaibi (2024) in their systematic review, KGs enable the explicit modeling of learning objectives, concept hierarchies, and prerequisite relationships that are often only implicit in traditional assessment practices. By encoding these semantic structures, educational systems can move beyond surface-level features like word count or sentence length, focusing instead on whether student writing accurately covers intended concepts and the relationships between them.

The construction of these educational knowledge graphs has evolved from manual, ontology-driven designs to more sophisticated “bottom-up” automated approaches. Recent advancements in NLP and Large Language Models (LLMs) have enabled pipelines that can automatically extract entities and relations from student essays with high precision



(Canal-Esteve & Gutiérrez, 2024). These semi-automatic workflows—where AI generates the graph and experts validate the results—significantly reduce the time required to build high-quality semantic networks while maintaining instructional control.

By integrating these KGs into the evaluation process, platforms like EduCompose can visualize the "Logic Map" of a student's work. This allows teachers to see not just what a student wrote, but how they connected specific course concepts. As noted by Canal-Esteve and Gutiérrez (2024), this capability supports a deeper level of diagnostic feedback, identifying where a student may have "conceptual gaps" or missing links in their understanding of the subject matter. This synthesis of semantic mapping and diagnostic assessment ensures that the evaluation is grounded in the actual conceptual depth of the student's work.

Large Language Models and Formative Feedback

Recent developments emphasize hybrid AI architectures that integrate Large Language Models (LLMs) with structured transformer-based classifiers and retrieval-augmented generation to improve reliability and reduce factual errors in educational feedback systems. These approaches ensure that LLM-generated outputs are grounded in more deterministic verification models, improving both accuracy and pedagogical stability (Zhang et al., 2024). This reflects the current direction toward safer, more controlled AI-assisted educational tools.

Earlier foundational studies emphasize that effective formative feedback must be specific, actionable, and directly aligned with student learning gaps. Hattie and Clarke (2024) stress that feedback should clearly identify performance deficiencies and guide revision, as generic comments provide limited learning value. Similarly, Heritage (2023) highlights that effective feedback must be precise and focused on improvement rather than general evaluation. These principles remain essential in AI-supported learning environments, where LLMs are designed to support, not replace, teacher judgment in delivering meaningful feedback.

Research in 2023 highlights both the strengths and limitations of LLM-based feedback systems in education. Kasneci et al. (2023) explain that LLMs significantly enhance grammar and style analysis by producing context-aware and human-like



explanations compared to traditional rule-based NLP systems. However, other findings caution that LLMs may produce “hallucinations,” or plausible but incorrect outputs, which can reduce the reliability of automated feedback systems (Zhang et al., 2023). These limitations show the need for cautious integration of generative models in academic writing evaluation.

Quantitative Evaluation Metrics

The effectiveness of AI-based educational systems must be evaluated using a multi-metric approach to ensure reliability, fairness, and alignment with human judgment. Recent research in 2024 emphasizes that relying on a single metric, such as simple accuracy, is insufficient for complex tasks like writing assessment where responses may be subjective or multifaceted. Instead, contemporary studies recommend combining Precision, Recall, and the F1-Score to assess how well systems correctly identify structural components while minimizing false positives and negatives (Yang et al., 2024). Furthermore, recent benchmarks for AI-generated academic feedback have reported BERTScore results of 0.885 and ROUGE-L scores of 0.652 (Zhang et al., 2024), indicating a strong alignment between machine-generated instructional advice and professional human expert judgments.

As established in 2023, the F1-Score is particularly valued in educational Natural Language Processing because it provides a balanced measure of performance, specifically when dealing with imbalanced datasets such as student-generated essays (Zhang et al., 2023). These metrics ensure that the system's output is not only technically accurate but also pedagogically sound. Modern evaluation frameworks highlight the importance of integrating these quantitative results with qualitative human oversight to ensure that system outputs remain interpretable, fair, and actionable for learners.

Beyond classification and generative metrics, agreement measures from earlier in this five-year period continue to provide a baseline for consistency. Measures such as Cohen's Kappa and error-based metrics like Root Mean Square Error (RMSE) are utilized to determine the level of alignment between machine-generated scores and human raters (Litman et al., 2021). Achieving a high level of agreement with human evaluators remains critical for establishing trust in AI-assisted assessment tools, particularly in high-stakes educational settings. This multi-metric strategy—integrating classification performance,



generative similarity, and human-machine agreement—provides a comprehensive baseline for validating AIED platforms, ensuring they support educators with transparency and rigor.

Synthesis

The literature and studies reviewed above provided the researchers with a broader understanding of automated essay evaluation, hybrid AI architectures, concept-based assessment, input modalities, academic integrity, and teacher workload, all of which are relevant to the development of EduCompose. These studies are similar to EduCompose because they also explore the use of artificial intelligence and natural language processing in improving writing assessment, feedback generation, and academic integrity (Tonmoy et al., 2025; Yang et al., 2024; Abu-Salih & Alotaibi, 2024). In particular, the reviewed literature supports the idea that automated tools can assist teachers in evaluating student writing more efficiently and more systematically (Magtalas & Eduvala, 2024; Litman et al., 2021).

However, EduCompose differs from most of the reviewed studies because it combines several diagnostic capabilities into one integrated, teacher-assisted system. While many of the studies focus on only one aspect of writing assessment, such as large language model reliability, OCR, automated scoring, knowledge graphs, or plagiarism detection, EduCompose synthesizes these functions into a single platform for essay evaluation (Kalai, 2025; Pawar et al., 2025; Schaller et al., 2024; Canal-Estevé & Gutiérrez, 2024; Khalil & Er, 2023). It also differs in that it does not aim to replace teacher judgment; instead, it presents diagnostic outputs as support for educators so that they can make the final assessment decisions based on both system results and their professional expertise (Holmes et al., 2023; Reich, 2023). In this way, EduCompose reflects a hybrid approach that balances automation with human oversight (Litman et al., 2021).

Overall, the reviewed studies established the technical feasibility of the individual components used in EduCompose, but they also revealed a gap in the literature: the absence of a unified, teacher-centered diagnostic platform designed specifically for essay evaluation. Existing systems often address isolated tasks rather than supporting the full workflow of essay analysis, feedback, and interpretation (Zhang et al., 2023; Abu-Salih & Alotaibi, 2024; Oducado et al., 2022). EduCompose addresses this gap by integrating multiple computational techniques into one explainable and practical system that helps



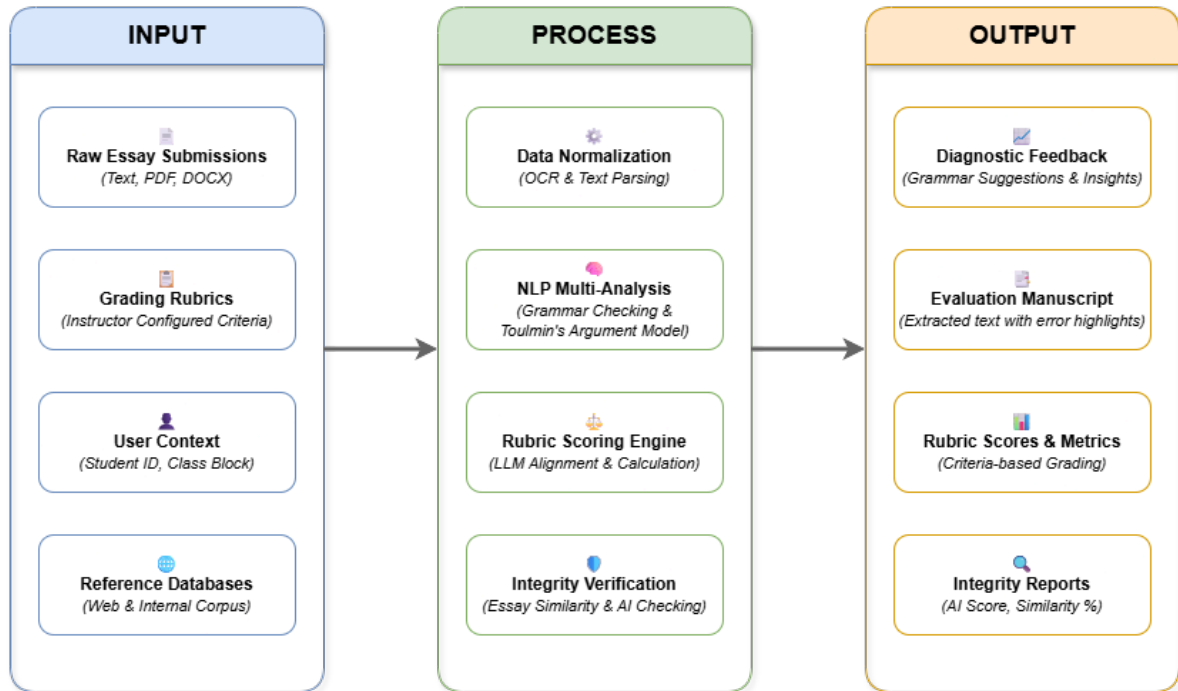
teachers evaluate essays more efficiently while still preserving their role as the primary decision-makers in the assessment process (Magtalas & Eduvala, 2024; Hattie & Clarke, 2024).

In summary, while the reviewed literature establishes the technical feasibility of individual components such as OCR, argument mining, and semantic mapping, there remains a lack of a unified diagnostic platform specifically tailored for the Philippine educational landscape. Most existing tools focus on autonomous grading which often excludes the teacher from the analytical loop. EduCompose fills this research gap by synthesizing Knowledge Graph-enhanced NLP and Toulmin's structural analysis into a single, teacher-centered dashboard. By bridging the divide between automated efficiency and pedagogical judgment, this study provides a more holistic and culturally relevant approach to essay evaluation, addressing both the technical accuracy of AI and the practical realities of teacher workload in local institutions.

Theoretical Framework

Figure 1.

Theoretical Framework



The Input–Process–Output (IPO) model serves as the theoretical foundation of this study, providing a structured framework for explaining how the EduCompose system transforms raw student essay data into meaningful analytical outputs. The IPO model is widely used in system development and research as it clearly illustrates the flow of data from initial input, through processing, to final output (McLeod, 2024; Laudon & Laudon, 2023).

In the input stage, the system receives various forms of data required for analysis. These include raw essay submissions in formats such as text, PDF, and DOCX; grading rubrics configured by instructors; user context such as student identification and class information; and reference databases, including web-based and internal corpora. These inputs serve as the foundational data that the system will analyze and evaluate.

The process stage represents the core functionality of EduCompose, where the input data is transformed through a series of computational procedures. First, data normalization is performed using OCR and text parsing to standardize the extracted content. This is followed by NLP multi-analysis, which includes grammar checking and argument structure evaluation based on Toulmin's Model. The system then applies a rubric scoring engine, aligning the analyzed text with predefined criteria to compute scores.



Lastly, integrity verification is conducted to assess essay originality through similarity checking and AI-based validation mechanisms. This stage demonstrates how the system processes raw input into structured and meaningful information.

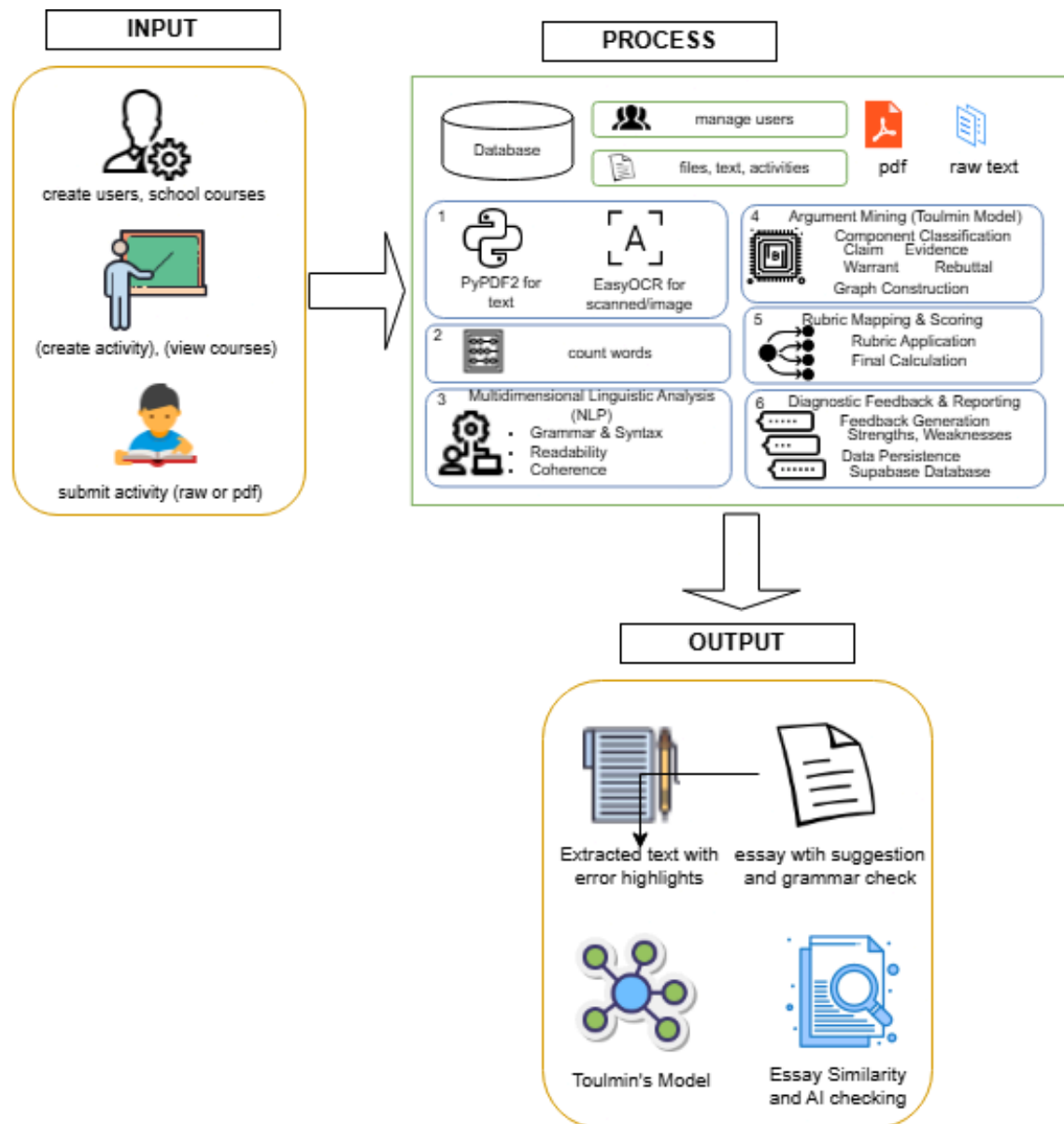
The output stage presents the results generated by the system after processing. These include diagnostic feedback, such as grammar suggestions and writing insights; an evaluation manuscript, which displays the processed text with highlighted issues; rubric scores and metrics, reflecting criteria-based grading; and integrity reports, which provide similarity percentages and AI detection indicators. These outputs are designed to support both teachers and students by delivering actionable insights and structured evaluation results.

Overall, the IPO model effectively represents how EduCompose converts diverse educational inputs into comprehensive outputs through a systematic and technology-driven process. It provides a clear visualization of how the system operates, ensuring that users can understand how data flows and is transformed within the platform (McLeod, 2024; Laudon & Laudon, 2023).

Conceptual Framework

Figure 2

Conceptual Framework



The conceptual framework for EduCompose, as illustrated in Figure 2, establishes a structured workflow designed to enhance teacher-assisted essay evaluation through a multi-stage digital pipeline. The process begins with the Data Input and Management



phase, where an administrator initializes the environment by creating users and school courses, while teachers set up specific activities within the Supabase-driven database. Students then interact with the system by submitting essays in either raw text or PDF format, which triggers the primary processing sequence. To ensure accessibility for all submission types, the system utilizes PyPDF2 for standard text extraction and EasyOCR for digitizing scanned or image-based handwritten work, effectively bridging the digital divide for students with limited device access.

Once the text is digitized, it enters the Multidimensional Analysis Engine, where the system performs several layers of automated scrutiny. This includes a linguistic analysis of grammar, syntax, readability, and coherence, alongside a structural evaluation based on the Toulmin Model to classify argumentative components like claims, evidence, and warrants. Simultaneously, the system generates a Knowledge Graph to visualize the conceptual organization and relational ideas within the student's writing. These analytical layers are then synthesized into rubric mapping and scoring, while also undergoing similarity and AI-content checking to maintain academic integrity.

The final stage of the framework focuses on Diagnostic Output and Feedback, delivering actionable evidence to the teacher via a specialized reporting interface. Educators are presented with the extracted text featuring error highlights, specific grammar suggestions, and a visual representation of the student's argumentative logic through the Toulmin Model and Knowledge Graph. This output is designed to strengthen, rather than replace, the teacher's expertise by providing deep diagnostic insights that inform targeted interventions. By persisting this data back into the Supabase database, the system ensures that long-term student progress can be tracked, ultimately reducing the manual burden on teachers while preserving high-quality formative assessment standards.



CHAPTER III

RESEARCH METHODOLOGY

This chapter discusses the methodologies utilized by the researchers in conducting the study. It presents the research design, population of the study, sampling design, data collection methods, and the statistical treatment of the data collected. These procedures guided the development and evaluation of EduCompose, a teacher-assisted essay evaluation system for Laguna University.

Research Design

This study employed a developmental and descriptive research design to guide the design, development, and evaluation of EduCompose. The developmental aspect of the study follows the principles of Design and Development Research (DDR), which involves the systematic design, development, and evaluation of instructional and non-instructional products to establish an empirical basis for their effectiveness. Recent literature highlights that DDR is appropriate for AI-driven educational systems as it supports iterative refinement within authentic learning environments (Tonmoy et al., 2025; Kalai, 2025).

In this study, the developmental component was implemented by first identifying the challenges encountered by teachers in essay evaluation. Based on these identified needs, a prototype system was designed and developed, integrating pre-trained Natural Language Processing (NLP) models, knowledge graph-based analysis, and a plagiarism-checking API. The system underwent iterative refinement through continuous testing and user feedback. Studies on large language model applications in education emphasize that integrating retrieval and verification mechanisms improves system reliability and reduces hallucination in educational feedback systems (Zhang et al., 2023; Tonmoy et al., 2025).

Complementing this, the descriptive aspect of the research was utilized to systematically describe and evaluate users' perceptions of the developed system. Descriptive quantitative methods were employed through survey questionnaires administered to teachers, students, and experts. Recent literature on AI-based educational systems highlights that descriptive evaluation is essential in assessing usability, functionality, and perceived usefulness without manipulating variables (Human-in-the-loop



Assessment with AI, 2025; Ethical AI for Teaching and Learning, 2023).

The integration of developmental and descriptive approaches enabled the researchers not only to design and refine EduCompose but also to evaluate how effectively the system met the needs and expectations of its intended users while maintaining alignment with ethical and pedagogical requirements in AI-assisted education (Toward Hybrid Teaching Intelligence, 2025; Ethical and Regulatory Challenges of Generative AI in Higher Education, 2025).

Population of the Study

The study was conducted at Laguna University, which was selected since it offers writing-intensive courses where English teachers regularly assess student essays and because the institution is the intended initial deployment site of EduCompose. The population of the study consisted of three groups directly related to the use and evaluation of the system: English teachers, students, and IT experts.

The first group was composed of English teachers from the College of Education and the College of Arts and Sciences who handle subjects with essay-based assessments such as Purposive Communication, The Contemporary World, and major English subjects. The College of Arts and Sciences has thirty-four (34) English teachers, while the College of Education has twenty (20) English teachers, for a total of fifty-four (54) English teachers. Because this number is relatively small and all qualified teachers are accessible, the researchers employed total enumeration (census) and included all fifty-four (54) English teachers as respondents in the study. Their participation was important in determining how the system could help reduce teacher workload and administrative burnout.

The second group consisted of students enrolled in writing-intensive subjects handled by these English teachers during the conduct of the study. For the purposes of sample size determination, the total number of officially enrolled students in Laguna University was 6,758, representing the student population from which respondents could be drawn. Because this population is large, the researchers determined the appropriate student sample size using Slovin's formula:



$$n = \frac{N}{1 + N \cdot e^2}$$

Where:

n = required sample size

N = population size

e = margin of error

Using $N = 6,758$ and a margin of error of $e = 0.10$ (10%), the computation is:

$$n = \frac{6,758}{1 + 6,758 (0.10)^2}$$

$$n = \frac{6,758}{1 + 6,758 (0.01)}$$

$$n = \frac{6,758}{1 + 67.58}$$

$$n = \frac{6,758}{68.58} \approx 98.56$$

Rounding up to the nearest whole number, the required sample size for the student group was ninety-nine (99) students. These 99 students served as the representative end-users who provided primary data in the form of essays and evaluated the clarity and usefulness of the system's feedback.

The third group was composed of three (3) IT experts outside of the institution who evaluated the system. Their assessment was essential in determining whether the developed system was technically sound, properly implemented, and capable of supporting the intended educational tasks. These experts were selected through purposive sampling based on their expertise in software development, AI-driven systems, and/or educational technology.

Overall, the study therefore included fifty-four (54) English teachers through total enumeration, ninety-nine (99) students determined using Slovin's formula from the 6,758-student population, and three (3) IT experts selected purposively, for a total of one



hundred fifty-six (156) respondents.

Table 1

Respondents of the Study

RESPONDENTS	NUMBER OF RESPONDENTS	PERCENTAGE
English Teachers	54	34.62%
Students	99	63.46%
IT Experts	3	1.92%
TOTAL	156	100%

Table 1 presents the categories of respondents in the study, consisting of English teachers, students, and IT experts. The teachers and students represent the primary target users of EduCompose in classroom essay evaluation, while the IT experts were included to assess the system's technical aspects. The respondents were selected based on their relevance to the objectives of the research, as they are the intended users and evaluators of the system, with teachers included through total enumeration, students through probabilistic sampling using Slovin's formula, and IT experts through purposive sampling.

Statistical Treatment of Data

To analyze and interpret the gathered data, the researchers utilized descriptive statistics, specifically percentage and weighted mean. The percentage distribution was applied to determine the proportion of each respondent group relative to the total number of respondents, while the weighted mean served as the primary tool to calculate the average ratings for each software quality characteristic under the ISO 25010:2023 standard. These numerical results were interpreted through a five-point Likert scale, allowing the researchers to objectively evaluate the functional suitability, performance, and overall effectiveness of the EduCompose system.



Percentage Distribution

The percentage distribution formula was used to determine the proportion of each category in relation to the total number of respondents or total data collected. It expresses each part as a percentage of the whole, making the data easier to interpret and compare. The formula is written as:

$$P = \frac{f}{N} \times 100$$

Where:

P = percentage

f = frequency (number of respondents in the group)

N = total number of respondents

Weighted Mean

The weighted mean was used to obtain the average of the responses for each statement in the survey questionnaire. This statistical tool is appropriate for Likert-type data, where each response option is assigned a numerical value and the average must reflect both the value and the frequency of responses. The general formula is shown below:

For a five-point Likert scale, this can be written as:

$$WM = \frac{(SA)(5) + (A)(4) + (MA)(3) + (D)(2) + (SD)(1)}{N}$$

Where:

SA	–	Strongly Agree
A	–	Agree
MA	–	Moderately Agree
D	–	Disagree
SD	–	Strongly Disagree



N – Total number of respondents

Likert Scale

A Likert scale was used to assess the respondents' level of agreement with each statement about EduCompose. The study adopted the following five-point scale:

Table 2

Likert Scale Interpretation Guide

Scale	Range	VERBAL INTERPRETATION
5	4.50 – 5.00	Strongly Agree
4	3.50 – 4.49	Agree
3	2.50 – 3.49	Moderately Agree
2	1.50 – 2.49	Disagree
1	1.00 – 1.49	Strongly Disagree

Data Collection Procedure

The research team conducted initial consultations with the thesis adviser to discuss the proposed system, clarify the objectives, and agree on the general direction of the project since Laguna University would serve as the research locale. During this phase, the preliminary features of EduCompose and the intended users were identified. The research instruments, particularly the survey questionnaires, were then drafted and submitted for content validation to ensure that the items were clear, relevant, and aligned with the research objectives.

After the instruments were validated, the researchers carried out informal interviews with selected English teachers from the College of Education and the College of Arts and Sciences to gather information about their current practices in checking essays and the difficulties they encounter in providing detailed feedback. Insights from these interviews helped refine the functional requirements of the system and confirm the need for diagnostic support in essay evaluation.

To support the design and justification of EduCompose, the research team also



reviewed existing thesis books and related studies from the Laguna University library and online sources, focusing on automated writing evaluation, AI-assisted systems, and local implementations of educational technologies. These documents contributed to the identification of relevant features and informed the Review of Related Literature.

For the system evaluation, the final phase of data collection involved administering survey questionnaires to the three groups of respondents: English teachers, students, and IT experts. The research team distributed the online survey link to all fifty-four (54) English teachers identified in the College of Education and the College of Arts and Sciences, to the randomly selected ninety-nine (99) students determined using Slovin's formula, and to the three (3) purposively selected IT experts. The participants were asked to use EduCompose by submitting or reviewing sample essays before answering the questionnaire.

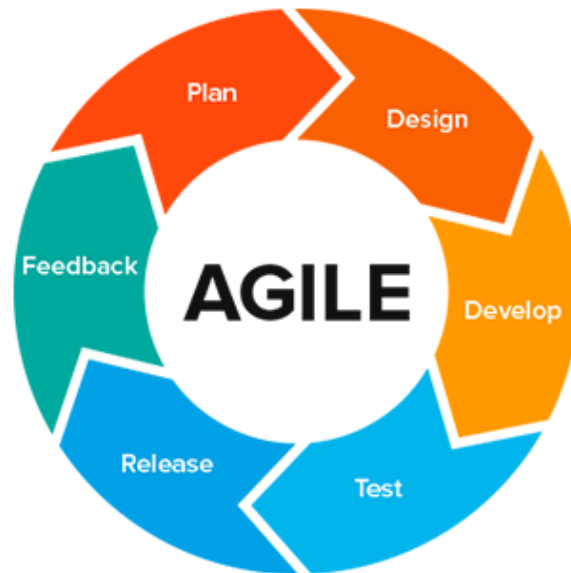
The instrument consisted of Likert-type items that measured perceptions of the system's usability, functionality, interface design, and overall usefulness. Responses were automatically recorded through the online form and later organized for statistical analysis using frequency, percentage, and weighted mean as described in the Statistical Treatment of Data section.

Software Methodology

The Agile Development Methodology was used in the development of EduCompose because it supports iterative design, continuous testing, and gradual refinement of system features. This approach is appropriate for the study since EduCompose integrates several complex components, including structural argument mining (Toulmin Mapping), diagnostic feedback generation, knowledge graph analysis, and plagiarism checking, which may require adjustments as development progresses. As part of the system's development under the Agile methodology, a Heuristic Rubric Mapping algorithm was implemented. This allows the system to automatically analyze textual data and map them into specific rubric criteria, ensuring that the automated feedback is both diagnostic and quantitative.

Figure 3

Agile Development Model



Agile Phases in EduCompose Development

Planning Phase

During the planning phase, the research team identified the main problems in essay evaluation that EduCompose should address, such as the need for faster feedback, better conceptual analysis, and support for plagiarism checking. The features of the system were prioritized based on the needs of English teachers, students, and IT experts. This phase ensured that the goals of the system were aligned with the actual requirements of the users before development began.

Software Project Schedule

The project adhered to an extended 16-week timeline (September 15, 2025 to April 22, 2026), culminating in a final defense scheduled for April 22, 2026. The schedule reflects the iterative nature of Agile development, with significant architectural additions occurring during the final phases in response to panel feedback, concluding with final refinement and



defense preparation through April 22,2026.

Figure 4

Gantt Chart

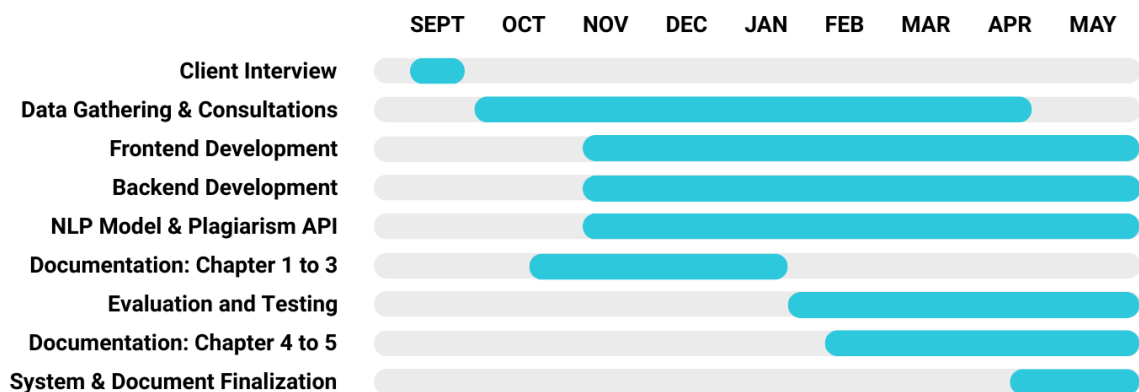


Figure 4 shows the project schedule for EduCompose follows a strategic nine-month timeline, beginning in September 2025 and concluding in May 2026. The initial phase focused on foundational research, starting with a Client Interview in September to establish system goals, followed by Data Gathering and Consultations in October. These early activities informed the drafting of the first half of the thesis, with Documentation for Chapters 1 to 3 spanning from mid-October through January. This sequence ensured that the theoretical framework and user requirements were fully established before the transition into intensive technical development.

The core engineering phase occurred concurrently between November and April, representing the most active period of the project. Frontend and Backend Development were executed simultaneously from November through March to build a cohesive user interface and database infrastructure. Parallel to this, the development of the NLP Model and Plagiarism API began in November and extended into April, allowing for the complex integration of semantic analysis and diagnostic tools. This overlapping schedule permitted iterative testing between the system's architecture and its analytical capabilities, ensuring that the software components functioned as a unified diagnostic platform.

The final quarter of the timeline was dedicated to validation and academic finalization. Evaluation and Testing commenced in February and ran through mid-April to



verify the system's diagnostic accuracy against teacher-led standards. As the technical build reached completion, the focus shifted to the final manuscript, with Documentation for Chapters 4 and 5 drafted between April and May to present the study's findings and analysis. The project concludes in May with System and Document Finalization, ensuring that both the EduCompose application and the complete thesis are polished and ready for institutional submission.

Design Phase

The design phase focused on organizing the overall structure of EduCompose, including the frontend interface, backend processing modules, database, and AI components. At this stage, the research team mapped how student essays would be submitted, processed, analyzed, and presented as diagnostic results. This phase helped establish a modular architecture that could support the system's essay analysis, knowledge graph generation, and reporting features.

Figure 5

Project Context Diagram

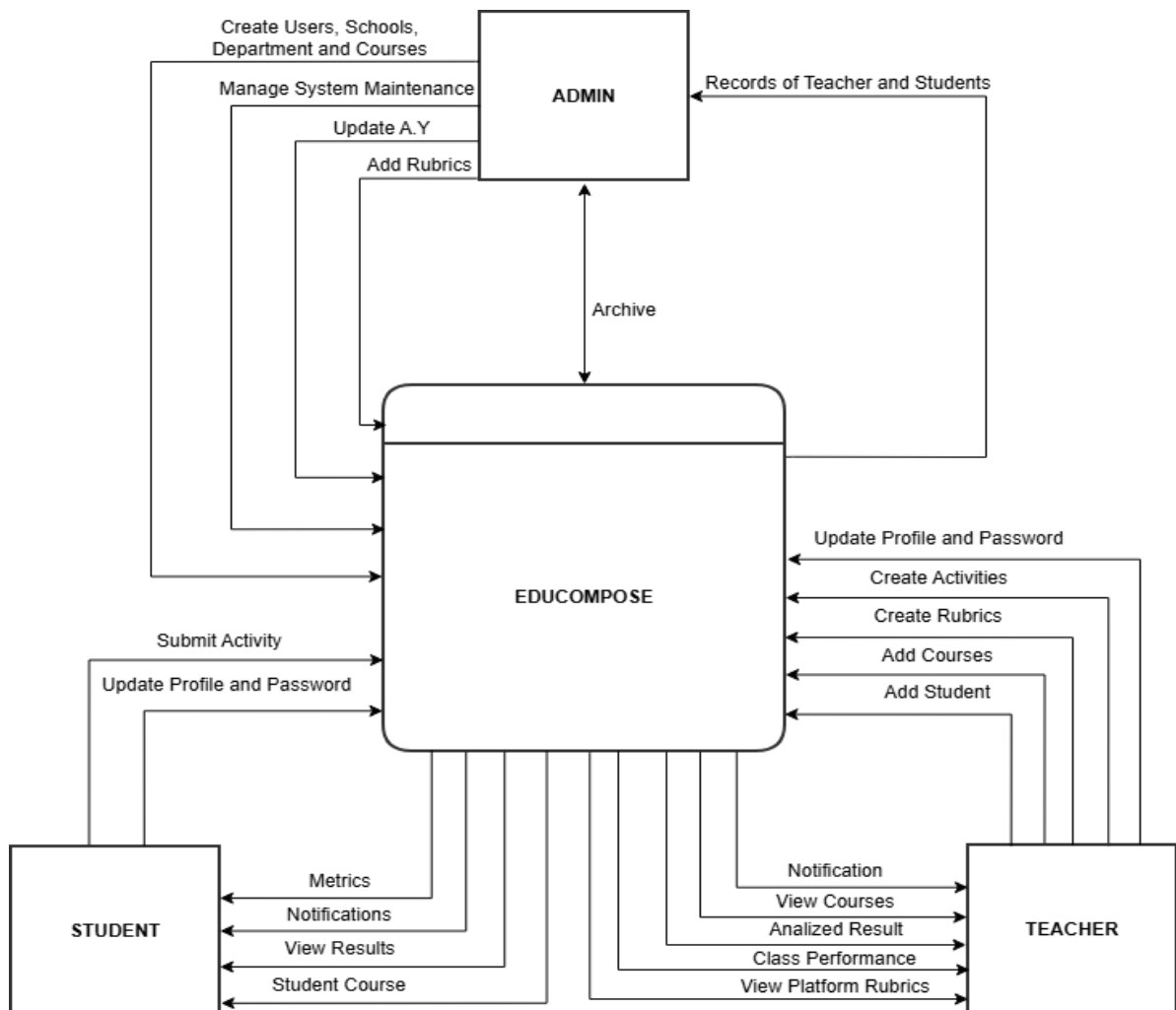


Figure 5 illustrates the high-level boundaries of the system and its interactions with three primary external entities: the Admin, the Teacher, and the Student. At the center of the architecture is the EduCompose engine, which serves as the primary processing hub for all data exchanges. The Admin entity is responsible for system-wide configurations, including the creation of users, schools, and departments, while receiving summarized records and managing the archival of data. The Teacher entity interacts with the system to design the educational environment by creating activities, adding courses, and defining



rubrics; in return, the system provides them with analyzed results, class performance metrics, and automated notifications. Meanwhile, the Student entity initiates the core data pipeline by submitting activities and updating their profiles, subsequently receiving diagnostic feedback, results, and specific course metrics from the system. This diagram highlights the system's role as a centralized diagnostic intermediary that facilitates a streamlined, evidence-based feedback loop between educators and learners.

Figure 6.

Admin Flowchart

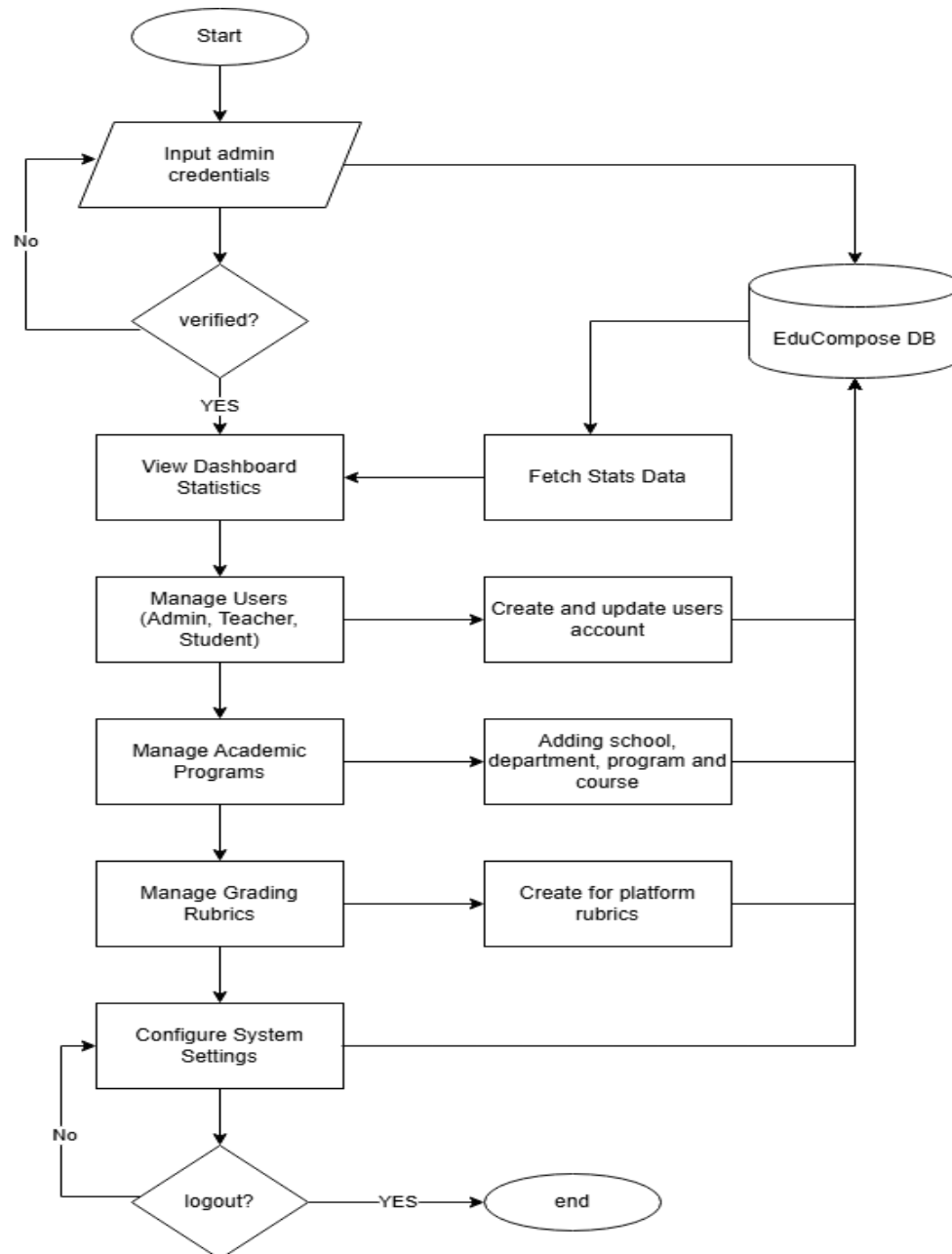


Figure 6 illustrates the admin process starts with a secure login that verifies admin credentials against the EduCompose DB. Upon successful verification, the administrator accesses a dashboard that displays high-level statistics and system data fetched from the database. The admin holds the primary responsibility for user management, including the creation and updating of admin, teacher, and student accounts. Beyond user oversight, the admin manages the academic structure by adding schools, departments, programs, and



courses, while also defining the grading rubrics used across the platform. All system configurations and management actions are recorded in the central database to ensure platform consistency.

Figure 7

Teacher Flowchart

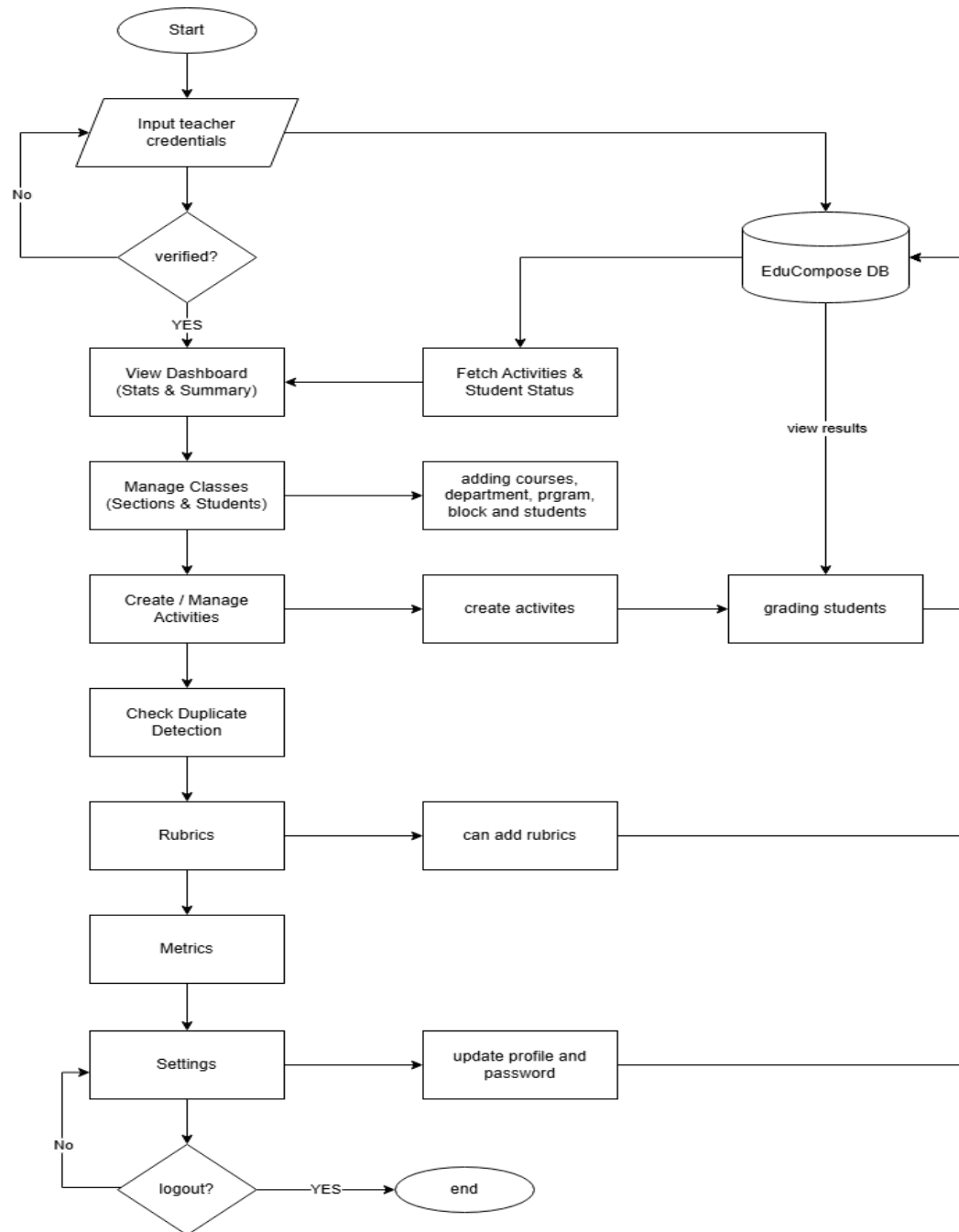


Figure 7 illustrates the teacher workflow initiates with a credential verification step that connects to the EduCompose DB to grant access. The teacher's dashboard provides a comprehensive view of activities and student statuses by fetching real-time data from the system. Teachers are equipped with tools to manage classes, which includes organizing departments, programs, and specific student blocks. The core of the teacher's role involves

creating and managing activities, which then moves into a grading phase where they can view student results and apply specific rubrics. Before final data entry, the system checks for duplicate submissions, ensuring that all metrics and profile updates are accurately reflected in the database.

Figure 8.

Student Flowchart

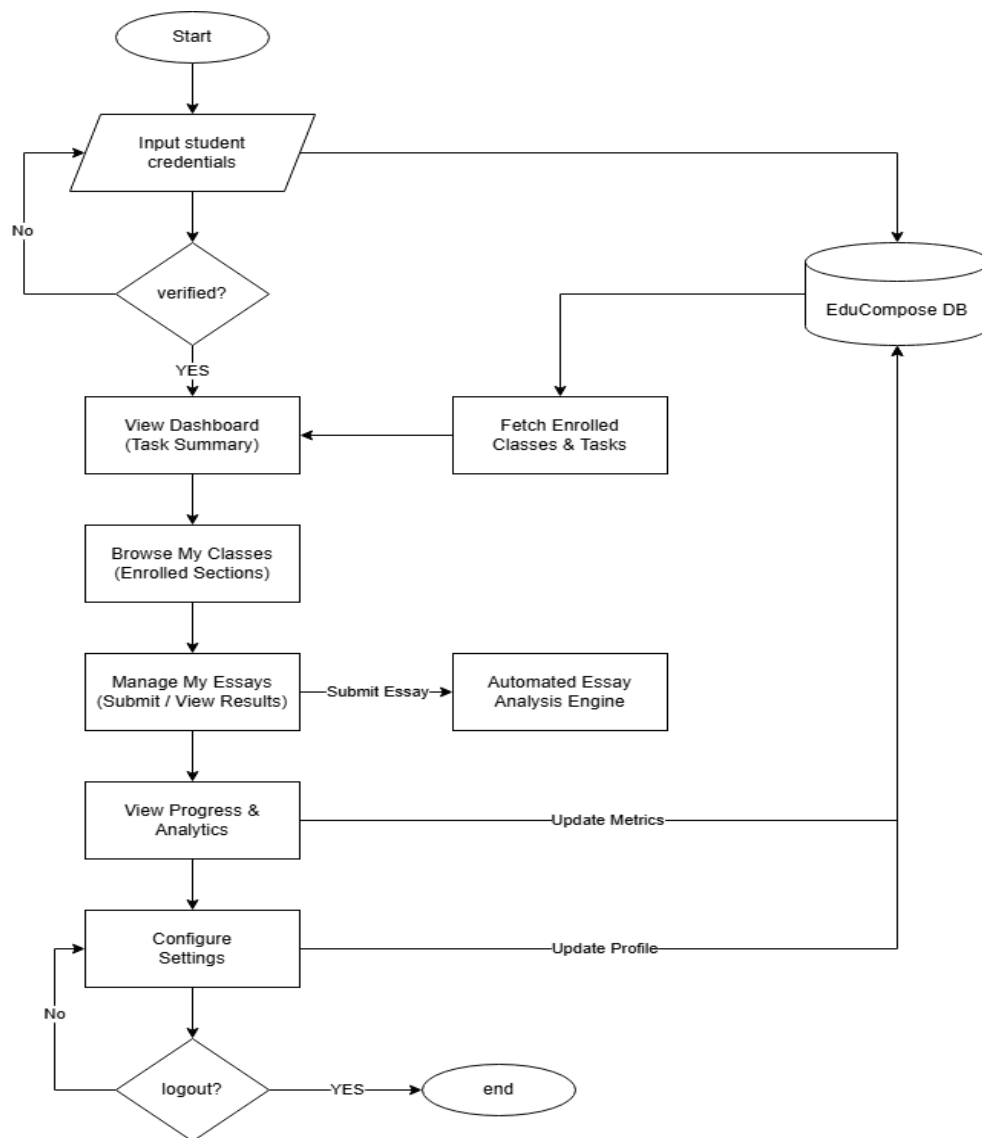


Figure 8 illustrates the student workflow begins when a user enters their credentials into the login interface, which are then verified against the EduCompose DB. Once

authenticated, the system fetches enrolled classes and tasks to populate the student's dashboard with a task summary. From here, the student can browse their enrolled sections and navigate to the essay management module. In this module, students submit their work to the Automated Essay Analysis Engine, which processes the text and provides results. Finally, the system allows the student to view progress analytics and configure profile settings, with all updates and metrics being synchronized back to the database before the user logs out.

Figure 9

Admin Data Flow Diagram

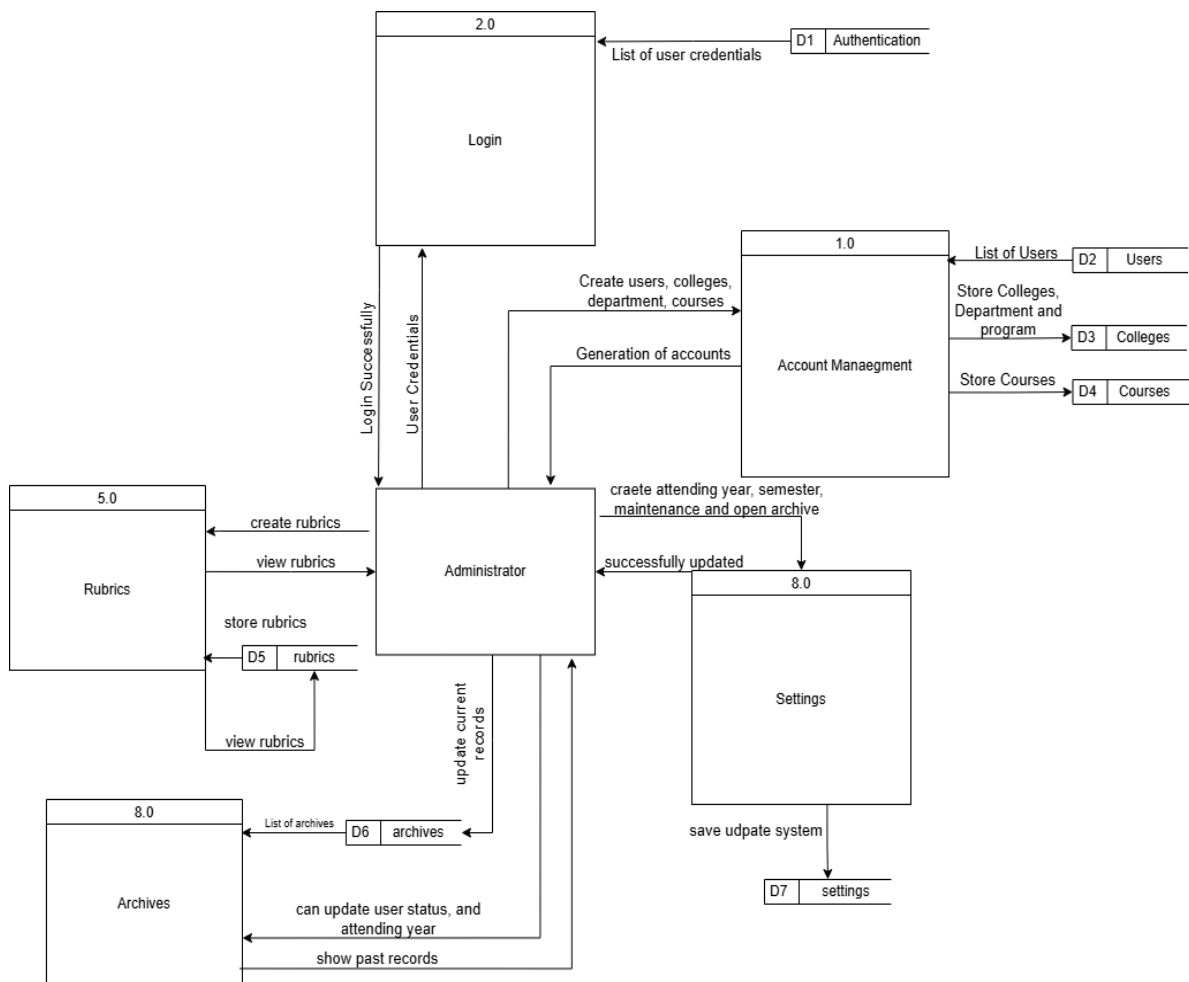


Figure 9 illustrates the administrator's interaction with the platform is initiated



through the **Login** process, utilizing the **Authentication (D1)** data store to validate administrative rights. Once authenticated, the admin manages the foundational data of the platform through **Account Management**, which involves creating users, colleges, and departments, and storing this information in the **Users (D2)**, **Colleges (D3)**, and **Courses (D4)** data stores. This centralized management ensures that all accounts and academic structures are properly generated and categorized for the entire institution.

In addition to user oversight, the administrator controls the evaluative and historical aspects of the system. In the **Rubrics** module, they create and view standardized criteria which are maintained in the **rubrics (D5)** data store. The admin also oversees the **Archives** section, where they can update user statuses and attending years or retrieve past records from the **archives (D6)** store. Finally, the **Settings** module allows the administrator to manage maintenance tasks, such as opening archives or updating semester details, ensuring all system configurations are saved to the **settings (D7)** data store.

Figure 10

Teacher Data Flow Diagram

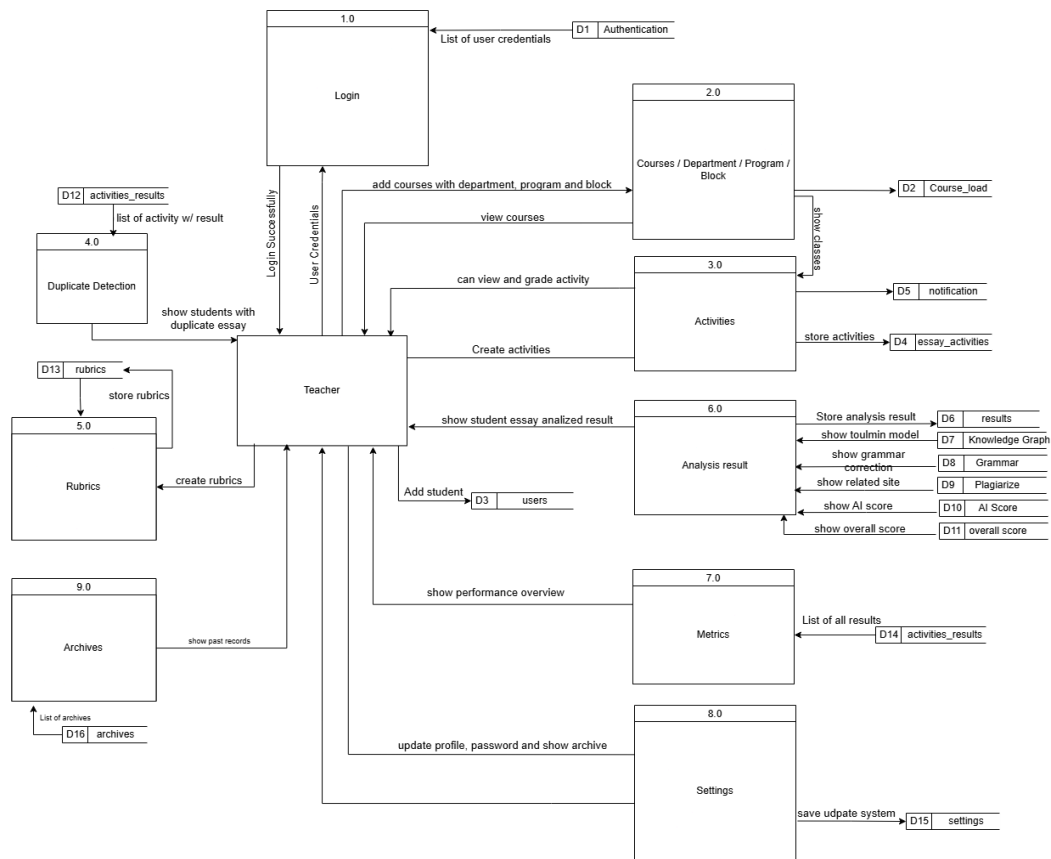


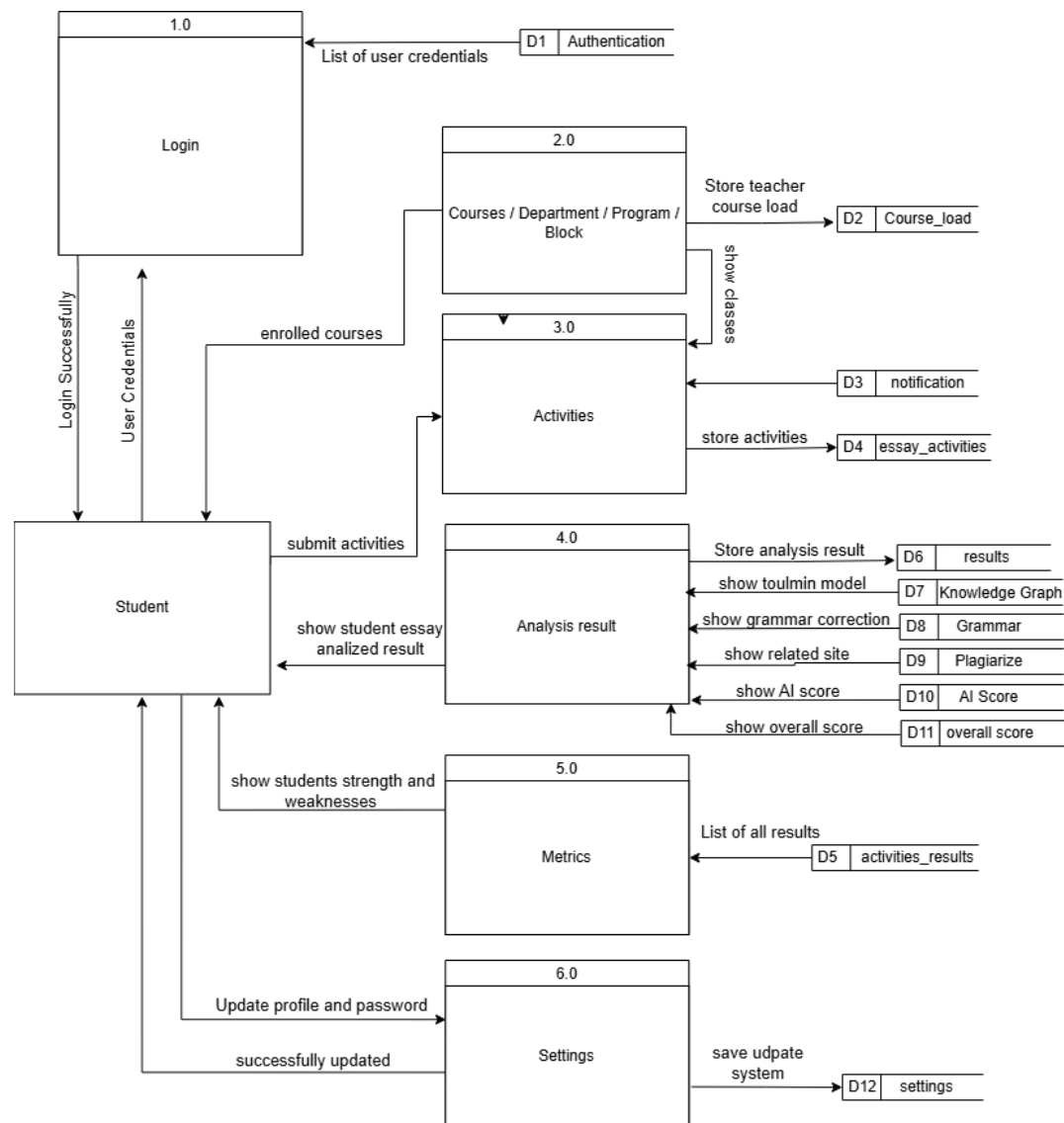
Figure 10 illustrates the teacher workflow commences with the authentication process, where user credentials are verified against the **Authentication (D1)** data store to grant access to the system. Once logged in, teachers manage the academic structure by adding and viewing details within the **Courses / Department / Program / Block** module, which stores the teacher's specific course load in **Course_load (D2)**. Teachers create and grade classroom tasks through the **Activities** process, triggering notifications in **notification (D5)** and recording activity details in **essay_activities (D4)**. To maintain academic integrity, the **Duplicate Detection** module identifies students with matching essays using the **activities_results (D12)** data.

Furthermore, teachers define evaluation standards in the **Rubrics** section, storing these criteria in the **rubrics (D13)** data store for future use. The **Analysis result** module provides detailed feedback on student work, pulling from multiple data points such as **Knowledge Graph (D7)**, **Grammar (D8)**, **Plagiarism (D9)**, and **AI Score (D10)** to show an overall score from **overall score (D11)**. Performance oversight is handled by the **Metrics**

process, which generates overviews from the **activities_results (D14)**. Finally, teachers can manage their account via **Settings**, saving system updates to **settings (D15)**, and accessing historical data through the **Archives** module which retrieves past records from **archives**.

Figure 11

Student Data Flow Diagram



The student journey begins at the **Login** module, where credentials are sent for verification against the **Authentication (D1)** data store. Upon successful entry, the student



interacts with the **Courses / Department / Program / Block** module to view their specific enrolled courses, which are populated based on the **Course_load (D2)** data. Students engage with the learning material by submitting tasks through the **Activities** process, which subsequently updates the **essay_activities (D4)** store and may trigger relevant **notifications (D3)**.

Once an activity is processed, students receive feedback via the **Analysis result** module, which displays a comprehensive breakdown including the Toulmin model from **Knowledge Graph (D7)**, **Grammar (D8)** corrections, and an **AI Score (D10)**. Academic growth is tracked in the **Metrics** section, where the system pulls from **activities_results (D5)** to highlight the student's specific strengths and weaknesses. The workflow concludes in the **Settings** module, allowing users to update their profile and password, with all system-level changes being recorded in the **settings (D12)** data store.

Figure 12

Entity Relationship Diagram

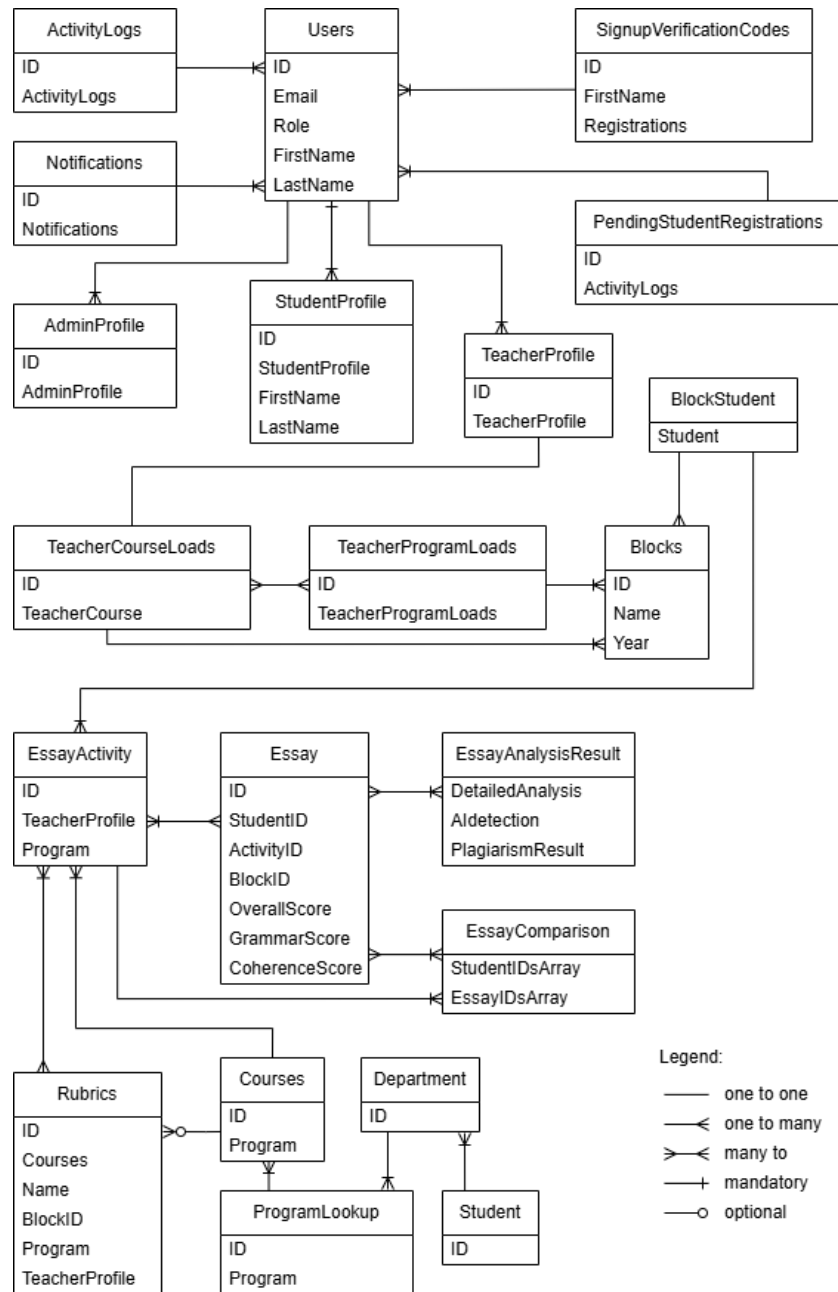


Figure 12 shows the Entity Relationship Diagram which serves as the database blueprint for EduCompose, detailing how data entities are structured and connected within the Supabase environment. The architecture is centered on a primary "Users" table that manages core authentication data like emails and roles. This central entity maintains strict



one-to-one relationships with specialized profile tables for admins, teachers, and students, ensuring data integrity. Furthermore, the diagram illustrates how essays and activities are linked through one-to-many relationships, allowing the system to track individual submissions within specific classroom blocks and programs.

Development Phase

The development phase involved the actual coding and integration of the modules needed for EduCompose. The frontend was built using React with TypeScript, powered by Vite as the build tool, and used React Router for navigation, Redux for state management, and Framer Motion for animations. The backend was developed in Python using the FastAPI framework served via Uvicorn, with Pydantic for data validation and SQLAlchemy for database interaction. The database and authentication layer was handled by Supabase. For natural language processing and essay analysis, the team used spaCy, NLTK, TextBlob, Sentence Transformers (built on PyTorch), and scikit-learn, while textstat handled readability scoring. Knowledge graph features made use of NetworkX and PyVis for construction and visualization. Additional services included OpenCV and Pillow for image processing, pdf2image and PyPDF2 for document handling, and Tesseract (via OCR service) for optical character recognition. The research team developed these components incrementally so that each could be tested and improved as it was added, ensuring the system could process essays, generate feedback, and display results in a way that was useful for teachers and students.

Testing Phase

The Testing Phase was conducted to verify whether EduCompose functions correctly and produces accurate results. Individual modules, such as the essay upload, NLP analysis, diagnostic feedback, and plagiarism-checking components, were tested through unit testing to ensure that they could process input data properly. Integration testing was then performed to confirm that all parts of EduCompose worked together as intended.

The system was tested by the researchers using sample student essays of varying lengths and error types. During testing, issues such as delayed processing for very long essays and minor user interface glitches were identified and corrected.



Development Phase

The Development phase involved launching the EduCompose system into a live production environment for end-user access. The front-end application was deployed using Vercel to ensure fast and responsive user interfaces, while the back-end services were hosted on Railway to handle secure API requests and server-side processing. The fine-tuned NLP model was stored and managed via Hugging Face to facilitate seamless text analysis capabilities. Finally, a dedicated .online domain was procured through Hostinger to provide a professional, unified, and accessible web address for the teachers and students of Laguna University.

Review Phase

In the review phase, the research team evaluated the system based on feedback from the thesis adviser, target users, and technical experts. In addition, comments and suggestions from the panel members during the initial defense were carefully considered, which led to several revisions before the re-defense. These revisions helped improve the interface, adjust system features, and refine the diagnostic output of EduCompose. This phase ensured that the system was better aligned with user expectations, panel recommendations, and the overall objectives of the study.

Hardware and Software Resources

Hardware Resources

Table 3.



Hardware and Specifications

Hardware	Specifications
Laptop 1 (Front-End)	RAM: 32.0 GB (31.3 GB usable) Processor: AMD Ryzen 5 5600H GPU: Nvidia RTX 3050 4GB Operating System: Windows 11 pro
Personal Computer 1 (Back-End)	RAM: 16GB Processor: Intel(R) Core(TM) i7-7700 GPU: NVIDIA GeForce GTX 1030 3GB Operating System: Windows 11 pro
Laptop 2 (Back-End)	RAM: 16.0 GB Processor: Intel I7- 6 th gen GPU: NVIDIA GeForce 930mx 2GB Operating System: Windows 11 pro
Laptop 3 (Front-End)	RAM: 8.00 GB Processor: Ryzen 3 7320U AMD Radeon Graphics Operating System: Windows 11 pro

Table 3 shows the specifications of the hardware devices used in the development of EduCompose. The main workstation used by the lead developer was a laptop with Windows 11 Pro, Intel Core i7 processor, 16 GB RAM, and NVIDIA GeForce 930MX graphics. Other devices, including a laptop with Windows 11 Pro, AMD Ryzen 5 5600H processor, 32 GB RAM, and NVIDIA RTX 3050 GPU, were also used as supporting hardware for additional development and testing tasks.

Software Resources

Table 4
Software and Tools



Category	Software/Tools
Programming Languages	Python 3.11+, JavaScript (ES6+), TypeScript, HTML5, CSS3
Version Control / Collaboration Tools	Git, GitHub
Web Framework	FastAPI (Backend), React 19 (Frontend), React Router (react-router-dom), Supabase (@supabase/supabase-js)
NLP & Text Processing	spaCy, NLTK, textstat, sentence-transformers
Machine Learning & AI	transformers, PyTorch, scikit-learn, Pandas, NumPy, SciPy
LLM APIs	OpenAI API (GPT-4o-mini), Google Gemini API (gemini-2.5-flash/pro)
Graph & Network Analysis	NetworkX
OCR & Document Processing	EasyOCR, pdf2image (with Poppler), PyPDF2, Pillow (PIL)
Plagiarism Detection	Copyscape API
Database	SQLAlchemy, Supabase
Authentication	python-jose, passlib
Frontend Framework	React 19, Vite
UI & Styling	Tailwind CSS, PostCSS, Autoprefixer, Framer Motion, Radix UI
Data Visualization	Chart.js, Recharts, react-force-graph-2d
Icons & Components	lucide-react
Code Editor & IDE	Visual Studio Code
Python Environment	venv (Python Virtual Environment)



Frontend Build Tool	Vite, npm (Node Package Manager)
Backend Package Manager	pip (Python Package Manager)
Backend Server	Uvicorn
Notifications	EmailJS
Deployment & Hosting	Vercel, Docker, Railway, Hugging Face, Hostinger

Table 4 presents the software resources and tools used in the development of EduCompose. The system was developed using Python 3.11+, JavaScript, TypeScript, HTML5, and CSS3, with FastAPI serving as the backend framework and React 19 as the frontend framework. Other tools such as Git and GitHub were used for version control and collaboration, while Visual Studio Code served as the primary code editor and development environment. The project also used Vite, npm, and pip to support frontend and backend building and package management.

For text processing, natural language processing, and machine learning, the system used spaCy, NLTK, textstat, sentence-transformers, transformers, PyTorch, scikit-learn, Pandas, NumPy, and SciPy. The application also integrated OpenAI API and Google Gemini API for language model support, NetworkX for graph analysis, EasyOCR and other document-processing tools for handling PDF and image inputs, and Copyscape API for plagiarism detection. In addition, SQLAlchemy, and Supabase were used for database management and authentication, while Tailwind CSS, Framer Motion, Radix UI, Chart.js, Recharts, and react-force-graph-2d were used for interface design and data visualization. The system was deployed using Vercel, Docker, Railway, Hugging Face, and Hostinger to make it accessible and functional online.



CHAPTER IV

RESULTS AND DISCUSSIONS

This chapter presents, analyzes, and interprets the data gathered from the system demonstrations and on-the-spot survey evaluations. It provides a comprehensive perspective on the developed software, EduCompose, detailing its performance as an AI-driven, teacher-assisted essay evaluation system for Laguna University. The findings are organized according to the specific objectives of the study to determine whether the system successfully meets the identified educational requirements and to serve as a guide for future software improvements.

1. Problems Encountered by English Language Teachers:

This section presents the assessment of English language teachers from the College of Education (COED) and College of Arts and Sciences (CAS) regarding the challenges they face in manual essay evaluation.

Table 5

Challenges Encountered in Manual Essay Evaluation

INDICATORS	WEIGHTED MEAN	VERBAL INTERPRETATION
1. Manual checking of student essays is highly time-consuming.	4.75	SA
2. Large class sizes make it difficult to provide detailed feedback for every student.	4.85	SA
3. Lack of specialized diagnostic tools to analyze the structural logic of an argument.	4.60	SA
4. Maintaining grading consistency across multiple sections is challenging.	4.40	A
5. Educators experience instructional fatigue due to heavy documentation workloads.	4.70	SA
Grand Mean	4.66	SA

Legend: SA – Strongly Agree (4.50-5.00), A – Agree (3.50 – 4.49), MA – Moderately Agree (2.50 – 3.49), D – Disagree (1.50 – 2.49), SD – Strongly Disagree (1.00 – 1.49), WM – Weighted Mean, VI – Verbal Interpretation.

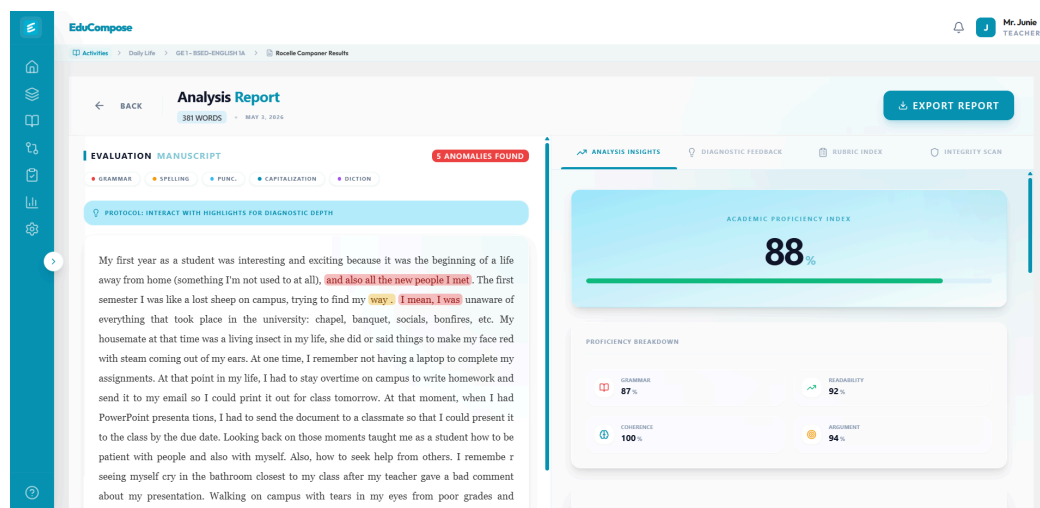
As shown in Table 5, the English language teachers at Laguna University identified significant challenges in the current manual essay evaluation process, resulting in a Grand Mean of 4.66 (Strongly Agree). The highest-rated indicators were the large class sizes (4.85) and the time-consuming nature of manual checking (4.75). These findings highlight a critical gap in the instructional workflow, where teachers are burdened by volume and lack the diagnostic tools (4.60) necessary for deep structural analysis. This data serves as the primary justification for the development of EduCompose, aiming to automate the analytical aspects of writing assessment and alleviate teacher burnout.

2. Develop EduCompose with the following features:

2.1.Extraction of multidimensional diagnostic metrics such as grammar, readability, coherence, and argument structure;

Figure 13

Multidimensional Diagnostic Metrics of the Essay



EduCompose was developed to analyze student essays and generate diagnostic feedback along several dimensions: grammar, readability, coherence, and argument structure. After an essay is submitted through the upload dashboard, the system applies natural language processing techniques to detect grammatical errors, inappropriate word

choices, and sentence-level issues. It then assigns a grammar score that indicates the severity and frequency of such problems.

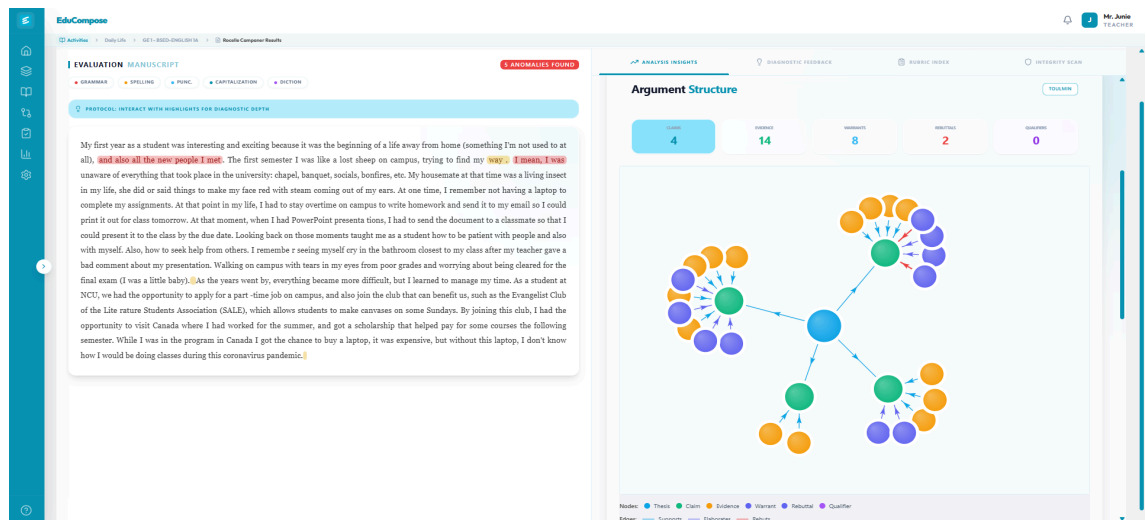
For readability, EduCompose calculates an estimate based on sentence length, word complexity, and paragraph structure. This score helps teachers quickly determine whether the essay is appropriate for the target grade level or if the language is too simple or too complex. In terms of coherence, the system evaluates how smoothly ideas connect within and across paragraphs, identifying abrupt transitions or missing links between sentences.

Finally, EduCompose attempts to recognize the argument structure of the essay by identifying claims, supporting evidence, and conclusions. It flags essays where claims are weakly supported or where the overall structure lacks a clear progression.

2.2. Semantic Network Visualizers (Knowledge Graphs);

Figure 14

Knowledge Graph visualization of an essay



To address the need for deeper conceptual analysis, EduCompose generates semantic network visualizations in the form of Knowledge Graphs. The system extracts key concepts from the essay (such as main topics, supporting ideas, and examples) and arranges them as nodes connected by edges that represent semantic relationships. This visualization allows teachers to see at a glance how well the student connects ideas and

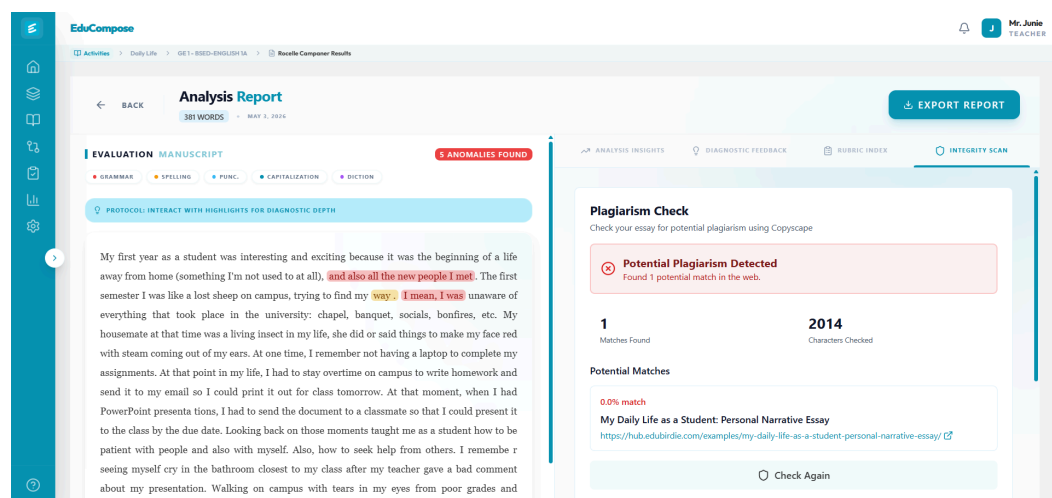
whether the essay presents a coherent network of related concepts rather than isolated statements.

Teachers who viewed the Knowledge Graphs during the demonstration commented that the visualizations made it easier to discuss the student's thinking process and to suggest ways to improve the logical flow of ideas. Students also found the graphs interesting and motivating, as they could see how their ideas were being interpreted and connected by the system.

2.3. Plagiarism and Similarity Checking;

Figure 15

Plagiarism and Similarity Checker of an essay



EduCompose also includes a plagiarism and similarity-checking module designed to help teachers identify copied or heavily paraphrased content. The system compares the submitted essay against a set of reference sources (or a database of prior student work, depending on the implementation) and returns a similarity score. Parts of the text that closely match external sources are highlighted so that teachers can review them more carefully.

In the demonstration, EduCompose was able to detect obvious copied passages and flag them with a higher similarity score. For paraphrased sections, the system focused on structural and lexical overlap, providing a moderate score that signals potential issues



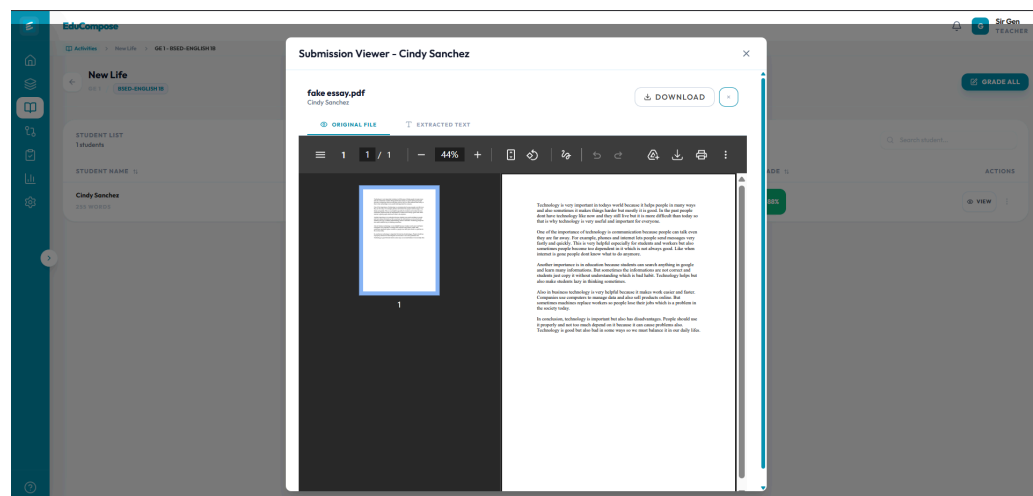
without automatically accusing the student of misconduct. This allows teachers to use their professional judgment when interpreting the results.

Respondents noted that the plagiarism-checking feature is valuable as it reduces the need for manual copy-and-paste searches on external tools. However, they also emphasized that the system should be used as a support tool rather than the sole basis for academic dishonesty decisions, since some similarities may arise from legitimate sources or common phrases.

2.4. Extraction of text using Optical Character Recognition (OCR);

Figure 16

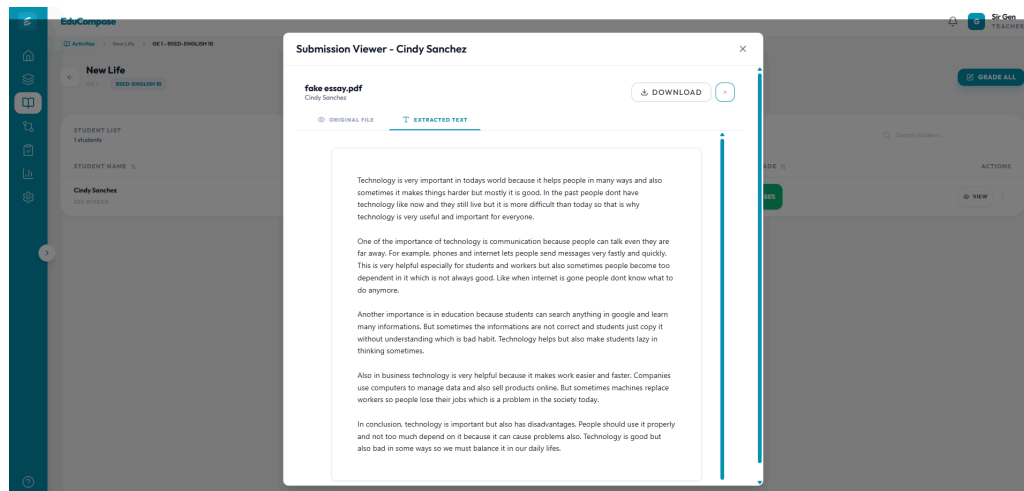
PDF File of the Student



The PDF file submitted by the student is displayed in the Submission Viewer of EduCompose, where the teacher can review the uploaded essay before grading. In the interface, the system shows the file name, the student's name, and a preview of the PDF document so that the teacher can examine the actual content of the submission

Figure 17

Extracted Text from the File



The extracted text view shows the content of the student's uploaded essay after it has been processed by the system.

- The proposed system was evaluated in terms of AI model performance using quantitative metrics: Precision, Recall, and F1-Score for argument categorization and grammar analysis, and ROUGE and BERTScore for the semantic quality of the generated feedback.

Table 6

Categorization Module Performance (Argument Components)

METRIC	SAMPLES	SCORE
Precision	28,859	0.7903
Recall	28,859	0.7901
F1 Score	28,859	0.7893

Table 6 presents the quantitative performance evaluation of the categorization module (DistilBERT) based on a sample size of 28,859 sentences. The module achieved a



balanced F1-score of 0.7893, indicating high reliability in identifying argumentative structures in student writing.

Table 7

LLM-Based Grammar and Style Analysis Performance

METRIC	SAMPLES	SCORE
Precision	25	0.94
Recall	25	0.89
F1 Score	25	0.91

Table 7 presents the performance of the LLM-powered grammar analysis module. Unlike traditional rule-based checkers, the system uses Large Language Models (LLMs) to identify nuanced stylistic and grammatical errors. When compared against manual human grading on a sample of 25 essays, the module achieved a high Precision of 0.94 and an F1-score of 0.91, proving its effectiveness in providing high-quality writing feedback.

Table 8

Feedback Generation Performance

METRIC	SAMPLES	SCORE
ROUGE-L	15	0.7245
BERTScore	15	0.9412
Semantic Similarity	15	0.89

Table 8 presents the performance evaluation of the feedback generation module. Using ROUGE-L and BERTScore, the system achieved scores of 0.7245 and 0.9412 respectively, indicating that the generated feedback is highly similar in content and meaning to professional academic standards.



4. Evaluate the overall performance of the developed EduCompose using the software quality standard ISO 25010 in terms of functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety.

4.1. Functional Suitability

Table 9

Evaluation of the respondents to the functional suitability of EduCompose

FUNCTIONAL SUITABILITY	STUDENTS	ENGLISH TEACHERS	IT EXPERTS	OVERALL WM	VI
1. The system provides all necessary functions for digital essay evaluation.	4.80	4.70	4.30	4.72	SA
2. The system accurately processes and analyzes the submitted text from the user.	4.70	4.60	4.30	4.63	SA
3. The system generates clear and accurate diagnostics useful to teachers.	4.80	4.80	4.60	4.79	SA
Average Weighted Mean	4.77	4.70	4.40	4.71	SA

Legend: SA – Strongly Agree (4.50-5.00), A – Agree (3.50 – 4.49), MA – Moderately Agree (2.50 – 3.49), D – Disagree (1.50 – 2.49), SD – Strongly Disagree (1.00 – 1.49), WM – Weighted Mean, VI – Verbal Interpretation.

Table 9 presents the evaluation of the respondents regarding the functional suitability of EduCompose. The statement "The system generates clear and accurate diagnostics useful to teachers" got the highest overall weighted mean of 4.79, with a verbal interpretation of Strongly Agree. The overall average weighted mean for functional suitability is 4.71 with a verbal interpretation of Strongly Agree. This shows that the respondents agreed that EduCompose has the necessary features and accuracy to process student essays.



4.2 Performance Efficiency

Table 10

Evaluation of the respondents to the performance efficiency of EduCompose

PERFORMANCE Efficiency	STUDENTS	ENGLISH TEACHERS	IT EXPERTS	OVERALL WM	VI
1. The system responds quickly to user actions and inputs.	4.70	4.50	4.30	4.58	SA
2. The system processes digital essays and AI diagnostics within an acceptable time.	4.60	4.60	4.00	4.56	SA
3. The system handles multiple users efficiently without lagging.	4.50	4.40	4.30	4.44	A
Average Weighted Mean	4.60	4.50	4.20	4.53	SA

Legend: SA – Strongly Agree (4.50-5.00), A – Agree (3.50 – 4.49), MA – Moderately Agree (2.50 – 3.49), D – Disagree (1.50 – 2.49), SD – Strongly Disagree (1.00 – 1.49), WM – Weighted Mean, VI – Verbal Interpretation

Table 10 shows the evaluation of the respondents on the performance efficiency of EduCompose. The statement "The system handles multiple users efficiently without lagging" got the lowest weighted mean of 4.44, with a verbal interpretation of Agree. However, the overall average weighted mean is 4.53 with a verbal interpretation of Strongly Agree. This indicates that the respondents agreed that the system uses resources properly and responds fast during the analysis process.



4.3 Compatibility

Table 11

Evaluation of the respondents to the compatibility of EduCompose

COMPATIBILITY	STUDENTS	ENGLISH	IT	OVERALL	
		TEACHERS	EXPERTS	WM	VI
1. The system is accessible across different web browsers.	4.90	4.80	4.60	4.83	SA
2. The system adapts smoothly to various screen sizes.	4.60	4.70	4.30	4.72	SA
3. The system handles consistent functionality across different devices.	4.80	4.80	4.60	4.79	SA
Average Weighted Mean	4.83	4.77	4.50	4.78	SA

Legend: SA – Strongly Agree (4.50-5.00), A – Agree (3.50 – 4.49), MA – Moderately Agree (2.50 – 3.49), D – Disagree (1.50 – 2.49), SD – Strongly Disagree (1.00 – 1.49), WM – Weighted Mean, VI – Verbal Interpretation

Table 11 presents the evaluation of the compatibility of the system. The highest rating was given to "The system is accessible across different web browsers" with a 4.83 weighted mean. The overall average weighted mean is 4.78 with a verbal interpretation of Strongly Agree. This shows that EduCompose works well across different web browsers and screen sizes.



4.4 Interaction Capability

Table 12

Evaluation of the respondents to the interaction capability of EduCompose

INTERACTION CAPABILITY	STUDENTS	ENGLISH TEACHERS	IT EXPERTS	OVERALL WM	VI
1. The dashboard interface is intuitive and easy to navigate.	4.90	4.70	4.30	4.76	SA
2. The system presents the diagnostic feedback and knowledge graph clearly.	4.80	4.80	4.60	4.79	SA
3. The system provides helpful prompts and clear error messages during use.	4.70	4.60	4.30	4.63	SA
Average Weighted Mean	4.80	4.70	4.40	4.73	SA

Legend: SA – Strongly Agree (4.50-5.00), A – Agree (3.50 – 4.49), MA – Moderately Agree (2.50 – 3.49), D – Disagree (1.50 – 2.49), SD – Strongly Disagree (1.00 – 1.49), WM – Weighted Mean, VI – Verbal Interpretation

Table 12 shows the evaluation of the respondents on interaction capability. The overall average weighted mean is 4.73 with a verbal interpretation of Strongly Agree. This means that both users and IT experts found the interface of the system and the visual feedback easy to use and understand.



4.5. Reliability

Table 13

Evaluation of the respondents to the reliability of EduCompose

RELIABILITY	STUDENTS	ENGLISH TEACHERS	IT EXPERTS	OVERALL WM	VI
1. The system operates smoothly without unexpected shutdowns or crashes	4.60	4.50	4.00	4.51	SA
2. The system consistently processes essay data accurately upon repeated use.	4.70	4.60	4.30	4.63	SA
3. The system handles unsupported file formats gracefully without breaking.	4.50	4.40	4.30	4.44	A
Average Weighted Mean	4.60	4.50	4.20	4.53	SA

Legend: SA – Strongly Agree (4.50-5.00), A – Agree (3.50 – 4.49), MA – Moderately Agree (2.50 – 3.49), D – Disagree (1.50 – 2.49), SD – Strongly Disagree (1.00 – 1.49), WM – Weighted Mean, VI – Verbal Interpretation

Table 13 displays the evaluation of the reliability of EduCompose. The overall average weighted mean is 4.53 with a verbal interpretation of Strongly Agree. This shows that the system runs smoothly and manages requests and user inputs without major errors.

4.6 Security



Table 14

Evaluation of the respondents to the security of EduCompose

SECURITY	STUDENTS	ENGLISH	IT	OVERALL	
		TEACHERS	EXPERTS	WM	VI
1. The system protects sensitive data based on user roles (Admin, Teacher, Student).	4.80	4.80	4.60	4.79	SA
2. User authentication and sessions are properly implemented and secure.	4.70	4.70	4.60	4.69	SA
3. The system prevents unauthorized users from accessing teacher controls.	4.90	4.80	4.60	4.83	SA
Average Weighted Mean	4.80	4.77	4.60	4.77	SA

Legend: SA – Strongly Agree (4.50-5.00), A – Agree (3.50 – 4.49), MA – Moderately Agree (2.50 – 3.49), D – Disagree (1.50 – 2.49), SD – Strongly Disagree (1.00 – 1.49), WM – Weighted Mean, VI – Verbal Interpretation

Table 14 shows the evaluation of the security of the system. The overall average weighted mean is 4.77 with a verbal interpretation of Strongly Agree. This means that all respondent groups agree that the authentication and database security measures protect student data and records properly.

4.7 Maintainability



Table 15

Evaluation of the respondents to the maintainability of EduCompose

MAINTAINABILITY	STUDENTS	ENGLISH	IT	OVERALL	
		TEACHERS	EXPERTS	WM	VI
1. The system's architecture is structured to support future updates.	4.60	4.50	4.60	4.56	SA
2. The integration of AI tools is built so that models can be updated seamlessly.	4.50	4.50	4.60	4.51	SA
3. The system's documentation allows for straightforward IT maintenance.	4.60	4.60	4.30	4.58	SA
Average Weighted Mean	4.57	4.53	4.50	4.55	SA

Legend: SA – Strongly Agree (4.50-5.00), A – Agree (3.50 – 4.49), MA – Moderately Agree (2.50 – 3.49), D – Disagree (1.50 – 2.49), SD – Strongly Disagree (1.00 – 1.49), WM – Weighted Mean, VI – Verbal Interpretation

Table 15 shows the evaluation of the maintainability of the system. The overall average weighted mean is 4.55 with a verbal interpretation of Strongly Agree. This confirms that the code and integration of models and external tools were built properly to allow future updates.

4.8 Flexibility



Table 16

Evaluation of the respondents to the flexibility of EduCompose

FLEXIBILITY	STUDENTS	ENGLISH TEACHERS	IT EXPERTS	OVERALL	
				WM	VI
1. The system easily accommodates different essay evaluation workflows.	4.70	4.60	4.30	4.63	SA
2. The architecture allows varying lengths and formats of digital submissions.	4.80	4.70	4.30	4.72	SA
3. The system's features can be expanded or customized in the future if needed.	4.70	4.70	4.60	4.69	SA
Average Weighted Mean	4.73	4.67	4.40	4.68	SA

Legend: SA – Strongly Agree (4.50-5.00), A – Agree (3.50 – 4.49), MA – Moderately Agree (2.50 – 3.49), D – Disagree (1.50 – 2.49), SD – Strongly Disagree (1.00 – 1.49), WM – Weighted Mean, VI – Verbal Interpretation

Table 16 presents the evaluation of the respondents on flexibility. The overall average weighted mean is 4.68 with a verbal interpretation of Strongly Agree. This shows that the software can be changed and expanded based on different academic assessment needs.

4.9 Safety



Table 17

Evaluation of the respondents to the safety of EduCompose

SAFETY	STUDENTS	ENGLISH	IT	OVERALL	
		TEACHERS	EXPERTS	WM	VI
1. The system prevents the accidental deletion of saved evaluation reports.	4.70	4.70	4.60	4.69	SA
2. The system issues warning prompts before critical actions (e.g., grading).	4.80	4.70	4.30	4.72	SA
3. The system complies with data safety policies for academic institutions.	4.80	4.80	4.60	4.79	SA
Average Weighted Mean	4.77	4.73	4.50	4.73	SA

Legend: SA – Strongly Agree (4.50-5.00), A – Agree (3.50 – 4.49), MA – Moderately Agree (2.50 – 3.49), D – Disagree (1.50 – 2.49), SD – Strongly Disagree (1.00 – 1.49), WM – Weighted Mean, VI – Verbal Interpretation

Table 17 shows the evaluation of the respondents on the safety of EduCompose. The overall average weighted mean is 4.73 with a verbal interpretation of Strongly Agree. This shows that the system prevents harmful actions and protects academic data.

Summary of Evaluation Results



Table 18

Overall Evaluation of EduCompose

CRITERIA	WEIGHTED MEAN	VERBAL INTERPRETATION
1. Functional Suitability	4.71	Strongly Agree
2. Performance Efficiency	4.53	Strongly Agree
3. Compatibility	4.78	Strongly Agree
4. Interaction Capability	4.73	Strongly Agree
5. Reliability	4.53	Strongly Agree
6. Security	4.77	Strongly Agree
7. Maintainability	4.55	Strongly Agree
8. Flexibility	4.68	Strongly Agree
9. Safety	4.73	Strongly Agree
Overall Average Weighted Mean	4.67	Strongly Agree

Table 18 shows the summary of the overall evaluation of EduCompose from the 43 respondents. Compatibility got the highest rating at 4.78 (Strongly Agree), while Performance Efficiency and Reliability got the lowest rating of 4.53 (Strongly Agree).

The overall computed average weighted mean for all ISO 25010 characteristics is **4.67** with a verbal interpretation of **Strongly Agree**. This overall result shows that EduCompose met its development goals and is very capable, efficient, and acceptable to be used as an essay evaluation tool.



CHAPTER V

CONCLUSIONS AND RECOMMENDATIONS

This chapter presents the summary of the entire study, the conclusions drawn from the evaluation phase, and the recommendations proposed for future system improvements.

Conclusions

The following are the conclusions drawn based on the findings of the study:

1. EduCompose successfully addressed teacher workload and time constraints by automating the mechanical aspects of essay evaluation, providing a centralized tool for instructional support.
2. The system fully achieved its functional goals, integrating multidimensional diagnostics, Knowledge Graph visualization, OCR text extraction, and plagiarism checking into a unified dashboard.
3. The integrated AI models proved technically reliable, achieving a 0.79 F1-score for argument classification and a 0.94 BERTScore for semantic feedback accuracy, aligning with professional standards.
4. The system reached a high level of software quality, obtaining an overall average weighted mean of 4.67 (Strongly Agree) based on the ISO 25010:2023 evaluation standards.

Recommendations

The following are recommended to improve and enhance the system for future development:

1. Improve the OCR model to accurately process handwritten essays, addressing the current focus on digital text.
2. Integrate Tagalog language processing to allow the system to evaluate Filipino-based essays and assignments.
3. Expand the system to accept bulk folder uploads of various file types like .docx and .pdf for faster class submissions.



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LAGUNA UNIVERSITY

Page 1 of 1

APPENDICES



APPENDIX A

Letters

April 15, 2026

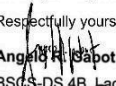
Dr. Rosalie D. Buban
Dean, College of Education
Laguna University
Brgy. Bubukal, Santa Cruz, Laguna

Dear Dr. Buban:

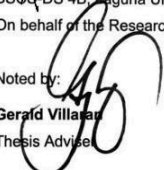
Good day. We, the 4th-year students from the College of Computing Studies, are currently conducting our undergraduate thesis titled **"EduCompose: Knowledge Graph-Enhanced NLP for Teacher-Assisted Essay Evaluation."** As we have now reached the system evaluation phase of our study, we respectfully request your approval to conduct a software quality evaluation survey. The instrument is a 20-item, 4-point Likert scale questionnaire adapted from the ISO/IEC 25010:2023 standard. We intend to distribute this to **seven (7)** selected faculty members under the College of Education who have expertise in essay evaluation. We plan to conduct this survey on **April 16, 2026**. The information gathered will be used solely for academic and research purposes to validate our developed prototype. Rest assured that confidentiality, privacy, and the ethical research guidelines of the university will be strictly observed in accordance with the Data Privacy Act of 2012. In line with this, we humbly seek your permission to allow us to distribute these questionnaires to your faculty members at their most convenient time. We will be grateful for your favorable consideration.

Thank you very much.

Respectfully yours,


Angelo R. Sapot
BSCS-DS 4B, Laguna University
On behalf of the Research Team

Noted by:


Gerald Villanar
Thesis Adviser

Recommending Approval:


Joselle A. Banocnoc
Dean, College of Computing Studies

Approved by:

Dr. Rosalie D. Buban

Dean, College of Education



LAGUNA UNIVERSITY

April 15, 2026

Ms. Charlotte Caponpon
Program Chair, College of Arts and Science
Laguna University
Brgy. Bubukal, Santa Cruz, Laguna

Dear Ms. Caponpon:

Good day. We, the 4th-year students from the College of Computing Studies, are currently conducting our undergraduate thesis titled **"EduCompose: Knowledge Graph-Enhanced NLP for Teacher-Assisted Essay Evaluation."** As we have now reached the system evaluation phase of our study, we respectfully request your approval to conduct a software quality evaluation survey. The instrument is a 20-item, 4-point Likert scale questionnaire adapted from the ISO/IEC 25010:2023 standard. We intend to distribute this to **eight (8)** faculty members under the General Education Department who have expertise in essay evaluation.

We plan to conduct this survey on **April 16, 2026**. The information gathered will be used solely for academic and research purposes to validate our developed prototype. Rest assured that confidentiality, privacy, and the ethical research guidelines of the university will be strictly observed in accordance with the Data Privacy Act of 2012.

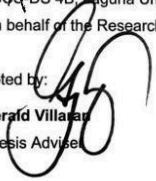
In line with this, we humbly seek your permission to allow us to distribute these questionnaires to your faculty members at their most convenient time. We will be grateful for your favorable consideration.

Thank you very much.

Respectfully yours,


Angel R. Capot
BSCE-DS 4B, Laguna University
On behalf of the Research Team

Noted by:


Gerald Villaluz
Thesis Adviser

Recommending Approval:


Joselle A. Banocnoc
Dean, College of Computing Studies

Approved by:


Ms. Charlotte Caponpon
Program Chair, College of Arts and Sciences



April 15, 2026

Agatha Quennie F. Ramos
Associate Software Engineer
Accenture
Makati City

Dear Ms. Ramos,

Greetings! We hope this letter finds you well. We, the student researchers from Laguna University taking up Bachelor of Science in Computer Science major in Data Science, are required to have three IT/CS experts to evaluate our developed software for the course Thesis 2.

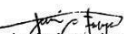
In line with this, we are asking for your expertise to assess our thesis titled "**EDUCOMPOSE: KNOWLEDGE GRAPH-ENHANCED NLP FOR TEACHER-ASSISTED ESSAY EVALUATION**" to ensure its standard quality. Another, we also humbly request a copy of your Curriculum Vitae for documentation purposes. Rest assured that the data collected will be handled with utmost confidentiality and will only be used for this purpose. If you have any concerns, please feel free to reach us at educompose04@gmail.com.

Thank you very much and we look forward to your support and cooperation.

Truly yours,


Genesis Delos Reyes
Student Researcher


Mishaal Catignas
Student Researcher


Julie Antopina
Student Researcher


Angela Gabot
Student Researcher

Noted by:


Gerald M. Villaran
Adviser



LAGUNA UNIVERSITY

April 13, 2026

Viner D. Ortañez
IT - MIS / Stradcom
Land Transportation Office (LTO)
Pila, Laguna

Dear Mr. Ortañez,

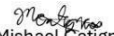
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Thank you very much and we look forward to your support and cooperation.

Truly yours,


Genesis Delos Reyes
Student Researcher


Mishaël Catignas
Student Researcher


Johnie Antopiha
Student Researcher


Angelo Gabot
Student Researcher

Noted by:


Gerald M. Villaran
Adviser



April 10, 2026

Raissa Ann D. Ortáñez
Packaged Software Associate
Global Financial Services Company
Makati City

Dear Ms. Ortáñez,

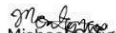
Greetings! We hope this letter finds you well. We, the student researchers from Laguna University taking up Bachelor of Science in Computer Science major in Data Science, are required to have three IT/CS experts to evaluate our developed software for the course Thesis 2.

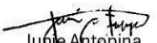
In line with this, we are asking for your expertise to assess our thesis titled "**EDUCOMPOSE: KNOWLEDGE GRAPH-ENHANCED NLP FOR TEACHER-ASSISTED ESSAY EVALUATION**" to ensure its standard quality. Another, we also humbly request a copy of your Curriculum Vitae for documentation purposes. Rest assured that the data collected will be handled with utmost confidentiality and will only be used for this purpose. If you have any concerns, please feel free to reach us at educompose04@gmail.com.

Thank you very much and we look forward to your support and cooperation.

Truly yours,


Genesis Delos Reyes
Student Researcher


Mishael Caignas
Student Researcher


Junie Altopina
Student Researcher


Angelo Gabot
Student Researcher

Noted by:


Gerald M. Villaran
Adviser



APPENDIX B

Certificates of IT Experts



TECHNICAL EXPERT'S CERTIFICATE OF TRUTH, ACCURATENESS, AND COMPLETENESS



EDUCOMPOSE: KNOWLEDGE GRAPH-ENHANCED NLP FOR
TEACHER-ASSISTED ESSAY EVALUATION

THESIS TITLE

GROUP MEMBERS:

Antopina, Junie A.
Catignas, Mishael T.
Delos Reyes, Genesis R.
Gabot, Angelo R.

I certify under my own judgement and conviction, based
from the criteria prescribed in the assessment made after
testing, evaluating and reasonable inquiry, the statements and
information contained in these document are true, accurate
and complete


Viner D. Ortanez

Technical Expert's Signature Over Printed Name

as

System Evaluator

DATE:





LAGUNA UNIVERSITY



TECHNICAL EXPERT'S CERTIFICATE OF TRUTH, ACCURATENESS, AND COMPLETENESS



EDUCOMPOSE: KNOWLEDGE GRAPH-ENHANCED NLP FOR
TEACHER-ASSISTED ESSAY EVALUATION

THESIS TITLE

GROUP MEMBERS:

Antopina, Junie A.
Catignas, Michael T.
Delos Reyes, Genesis R.
Gabot, Angelo R.

I certify under my own judgement and conviction, based from the criteria prescribed in the assessment made after testing, evaluating and reasonable inquiry, the statements and information contained in these document are true, accurate and complete



Agatha Quennie F. Ramos

Technical Expert's Signature Over Printed Name

as

System Evaluator

DATE: _____



LAGUNA UNIVERSITY



TECHNICAL EXPERT'S CERTIFICATE OF TRUTH, ACCURATENESS, AND COMPLETENESS



EDUCOMPOSE: KNOWLEDGE GRAPH-ENHANCED NLP FOR
TEACHER-ASSISTED ESSAY EVALUATION

THESIS TITLE

GROUP MEMBERS:

Antopina, Junie A.
Catignas, Michael T.
Delos Reyes, Genesis R.
Gabot, Angelo R.

I certify under my own judgement and conviction, based
from the criteria prescribed in the assessment made after
testing, evaluating and reasonable inquiry, the statements and
information contained in these document are true, accurate
and complete


Raissa Ann D. Ortanez

Technical Expert's Signature Over Printed Name

as

System Evaluator

DATE: _____



Appendix C

Title Proposal



LAGUNA UNIVERSITY
College of Computing Studies
Laguna Sports Complex, Brgy. Bubukal, Santa Cruz, Laguna

CAPSTONE 1 AND CS THESIS 1 PROPOSED TOPICS/ TITLES

GROUP NO. 1

STUDENT NAMES: ANTOPINA, JUNIE A. (Leader)
GABOT, ANGELO R.
DELOS REYES, GENESIS R.
CATIGNAS, MISHAEL T.

ADVISER: GERALD VILLARAN

DATE SUBMITTED: September 11, 2025

SUBMITTED TO: JOSELLE A. BANOCNOC, Course Instructor

*This paper is intended to discuss the description of the study as the requirement for the topic/ title proposals. It must contain the rationale of the study, research objectives, research design, research locale, and target respondents. **Student researchers must prepare the literature review that they have prepared to support the discussion.***



TITLE 1: CivicSense: Graph-Based Models and Temporal Analysis for Barangay Complaint Intelligence

I. INTRODUCTION

Barangays receive lots of citizen complaints (noise, garbage, flooding, broken lights). Many of these are duplicates, vague, or late. That wastes staff time and delays fixes. The core idea: treat complaints not just as isolated texts but as **items in a network** (linked by similarity, location, reporter, date). Using graph methods helps find clusters of related reports (duplicates or the same problem reported by different people). Adding temporal analysis lets the system detect repeating patterns (weekend garbage spikes, seasonal flooding), which helps planning (e.g., allocate cleanup staff early). In short: move from manual triage to **data-driven prioritization**.

II. RATIONALE OF THE STUDY

- **Current reality:** Many LGUs use simple forms or spreadsheets. These capture data but don't analyze. Keyword matching misses synonyms, typos, and context (e.g., "water overflow" vs "flooded street" are related but not the same words).
- **Why graph approaches:** Complaints are naturally linked (same location, same time window, same reporter, similar text). Graphs let you encode those relations and detect communities of related incidents even if wording differs.
- **Why temporal analysis:** Some issues are episodic. If you can forecast a surge (e.g., dengue-related complaints rise after rainy weeks), the barangay can act proactively.
- **Practical outcome:** The system gives prioritized lists (clusters + predicted hot categories), improving response times and resource allocation.

III. RESEARCH OBJECTIVES (and why each one)

General objective: Build and evaluate a system that groups complaints using graph-based methods and detects temporal trends for better barangay response.

Specific objectives (with rationale & measurable outcome):

- **Collect & preprocess complaint texts** — so models have clean input (outcome: cleaned dataset, sampling plan).
- **Build a topic categorization model (NLP)** — baseline needed to group complaints by type (accuracy/F1 target).



- **Design graph-based clustering for duplicate detection** — show improvement over naive deduplication (compare ARI/NMI or manual labeling).
- **Apply temporal analysis / forecasting on category counts** — measure forecast accuracy (MAE/RMSE) and show usefulness in a small scenario.
- **Usability evaluation with barangay staff** — measure perceived usefulness and time saved (survey + observation).

IV. SCOPE AND LIMITATIONS

- **Scope:** Only text complaints (forms, chat, short messages). Focus on grouping (duplicates), categorization, and trend detection for a small number of categories (e.g., sanitation, drainage, public lighting, safety). Prototype, not a full municipal roll-out.
- **Limitations:** Results depend on data quality and volume; models trained on 1–2 barangays may not generalize. Does not handle multimedia complaints (photos, voice) in the first version. Privacy concerns must be handled (anonymize names, locations where necessary).

V. RESEARCH LOCALE

Pick 1 barangay (or 2 neighboring barangays) with enough complaint traffic and cooperative officials. Choose somewhere where logs exist (paper or digital) and where you can run a small user study. Explain to the panel you'll seek permission to access anonymized logs and run a short pilot.

VI. TARGET RESPONDENTS

- **Primary respondents:** barangay officials/staff who process complaints — for usability testing and interviews.
- **Secondary respondents:** residents who submit complaints (can collect ~100–500 example complaints via a short survey or simulated submissions). Always state you will obtain consent and anonymize data.



TITLE 2: EduCompose: Knowledge Graph–Enhanced NLP for Teacher-Assisted Essay Evaluation

I. INTRODUCTION

Essay writing remains a fundamental instrument for evaluating student learning across senior high school and college levels. However, teachers continually face the challenge of assessing numerous essays while ensuring fair, detailed, and consistent feedback. Evaluating both the surface-level features of writing (such as grammar and readability) and the more complex, higher-order skills (including coherence, logical structure, and conceptual understanding) is an arduous and time-consuming task.

Existing tools like grammar checkers and plagiarism detectors primarily address technical correctness and originality but offer limited pedagogical insight. These systems are designed mainly for student use and focus on fixing mechanical errors rather than supporting educators in diagnosing underlying weaknesses in writing development. As a result, the pressing need for teacher-oriented tools that enhance instructional feedback, rather than replace human judgment, remains unfulfilled.

EduCompose introduces an innovative approach by combining Natural Language Processing (NLP) and Knowledge Graph technologies within a teacher-centered evaluation framework. Rather than scoring essays or providing direct feedback to students, the system analyzes written texts to generate interpretive reports on aspects of writing such as grammar, readability, coherence, and argument strength. These reports empower teachers to deliver personalized, evidence-based guidance appropriate to each student's needs. By emphasizing diagnostic evaluation over automated grading, EduCompose fosters an AI-human partnership that strengthens authentic teaching practices and promotes deeper student learning.

II. RATIONALE OF THE STUDY

Teacher-Centered Gap: Current AI writing tools like Grammarly and Quillbot are student-oriented, creating a dependency on instant automated feedback. EduCompose,



in contrast, places teachers at the core of the evaluation process, aiding rather than replacing them.

Efficiency: Teachers dedicate extensive hours to essay assessment, often repeating low-level feedback. EduCompose streamlines this process by automating preliminary analysis—such as grammatical correctness, readability indices, and topic coherence—allowing teachers to concentrate on higher-level mentorship and formative evaluation.

Educational Value: The system emphasizes teacher empowerment through AI support. Instead of diminishing instructional value, EduCompose amplifies teachers' capacity to provide authentic, meaningful feedback grounded in linguistic and conceptual evidence.

Research Gap: Although numerous NLP educational applications exist, most are either student-driven or limited to surface-level feedback. Few, if any, integrate Knowledge Graphs to map conceptual relationships or evaluate argumentation quality within essays—a feature EduCompose directly addresses.

III. RESEARCH OBJECTIVES

General Objective

To design and develop a prototype essay evaluation system that employs Natural Language Processing and Knowledge Graphs to support teachers in diagnosing and addressing student writing challenges.

Specific Objectives

- To create a functional system prototype capable of processing essay inputs and identifying key indicators of writing quality, including grammar, readability, coherence, and argumentation.
- To integrate knowledge-based representations that reveal conceptual structures and gaps within student essays for a deeper understanding of writing development.
- To develop a teacher-oriented interface that presents diagnostic findings in clear, actionable formats suitable for classroom use.
- To gather insights from teacher feedback during early validation to refine the system's usability, accuracy, and educational value.



IV. SCOPE AND LIMITATIONS

Scope:

- The prototype focuses solely on teachers and will not be directly accessible to students.
- Analyses are limited to English expository and argumentative essays.
- The system's diagnostic report covers grammar issues, readability assessments, conceptual coherence, and argumentation patterns.

Limitations:

- The system will not assign grades or holistic scores. Rather, it highlights potential writing weaknesses for instructional support.
- Knowledge Graph analysis will rely on preexisting data sources such as ConceptNet, WordNet, and domain-specific ontologies.
- The system serves as a teacher-assistive tool, not a replacement for professional judgment or pedagogical experience.

V. RESEARCH LOCALE

The study will be conducted in selected partner schools or universities where English teachers actively engage in essay evaluation and writing instruction. These institutions will contribute sample essays for analysis as well as teacher participants for system validation.

VI. TARGET RESPONDENTS

Primary Respondents:

Senior high school and college English teachers with experience evaluating expository and argumentative essays. They will provide annotations, validate analytic reports, and assess the prototype's instructional effectiveness.

Secondary Respondents (Indirect):

Students whose essays will serve as data inputs for testing and validation. They will not interact directly with the system, but their written works will form the primary dataset for prototype evaluation.



TITLE 3: Voice-to-Policy: Hybrid AI Framework for Translating Sentiment into Governance Recommendations

I. INTRODUCTION

Organizations collect public sentiment from complaints, social media, and surveys. Conventional sentiment analysis stops at "positive/negative." Policymakers want **actionable advice** (what to fix, where, which department to mobilize). Voice-to-Policy combines ML for detecting what people are upset about (aspects and sentiment) with a simple symbolic mapping that suggests practical actions (policy-level suggestions), creating a bridge from raw public voice to decision-ready options.

II. RATIONALE OF THE STUDY (detailed reasoning)

- **Problem:** Policymakers get sentiment dashboards but not recommended next steps. A negative spike in "transport" sentiment doesn't say whether to increase buses, fix traffic lights, or re-route tricycles.
- **Hybrid solution:** Use ML to extract *aspects* (transport, sanitation) and sentiment strength, then use a ruleset or symbolic map (co-created with domain experts) that translates common aspect+sentiment patterns into candidate actions.
- **Why this helps:** Produces short, explainable recommendations that can be validated by human staff, improving responsiveness and trust.

III. RESEARCH OBJECTIVES (and why each matters)

General objective: Build and validate a hybrid system that turns text-based public sentiment into recommended actions for local governance.

Specific objectives:

- **Collect a dataset** of complaints/tweets/survey responses labeled for aspect and sentiment.
- **Train an aspect-based sentiment model** (baseline: rule-based / classical; advanced: fine-tuned BERT).
- **Create a symbolic mapping** (simple rule engine) that maps aspect+severity into candidate actions.
- **Evaluate for accuracy and relevance** (compare to expert suggestions).
- **Pilot with LGU staff** to get feedback and iterate.

IV. SCOPE & LIMITATIONS



- **Scope:** Focus text-based opinions in specific domains (e.g., transport, sanitation, health services) and produce non-binding recommendations.
- **Limitations:** Recommendations are only as good as the mapping rules and data; political sensitivity and privacy concerns must be handled carefully. The system provides *suggestions* for human decision-makers, not automated policy changes.

V. RESEARCH LOCALE

A municipality or city with accessible public opinion sources (official complaint channels or public social media streams). Start with a single topic area to keep scope manageable (e.g., transportation complaints).

VI. TARGET RESPONDENTS

- **Policymakers/LGU staff** who review recommendations.
- **Citizens** whose complaints/sentiments will be used (anonymized).



APPENDIX D

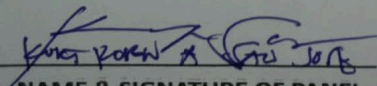
Comment Sheet

Antopina, Junie	Catignas, Mishael
Delos Reyes, Genesis	Gabot, Angelo
EduCompose: Knowledge Graph-Enhanced NLP for Teacher-Assisted Essay Evaluation	
SUGGESTIONS	
system: - The system must have ways on reading the handwriting of students because it's part of statement of the problem 1.3. - The student must have account or there should be a feedback using emails - The plagiarism must have more accurate results - The student who submitted first or the date must be included in the similarity index - The OCR for the manually written essay must be also included in the system - The system must have statistical measurement that will measure if the tools / model you used was effective or not. Documents: 1. the statement of the problem & objective does not align. make it shorter because it has lots of statements & specific objective that does not need to be included. 2. specify the limitation of the study for example pay per use or credit based model. don't just define the technology. include the limitation of that technology on your system. 3. The RRL must be in chronological order. 4. include all the ^{english} teachers because you only have 20 teachers as your population 5. the ITO must be change. you must check the last year document. It must have the flow of all technologies that has been used.	
Boa May M. Belgamino NAME & SIGNATURE OF PANEL	
01/05/2026 DATE	

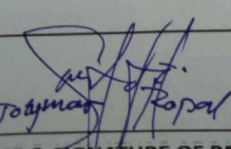


January 5, 2026	Time	1:00pm to 2:00pm
1	Program	BSCS-DS 4B
Antopina, Junie	Catignas, Mishael	
Delos Reyes, Genesis	Gabol, Angelo	
EduCompose: Knowledge Graph-Enhanced NLP for Teacher-Assisted Essay Evaluation		

SUGGESTIONS
BATCH UPLOAD - x (PROGRAM, SECTION) HIGHLIGHT INCORRECT WORDS, PUNCTUATION AND STUDENT ERRORS CONTEXT DETECTION / PLAGIARISM CHECKER FILTERING OVERALL EVALUATION FOR AT-RISK STUDENTS BATCH EXPORT / REPORT GENERATION ARCHIVE STUDENT ACCOUNT OCR SINCE YOU'LL HAVE A NEW USER ACCOUNT, FURTHER COMMENTS ON NEXT PRESENTATION

 NAME & SIGNATURE OF PANEL	<u>1-5-26</u> DATE
--	-----------------------



	January 5, 2026	Time	
	1	Program	BSCS-DS 4B
ers	Antopina, Junie		Catignas, Mishael
	Delos Reyes, Genesis		Gabot, Angelo
Title	EduCompose: Knowledge Graph-Enhanced NLP for Teacher-Assisted Essay Evaluation		
E	SUGGESTIONS		
	Profile Batch Upload Provide Accuracy of OCT Validation test management Duplication if Function move inside Archive Total grade in ONE button Student test Per-Sem or per grading if Sr. High is your client Identify if it's copy and paste Filter Report generation Due date notif Reminder Re Submit Re Upload Comment will be added upon next presentation		
Joyman 			106-2026
NAME & SIGNATURE OF PANEL			DATE



APPENDIX E

Thesis Advising Form

LU-AA-FO-10 rev.0 10132020



LAGUNA UNIVERSITY
Research and Development Center
Laguna Sports Complex, Brgy. Bubukal
Santa Cruz, Laguna
THESIS ADVISING FORM



Group No	CSB1	Sem, AY	1st Sem, 2025-2026
Department	College of Computing Studies	Program	Computer Science

We agree to engage in adviser-advisee relationship in consonance to a thesis/capstone project/feasibility study/ project study that will be written from 2025 to 2026

We agree to follow a work schedule. In the event the adviser fails to honor the terms of the Agreement while the research is implemented, he/she shall not receive the pay equivalent to the service rendered.

We agree that the academic partnership leads to publication of the manuscript in a reputable peer-reviewed journal with the Adviser as the co-author and the advisee, the primary author. In such case, the publication fee shall be paid by the advisee or could be sourced out from sponsors. Both parties will implement the corrections during the peer review process.

We agree that the paper is presented in a public forum by the Advisee if available at such an opportunity or by the Adviser if the Advisee is no longer around. Only the name of the oral presenter shall be submitted to the Conference organizer.

We agree to follow stringent quality assurance measures such as professional editing, plagiarism, grammar and readability tests, reference checking, intercontinental literature sources, Statistical software processing if applicable, among others.

We agree to use various forms of contact such as face to face, online skype and facetime, viber and mobile, email, social network such as Facebook messenger, LinkedIn, among others.

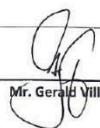
We agree to dress formally and prepare adequately for the formal oral presentation both in the oral defense panel and public performances.

We agree to be open, honest and frank with each other to expedite the work.

We agree to give time to complete this work by following the timelines in the stages of the research and official time for the public and professional duties of the adviser.

We agree to inform each other of developments such as submission to the Dean must bear the signature of the adviser as an endorsement, and whoever is going to present the paper in public forum, and presentation of the article to a particular journal.

Signed this 23rd of September in the year of our Lord two thousand and twenty-five in Laguna University.

Antopina, Junie A. Catignas, Mishael T. Delos Reyes, Genesis R. Gabot, Angelo R.	 Mr. Gerald Villaran
Advisees	Adviser



APPENDIX F

Advising Monitoring Form



LAGUNA UNIVERSITY
Research and Development Center
Laguna Sports Complex, Brgy. Bubukal
Santa Cruz, Laguna



LU-AA-FO-11 rev.0 10132020

Program	Bachelor of Science in Computer Science with specialization in Data Science
Research Title	EduCompose: Knowledge Graph-Enhanced NLP For Teacher-Assisted Essay Evaluation
Researchers	Antopina, Junie Catignas, Mishael Delos Reyes, Genesis Gabot, Angelo
Thesis Adviser	Mr. Gerald Villaran

Date	Chapter Page	Suggestions	Signature of Adviser
September 19, 2025	Chapter 1-2	Gather Data that from suggested RRLs	
October 2, 2025	N/A (System)	NLP Checking and Consultation. - Use K-graph	
October 7, 2025	Chapter 1	Checking of Chapter 1, some revisions suggested	
November 17, 2025	Chapter 3	Create and show the analysis for the evaluation of the essay Knowledge Graph enhancement suggestions	
November 24, 2025	Chapter 1-3	Chapter 1-3 Checking (revisions for chapter 3 like Model change from Waterfall to Agile) Use powerpoint slides from Google Collab to explain the NLP used properly	



APPENDIX G

System Evaluation (Clients, Test Users)

EDUCOMPPOSE: KNOWLEDGE GRAPH- ENHANCED NLP FOR TEACHER- ASSISTED ESSAY EVALUATION SURVEY

Warm Greetings!

We are 4th-year BSCS-Data Science students from Laguna University, currently conducting our undergraduate thesis titled "EduCompose: Knowledge Graph-Enhanced NLP for Teacher-Assisted Essay Evaluation".

We sincerely appreciate your participation in testing our web application. As a participant who has explored the system, your feedback is essential in helping us evaluate the software quality, usability, and overall effectiveness of our platform. Your insights provide the critical data needed to refine our research and improve the system's performance.

misheefcaginas@gmail.com [Switch account](#)

Not shared

* Indicates required question

System Testing Instructions:

To evaluate the system accurately, please ensure you have visited our platform and explored its core functionalities prior to answering the questions.

[educompose.online](#) (for the teacher end)

If you'd like to test the teacher-end interface, please try creating a new account to experience the full registration and login flow.

Teacher Acc:
email: solimanmyra333@gmail.com
pass: Password123!

[educompose.online/student_login](#) (for the student end)

Student Acc:
ID: 201-2003
Pass: 2003-01-01
ID: 201-2004
Pass: 2004-01-10
ID: 201-2005
Pass: 2005-01-01
ID: 201-2006
Pass: 2006-01-01

Please avoid using the same account in different devices for it will not work.

Survey Instructions:

Based on your experience with the platform, please evaluate its software quality according to the ISO/IEC 25010:2023 standards. This survey will only take approximately 3 to 5 minutes to complete. If you encounter any technical issues, please contact the research team at educompose04@gmail.com

Evaluator Details

Email Address (optional)

Your answer

Full Name (optional)

Your answer

Role *

- ☒ English Language Teacher
- ☐ IT Expert / Evaluator
- ☐ Test User

ISO/IEC 25010:2023 System Evaluation

Please evaluate the EduCompose system based on the following criteria. Read each statement carefully and select the response that best represents your experience, where:

- 5 = Strongly Agree
4 = Agree
3 = Moderately Agree
2 = Disagree
1 = Strongly Disagree

1. Functional Stability *

	5 - Strongly Agree	4 - Agree	3 - Moderately Agree	2 - Disagree	1 - Strongly Disagree
The system provides all necessary functions for digital essay evaluation.	☐	☐	☐	☐	☐
The system accurately extracts and analyzes diagnostic metrics from the text.	☐	☐	☐	☐	☐
The system produces useful and relevant diagnostic results for teacher review.	☐	☐	☐	☐	☐

3. Compatibility

	5 - Strongly Agree	4 - Agree	3 - Moderately Agree	2 - Disagree	1 - Strongly Disagree
The system operates properly across different web browsers.	☐	☐	☐	☐	☐
The system integrates smoothly with external services.	☐	☐	☐	☐	☐
The system maintains consistent functionality across different screen sizes.	☐	☐	☐	☐	☐



2. Performance Efficiency

*	5 - Strongly Agree	4 - Agree	3 - Moderately Agree	2 - Disagree	1 - Strongly Disagree
The system responds quickly to user actions and inputs.	☐	☐	☐	☐	☐
The system processes digital essays and generates results within an acceptable time.	☐	☐	☐	☐	☐
The system handles multiple tasks efficiently without lagging.	☐	☐	☐	☐	☐

4. Interaction Capability (Usability)

*	5 - Strongly Agree	4 - Agree	3 - Moderately Agree	2 - Disagree	1 - Strongly Disagree
The dashboard interface is intuitive and easy to navigate.	☐	☐	☐	☐	☐
The system presents the diagnostic feedback and Knowledge Graphs clearly.	☐	☐	☐	☐	☐
The system provides helpful prompts and clear error messages during use.	☐	☐	☐	☐	☐

5. Reliability

*	5 - Strongly Agree	4 - Agree	3 - Moderately Agree	2 - Disagree	1 - Strongly Disagree
The system operates smoothly without unexpected errors or crashes.	☐	☐	☐	☐	☐
The system consistently produces reliable diagnostic results upon repeated use.	☐	☐	☐	☐	☐
The system handles unsupported file formats gracefully without breaking.	☐	☐	☐	☐	☐

6. Security

*	5 - Strongly Agree	4 - Agree	3 - Moderately Agree	2 - Disagree	1 - Strongly Disagree
The system properly restricts access based on user roles (Admin, Teacher, Student).	☐	☐	☐	☐	☐
The system protects user accounts with proper authentication and login requirements.	☐	☐	☐	☐	☐
The system ensures that submitted student essays and data remain confidential.	☐	☐	☐	☐	☐



7. Maintainability

*

	5 - Strongly Agree	4 - Agree	3 - Moderately Agree	2 - Disagree	1 - Strongly Disagree
The system's architecture is structured to support future updates.	☐	☐	☐	☐	☐
Modifications or improvements can be applied without causing major disruptions.	☐	☐	☐	☐	☐
The system is organized in a way that allows for straightforward IT maintenance.	☐	☐	☐	☐	☐

8. Flexibility

	5 - Strongly Agree	4 - Agree	3 - Moderately Agree	2 - Disagree	1 - Strongly Disagree
The system easily accommodates different teacher evaluation workflows.	☐	☐	☐	☐	☐
The system can handle varying lengths and styles of digital text submissions.	☐	☐	☐	☐	☐
The system's features can be expanded or customized in the future if needed.	☐	☐	☐	☐	☐

9. Safety

	5 - Strongly Agree	4 - Agree	3 - Moderately Agree	2 - Disagree	1 - Strongly Disagree
The system prevents the accidental deletion or loss of evaluation reports.	☐	☐	☐	☐	☐
The system avoids harmful operations that could lead to data or security risks.	☐	☐	☐	☐	☐
The system operates safely within the intended academic and ethical guidelines.	☐	☐	☐	☐	☐



LAGUNA UNIVERSITY



IT EXPERTS

System Client and Technical Expert Assessment

Capstone Project / Thesis Title:

Respondent Information

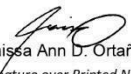
1. Name (Optional):

For each criterion, rate the system by checking the corresponding weight you deemed appropriate:

5 = Strongly Agree 4 = Agree 3 = Moderately Agree 2 = Disagree 1 = Strongly Disagree

Functional Stability	5	4	3	2	1
The system provides all necessary functions for digital essay evaluation.	☒				
The system accurately extracts and analyzes diagnostic metrics from the text.	☒				
The system produces useful and relevant diagnostic results for teacher review.	☒				
Performance Efficiency	5	4	3	2	1
The system responds quickly to user actions and inputs.	☒				
The system processes digital essays and generates results within an acceptable time.	☒				
The system handles multiple tasks efficiently without lagging.	☒				
Compatibility	5	4	3	2	1
The system operates properly across different web browsers.	☒				
The system integrates smoothly with external services.	☒				
The system maintains consistent functionality across different screen sizes.	☒				
Interaction Capability	5	4	3	2	1
The dashboard interface is intuitive and easy to navigate.	☒				
The system presents the diagnostic feedback and Knowledge Graphs clearly.	☒				
The system provides helpful prompts and clear error messages during use.	☒				
Reliability	5	4	3	2	1
The system operates smoothly without unexpected errors or crashes.	☒				
The system consistently produces reliable diagnostic results upon repeated use.	☒				
The system handles unsupported file formats gracefully without breaking.	☒				
Security	5	4	3	2	1
The system properly restricts access based on user roles (Admin, Teacher, Student).	☒				
The system protects user accounts with proper authentication and login requirements.	☒				
The system ensures that submitted student essays and data remain confidential.	☒				
Maintainability	5	4	3	2	1
The system's architecture is structured to support future updates.	☒				
Modifications or improvements can be applied without causing major disruptions.	☒				
The system is organized in a way that allows for straightforward IT maintenance.	☒				
Flexibility	5	4	3	2	1
The system easily accommodates different teacher evaluation workflows.	☒				
The system can handle varying lengths and styles of digital text submissions.	☒				
The system's features can be expanded or customized in the future if needed.	☒				
Safety	5	4	3	2	1
The system prevents the accidental deletion or loss of evaluation reports.	☒				
The system avoids harmful operations that could lead to data or security risks.	☒				
The system operates safely within the intended academic and ethical guidelines.	☒				

Note: Overall numeric score will serve as a guide for the panel's final evaluation. If an essential criterion is rated unsatisfactory, you may decide to reject the program even if some of its features are attractive


Raissa Ann D. Ortañes
Signature over Printed Name
Date: _____



System Client and Technical Expert Assessment

Capstone Project / Thesis Title:

Respondent Information

1. Name (Optional):

For each criterion, rate the system by checking the corresponding weight you deemed appropriate:

5 = Strongly Agree 4 = Agree 3 = Moderately Agree 2 = Disagree 1 = Strongly Disagree

Functional Stability	5	4	3	2	1
The system provides all necessary functions for digital essay evaluation.	/				
The system accurately extracts and analyzes diagnostic metrics from the text.	/				
The system produces useful and relevant diagnostic results for teacher review.	/				
Performance Efficiency	5	4	3	2	1
The system responds quickly to user actions and inputs.	/				
The system processes digital essays and generates results within an acceptable time.	/				
The system handles multiple tasks efficiently without lagging.	/				
Compatibility	5	4	3	2	1
The system operates properly across different web browsers.	/				
The system integrates smoothly with external services.	/				
The system maintains consistent functionality across different screen sizes.	/				
Interaction Capability	5	4	3	2	1
The dashboard interface is intuitive and easy to navigate.	/				
The system presents the diagnostic feedback and Knowledge Graphs clearly.	/				
The system provides helpful prompts and clear error messages during use.	/				
Reliability	5	4	3	2	1
The system operates smoothly without unexpected errors or crashes.	/				
The system consistently produces reliable diagnostic results upon repeated use.	/				
The system handles unsupported file formats gracefully without breaking.	/				
Security	5	4	3	2	1
The system properly restricts access based on user roles (Admin, Teacher, Student).	/				
The system protects user accounts with proper authentication and login requirements.	/				
The system ensures that submitted student essays and data remain confidential.	/				
Maintainability	5	4	3	2	1
The system's architecture is structured to support future updates.	/				
Modifications or improvements can be applied without causing major disruptions.	/				
The system is organized in a way that allows for straightforward IT maintenance.	/				
Flexibility	5	4	3	2	1
The system easily accommodates different teacher evaluation workflows.	/				
The system can handle varying lengths and styles of digital text submissions.	/				
The system's features can be expanded or customized in the future if needed.	/				
Safety	5	4	3	2	1
The system prevents the accidental deletion or loss of evaluation reports.	/				
The system avoids harmful operations that could lead to data or security risks.	/				
The system operates safely within the intended academic and ethical guidelines.	/				

Note: Overall numeric score will serve as a guide for the panel's final evaluation. If an essential criterion is rated unsatisfactory, you may decide to reject the program even if some of its features are attractive


Viner D. Ortalez
Signature over Printed Name

Date: _____



System Client and Technical Expert Assessment

Capstone Project / Thesis Title:

Respondent Information

1. Name (Optional):

For each criterion, rate the system by checking the corresponding weight you deemed appropriate:

5 = Strongly Agree 4 = Agree 3 = Moderately Agree 2 = Disagree 1 = Strongly Disagree

Functional Stability	5	4	3	2	1
The system provides all necessary functions for digital essay evaluation.	/	/			
The system accurately extracts and analyzes diagnostic metrics from the text.	/	/			
The system produces useful and relevant diagnostic results for teacher review.	/	/			
Performance Efficiency	5	4	3	2	1
The system responds quickly to user actions and inputs.	/	/			
The system processes digital essays and generates results within an acceptable time.	/	/			
The system handles multiple tasks efficiently without lagging.	/	/			
Compatibility	5	4	3	2	1
The system operates properly across different web browsers.	/	/			
The system integrates smoothly with external services.	/	/			
The system maintains consistent functionality across different screen sizes.	/	/			
Interaction Capability	5	4	3	2	1
The dashboard interface is intuitive and easy to navigate.	/	/			
The system presents the diagnostic feedback and Knowledge Graphs clearly.	/	/			
The system provides helpful prompts and clear error messages during use.	/	/			
Reliability	5	4	3	2	1
The system operates smoothly without unexpected errors or crashes.	/	/			
The system consistently produces reliable diagnostic results upon repeated use.	/	/			
The system handles unsupported file formats gracefully without breaking.	/	/			
Security	5	4	3	2	1
The system properly restricts access based on user roles (Admin, Teacher, Student).	/	/			
The system protects user accounts with proper authentication and login requirements.	/	/			
The system ensures that submitted student essays and data remain confidential.	/	/			
Maintainability	5	4	3	2	1
The system's architecture is structured to support future updates.	/	/			
Modifications or improvements can be applied without causing major disruptions.	/	/			
The system is organized in a way that allows for straightforward IT maintenance.	/	/			
Flexibility	5	4	3	2	1
The system easily accommodates different teacher evaluation workflows.	/	/			
The system can handle varying lengths and styles of digital text submissions.	/	/			
The system's features can be expanded or customized in the future if needed.	/	/			
Safety	5	4	3	2	1
The system prevents the accidental deletion or loss of evaluation reports.	/	/			
The system avoids harmful operations that could lead to data or security risks.	/	/			
The system operates safely within the intended academic and ethical guidelines.	/	/			

Note: Overall numeric score will serve as a guide for the panel's final evaluation. If an essential criterion is rated unsatisfactory, you may decide to reject the program even if some of its features are attractive


Agatha Queranie F. Ramos
Signature over Printed Name

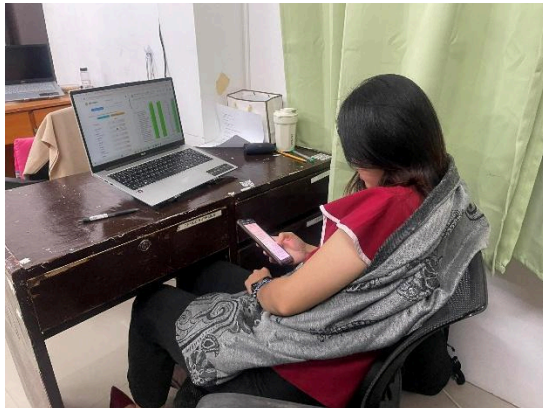
Date: _____



APPENDIX H

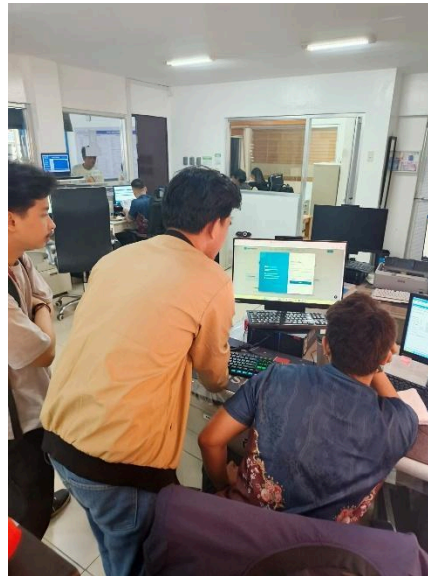
Photo Documentation

Software Testing with Users





With IT Experts



Michael Catignas (Presenting)

Analysis Results

Analysis Report

EXPORT REPORT

EVALUATION MANUSCRIPT

Passing an upcoming defense requires preparation, confidence, and clear communication. First, you must fully understand your research. Know your objectives, methods, and results, and be ready to explain why you made certain decisions. Panelists often test your reasoning, so it is important to justify your work clearly and logically. Next, anticipate possible questions, identify weak points or limitations in your study and prepare answers in advance. Practicing with classmates or friends can help you become more confident and improve how you respond. This also helps you organize your thoughts and speak more clearly. Presentation skills are equally important. Keep your slides simple and focus on key ideas instead of too much detail. Speak clearly, maintain eye contact, and avoid reading directly from your slides. Confidence in delivery leaves a strong impression. During the defense, stay calm and composed. Listen carefully to questions, think before

ACADEMIC PROFICIENCY INDEX

89%

PROFICIENCY BREAKDOWN

Category	Score
Knowledge	100%
Reasoning	89%
Communication	100%
Accuracy	97%

5:26 PM | xrb-wuqj-cwv

Participants in the video conference:

- Michael Catignas
- Michael Catignas
- Michael Catignas
- Michael Catignas
- Michael Catignas



APPENDIX I

Curriculum Vitae

IT Experts



Packaged App Development Associate

Agatha Quennie F. Ramos



Contact

+63(9765994744
agatha.quennie.ramos@accenture.com

Profile

A computer science graduate with knowledge in programming, web development and databases. Skilled in ReactJS And Angular. Experienced as a working student, demonstrating strong time management, adaptability, and a solid work ethic. Completed several academic projects and eager to learn and grow in a software development role. A quick learner with strong problem-solving abilities, ready to contribute to a team.

Professional Skills and qualifications

- Frontend Development:** React, Vue, HTML, CSS, JavaScript, TypeScript
- Backend Development:** Node.js, Express.js
- Tools:** Git and Android Studio
- Soft Skills:** Time Management, Adaptability, Problem Solving

Relevant Work Experience

Accenture

Project: CIO – APAC Payroll (March 2025-present)

Position: Developer

- Developed and maintained an ERM application using Angular for the front end and C# for the back end, designed to securely archive large volumes of payroll data with advanced search and download features, improving data management efficiency and ensuring compliance with strict data privacy standards through AWS storage.
- Built and deployed a seamless onboarding automation feature, now actively used in production

Front-end Development Bootcamp – VueJS (Feb 2025-March2025)

- Developed a responsive e-commerce web application using CSS, ensuring cross-device compatibility.
- Built a contact management web application with VueJS implementing a JSON-based database for efficient contact storage and retrieval.

Back-end Development Bootcamp – Node.js (Jan 2025 – Feb 2025)

- Built backend applications with Node.js and MongoDB, managing data storage, retrieval, and API integration.
- Performed API testing with Postman, unit testing with Jest, and ensured code quality through SonarQube analysis.

Front-end Development Bootcamp – ReactJS (Sept 2024 – Oct 2024)

- Developed a responsive e-commerce web application using HTML and CSS, ensuring cross-device compatibility.
- Built a contact management web application with ReactJS and Tailwind CSS, implementing a JSON-based database for efficient contact storage and retrieval.

Accenture Technology Analytics School (ATAS)

- Understanding the Software Development Life Cycle

- Gaining knowledge to Prototyping and Testing

- Understanding Agile Case Study

Non-Accenture

STMicroelectronics, Inc.

Role: Incoming Quality Inspector / Production Operator II

- Inspected incoming materials, documented defects, used measurement tools, and coordinated with suppliers to resolve quality issues while suggesting process improvements.

- As a Production Operator II, Operated testing machines, troubleshoot errors, and used software tools to ensure product quality before final production

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Viner Ortañez

IT Personnel

Cavinti, Laguna | vinerortanez1996@gmail.com | 0951-759-3149

Professional Summary

Results-driven IT professional with over 8 years of experience supporting Management Information Systems (MIS) and Stradcom operations. Proven ability to maintain system reliability, resolve technical issues efficiently, and support business-critical processes. Adept at troubleshooting, system monitoring, and delivering consistent IT support in fast-paced environments.

Professional Experience

IT Personnel (MIS / Stradcom)

2018 – Present

- Deliver end-to-end technical support for hardware, software, and network issues, ensuring minimal disruption to operations
- Manage and maintain MIS and Stradcom systems to support daily organizational processes
- Diagnose and resolve system errors, reducing downtime and improving system efficiency
- Monitor system performance and recommend improvements to enhance reliability
- Support data processing, report generation, and system documentation
- Collaborate with cross-functional teams to address IT-related concerns and implement solutions
- Ensure compliance with internal IT procedures and operational standards

Education

Bachelor of Science in Information Technology (BSIT)

Northeastern University

2014 – 2018

High School Diploma

Cavinti Integrated National High School

2010 – 2014

Core Competencies

- IT Support & Technical Troubleshooting
- MIS & Stradcom System Management
- Network & Hardware Maintenance
- System Monitoring & Performance Optimization
- Data Management & Reporting
- Incident Resolution & User Support
- Documentation & Process Improvement

Technical Skills

- Operating Systems: Windows Environment
- Basic Networking & Connectivity
- Hardware Diagnostics & Maintenance
- System Troubleshooting & Support
- Data Encoding & Report Generation

References

Available upon request



RAISSA ANN D. ORTANEZ

Application Developer

Address: Las Piñas, Philippines
Phone: +639760970480
Email: ortanezraissa@gmail.com
LinkedIn: raissaortanez

SUMMARY

Detail-oriented Application Developer with experience in mainframe testing, data migration, and integration development. Skilled in debugging transaction services using Micro Focus Eclipse, developing and scheduling JCL jobs, and conducting system testing to ensure quality and reliability. Strong foundation in Java, COBOL, and web development, with hands-on experience across financial services and technology projects. Adept at cross-team collaboration, documentation, and delivering efficient technical solutions.

WORK EXPERIENCE

Accenture

July 2024 – Current

Testing Team | Global Financial Services Company | October 2025 – Current

- Performed testing and debugging of transaction services using **Micro Focus Eclipse**.
- Assisted the data migration team by conducting **JCL scans** for analysis and validation.
- Supported the scheduler team in **scheduling** and managing JCL jobs to ensure smooth batch operations.

Data Migration Team | Automotive Financial Services Company | November 2024 – May 2025

- Developed JCL Jobs for migrating data from **Teradata to BigQuery**.
- Conducted Pre-Analysis of Teradata jobs
- Performed Peer Reviews, Unit Testing, and Documentation to maintain code quality and ensure smooth project execution.
- Actively contributed to the engagement team, assisting in planning and coordinating project team activities.

CodeActive

December 2023 – April 2024

Junior Technical Consultant

- Key contributor to the integration development project using **Celigo**.
- Created technical documentation for the integration process.
- Conducted integration testing to ensure system functionality.
- Created custom fields in **Oracle NetSuite** and **Pipedrive** to match data mapping.

The Umonics Method

March – June 2023

IT Team Lead Intern | Internship

- Delegated tasks to members as their Team Lead and guided the new interns.
- Supervised projects assigned to our team and made sure that we are on track with the arranged timeline.
- Assisted with the company's WordPress Website Development.
- Performed QA/QC to make sure that the team are properly following the company's SOP.
- Helped boost the online presence of our company's website by utilizing Google Ads and Google Search Console for SEO and encoded the data for analyzation.

EDUCATION

Technological University of the Philippines

2019 – 2023

Bachelor of Science in Information Technology

- Thesis on "**UrbanFresh: Android Application for Fruits and Vegetables Marketplace**"

BOOTCAMPS

- AS400 RPG Bootcamp** | Top performer | Skills: **RPGLE, SQLRPGLE, CLP/CLLE** | September 2025
- COBOL Mainframe Bootcamp** | Awarded "Best in Logic" | Top performer | Skills: **JCL, COBOL** | August – September 2024
- Java Back-End Development & Spring Boot** | Skills: **Java Core, Spring Boot, REST APIs, MySQL** | September – October 2024
- Mainframe Assembler (ASM) Bootcamp** | Top performer | Skills: **Assembly, Debugger Cassi** | October 2024
- Fullstack Web Development Bootcamp** | Online course certification | Skills: **Frontend, Backend, Frameworks, Workflow** | 2023

References: Available upon request.





Researchers

LU:AA-FO-61 rev.0 09012021 / Page2

- **"Generative AI: From Concept to Creation"** Issued on: 15 APR 2025 Issued by: Udemy | Authorized by: Udemy Verify: ude.my/UC-7f02681b-e6d1-4b30-8a95-9e5d8d1fa5e3
- **"Certified Quantum Computing Expert"** Issued on: 15 APR 2025 Issued by: Udemy | Authorized by: Udemy Verify: ude.my/UC-41829106-0e5e-4e87-990b-25fd2cc0531a
- **"Azure Cloud Security with Microsoft Azure For Beginners"** Issued on: 15 APR 2025 Issued by: Udemy | Authorized by: Udemy Verify: ude.my/UC-ab4ef6f7-173b-46a3-b738-089ef7879529
- **"Empowering Communities through Information and Communications Technology"** Training conducted by the Department of Information and Communications Technology - Region IVA Rizal, held on June 21, 2024, via zoom.us platform.
- **"Expand Your E-commerce Reach Globally with AI"** Training conducted by the Department of Information and Communications Technology - Region IVA Rizal, held on June 20, 2024, via zoom.us platform.
- **"AI Awakening: Unleashing Genius in Gen Z and Gen Alpha"** Training conducted by the Department of Information and Communications Technology - Region IVA Rizal, held on June 19, 2024, via zoom.us platform.
- **"CS Training Seminar- MS Copilot"** Laguna University College of Computing Studies, Laguna University, Sports Complex Sta. Cruz, Laguna, May 17, 2024
- **"Game Development"** held at Laguna University - RLH Building Audio-Visual Room. Given this day 17th of May 2024 at Laguna University, Brgy. Bubukal, Sta. Cruz, Laguna.
- **"Software Engineering Essentials"** Issued on: 08 NOV 2024 Issued by: Coursera | Authorized by: IBM Verify: <https://www.credly.com/go/YAzT96NL>
- **"Introduction to HTML, CSS, & JavaScript"** Issued on: 25 DEC 2024 Issued by: Coursera | Authorized by: IBM Verify: <https://www.credly.com/go/7FuRe54s>
- **"N-Able (N-Central) Revolutionize Service Delivery,"** Laguna University College of Computing Studies, Caliraya Resorts Club, Lumban, Laguna, March 23-24, 2023
- **"Virtualization: Harnessing the Power of Virtual Machines"** Laguna University College of Computing Studies, Caliraya Resorts Club, Lumban, Laguna, March 23-24, 2023
- **"Securing Your Digital Footprint: Understanding Cybersecurity"** Laguna University College of Computing Studies, Caliraya Resorts Club, Lumban, Laguna, March 23-24, 2023

Award(s) and Commendation(s)

- **Academic Honors (With Honors)** AMA Computer College | 2020
- **Academic Honors (With Honors)** Dayap National High School | 2018

Skill(s)/Talent(s)

Front-end Development

- HTML5, CSS3, JavaScript
- React and Vite (Basic to Intermediate)
- Responsive web design
- UI/UX fundamentals for user-friendly interfaces

Back-end Development

- Basic Back-end Development using Python and Node.js
- Implementing basic CRUD operations
- Basic Database Operations using MySQL
- Familiarity with Firebase and Supabase



Creative and Design Skills

- Artistic skills in drawing, coloring and visual design
- Sense in color harmony, layout, layout and aesthetics
- Designing web interfaces

Tools and Technologies

- Visual Studio Code
- XAMMP
- Pycharm
- Google Colab
- Git and Github
- Design Tools (Figma/Canva)
- Proficient in using Microsoft Office Tools (MS word, Excel, Powerpoint)

Professional Skills

- Adaptability and Flexibility
- Ability to collaborate with a team.
- Goal-oriented
- Problem-solving ability
- Attention to details
- Time Management

Employment Details

Student Assistant

Community Extension Services Unit
Laguna University, Sta. Cruz, Laguna
November 2023- February 2024

Duties and Responsibilities: Performing daily tasks assigned by the supervisor

List of Accomplishment and Contributions

- Edited the CESU Manual for clarity and formatting.
- Designed and edited image layouts for the CESU webpage.
- Provided administrative support, including document printing and technical assistance for online meetings.

Factory Worker

Production Department
JBC Food Corporation
June 2023- August 2024

Duties and Responsibilities: Managed the continuous feeding of snack products into production machinery to ensure an uninterrupted flow along the conveyor system.

: Observed safety and sanitation protocols while handling food products in a high-volume manufacturing environment.

List of Accomplishment and Contributions

- Improved operational efficiency by maintaining a steady flow of materials, which reduced machine downtime.



- Consistently met daily production targets through effective physical coordination and material handling.
- Contributed to a clean and organized workspace by efficiently managing the movement of goods between production stages.

Operator

Assembly Department
Canon Business Machines (Philippines). Inc.
October 2021- March 2022

Duties and Responsibilities: Applied thermal pads to critical electronic components to ensure proper heat dissipation and hardware longevity.

: Operated specialized testing equipment to conduct functional checks and verify the integrity of assembled products.

: Performed precise mechanical assembly of printer units, specifically focusing on the secure fastening of side panels and internal components.

List of Accomplishment and Contributions

- Maintained a high standard of assembly accuracy, ensuring that all physical components and thermal interfaces met strict quality specifications.
- Contributed to streamlined production flow by efficiently managing the scanning and logging process for high volumes of printer units.
- Consistently achieved daily output targets while maintaining zero-defect rates during the testing phase.
- Demonstrated technical proficiency in using industrial assembly tools and diagnostic testers.

I hereby certify that the above information is true and correct.

JUNIE A. ANTOPINA
PRINTED NAME OVER SIGNATURE

December 23, 2025
DATE

☐ I agree to the collection and processing of my data for the purpose of facilitating the application as student intern in accredited companies. I understand that my personal information is protected by RA 10173, Data Privacy Act of 2012, and that I am required to provide truthful information.



LAGUNA UNIVERSITY

Academic Affairs Division

Laguna Sports Complex, Brgy. Bubukal, Santa Cruz, Laguna
Tel. Nos. (049) 576-4359

STUDENT INTERN CURRICULUM VITAE

Note: Do not change FONT STYLE and SIZE. For ITEMS not applicable just write NOT APPLICABLE.



MISHAEL TENGO CATIGNAS

Greenfields Subdivision Brgy. Sampaloc Pagsanjan, Laguna

catignasmishael@gmail.com

+639458967327

Education

Baccalaureate: BS Computer Science with Specialization in Data Science
Laguna University
Laguna Sports Complex Brgy. Bubukal, Sta. Cruz, Laguna
Inclusive dates (2020 – Present)

Secondary Education: With Honors
Laguna Senior High School
Provincial Capitol Compound, Santa Cruz, Laguna
Inclusive dates (2018 – 2020)

With Honors
Pedro Guevara Memorial National High School
P. Guevara Ave., Brgy. Poblacion I, Santa Cruz, Laguna
Inclusive dates (2014 – 2020)



Elementary Education: With Honors

Francisco Benitez Memorial School
B. Cosme St., Pob. II, Pagsanjan, Laguna
Inclusive dates (2008 – 2014)

Conference(s), Training(s), Seminar(s) Attended

- **"Certified Quantum Computing Expert"**, Silicofeller Quantum Academy, Udemy, April 15, 2025
- **"Azure Cloud Security with Microsoft Azure for Beginners"**, Muharrem AYDIN, Udemy, April 14, 2025
- **"Generative AI: From Concept to Creation"**, Proton Expert Systems & Solutions, Udemy, April 14, 2025
- **"CS Training Seminar- MS Copilot"**, Laguna University College of Computing Studies, Laguna University, Sports Complex Sta. Cruz, Laguna, May 17, 2024
- **"CS Training Seminar- Game Development"**, Laguna University College of Computing Studies, Laguna University, Sports Complex Sta. Cruz, Laguna, May 17, 2024
- **"Python Tutorial by YTR"** Youth the Rock Organization, March 29, 2024
- **"Shaping Tomorrow: AI Chatbots and Emerging Frontiers in Future Projects"** Laguna University College of Computing Studies, Rizal Re-Creation Center, Rizal, Laguna, March 19, 2024
- **"The Limitless Power of AI into Humanity"** Laguna University College of Computing Studies, Rizal Re-Creation Center, Rizal, Laguna, March 18, 2024
- **"Empowering Communities through Information and Communication Technology"**, Department of Information and Communications Technology Region IV-A, June 21, 2024
- **"Expand your E-commerce Reach Globally with AI"**, Department of Information and Communications Technology Region IV-A, June 20, 2024
- **"AI Awakening: Unleashing Genius in Gen Z and Gen Alpha"**, Department of Information and Communications Technology Region IV-A, June 19, 2024
- **"N-Able (N-Central) Revolutionize Service Delivery,"** Laguna University College of Computing Studies, Caliraya Resorts Club, Lumban, Laguna, March 23-24, 2023
- **"Virtualization: Harnessing the Power of Virtual Machines"** Laguna University College of Computing Studies, Caliraya Resorts Club, Lumban, Laguna, March 23-24, 2023
- **"Securing Your Digital Footprint: Understanding Cybersecurity"** Laguna University College of Computing Studies, Caliraya Resorts Club, Lumban, Laguna, March 23-24, 2023



Award(s) and Commendation(s)

- Academic Honors (With Honors) Laguna Senior High School | 2020
- Academic Honors (With Honors) Pedro Guevara Memorial National High School | 2018

Skill(s)/Talent(s)

IT Support & Troubleshooting

- Computer Hardware Assessment & Maintenance
- Operating System Installation & Reformatting (Windows 10/11)
- Network Connectivity Troubleshooting (LAN, Wi-Fi configuration)
- Software Installation, Patching, and Driver Updates

Data Management & Office Productivity

- Proficient in Microsoft Office Suite (Word, Excel, PowerPoint)
- Data Entry, File Management & System Organization
- Basic Database Management Concepts
- Google Workspace (Docs, Sheets, Drive) proficiency

Creative & Design Skills

- Visual Design & Layouting (Canva / Figma)
- Basic Photo Editing
- UI/UX Appreciation

Professional Attributes

- Strong problem-solving skills for technical issues
- Patient and service-oriented (Customer Support mindset)
- Adaptable to fast-paced office environments
- Eager to learn new office protocols and systems

Employment Details

Student Assistant

Laguna University Library

Laguna University, Sta. Cruz, Laguna

November 2023- January 2024

Duties and Responsibilities: Performing daily tasks assigned by the supervisor

List of Accomplishment and Contributions

- Encoding of Book information
- Sorting Books in the Shelves
- Cleaning Computers in the Lab

**Work Immersion Program**

Bulwagana Cong. Benjamin C. Agarao Jr. Brgy. Poblacion I Hall

A. Teodoro St, Santa Cruz, Laguna

January 13, 2020 – January 30, 2020

Duties and Responsibilities: Help staff with daily tasks and office activities

Performing daily tasks assigned by the supervisors

List of Accomplishment and Contributions

- Potting fertilized soil
- Sorting Documents

I hereby certify that the above information is true and correct.

MISHAEL T. CATIGNAS

PRINTED NAME OVER SIGNATURE

December 23, 2025

DATE

☐ I agree to the collection and processing of my data for the purpose of facilitating the application as student intern in accredited companies. I understand that my personal information is protected by RA 10173, Data Privacy Act of 2012, and that I am required to provide truthful information.



LAGUNA UNIVERSITY

Academic Affairs Division

Laguna Sports Complex, Brgy. Bubukal, Santa Cruz, Laguna

Tel. Nos. (049) 576-4359

STUDENT INTERN CURRICULUM VITAE



ANGELO RAGASA GABOT

826 Sitio Uno National Road, Barangay Binan, Pagsanjan, Laguna,

angelogabot30@gmail.com

0969-268-7560

Education

Baccalaureate: Bachelor of Science in Computer Science
with Specialization in Data Science
Laguna University
Laguna Sports Complex, Brgy. Bubukal, Santa Cruz, Laguna
2021 – Present

Secondary Education: St. Nicholas Academy of Castillejos Inc.
Bugallon Road, National Highway, Castillejos, Zambales
2018 – 2020

St. Nicholas Academy of Castillejos Inc.
Bugallon Road, National Highway, Castillejos, Zambales
2014 – 2018

Elementary Education: Castillejos Elementary School
12 National Road, Castillejos, Zambales
2008 – 2014



Conference(s), Training(s), Seminar(s) Attended

"Strengthening Student Well-Being: Addressing Academic Stress and Mental Health Challenges" Guidance and Counseling Office, via Zoom Meeting, October 8, 2025

"CS Trainings: Exploring the Fundamentals of Game Development" Laguna University College of Computing Studies, Laguna Sports Complex. Sta. Cruz, Laguna, May 17, 2024

"CS Training Seminar- MS Copilot" Laguna University College of Computing Studies, Laguna University, Sports Complex Sta. Cruz, Laguna, May 17, 2024

"Revitalizing the 21st Century Learners for Industry Demand." Laguna University College of Computing Studies, Caliraya Resorts Club, Brgy. Lewin, Lumban, Laguna, March 23-24, 2023

"Securing the Digital Footprint: Understanding Cybersecurity" Laguna University College of Computing Studies, Caliraya Resorts Club, Brgy. Lewin, Lumban, Laguna, March 23-24, 2023

"N-Able Revolutionize Service Delivery" Laguna University College of Computing Studies, Caliraya Resorts Club, Brgy. Lewin, Lumban, Laguna, March 23, 2023

"Virtualization: Harnessing the Power of Virtual Machines" Laguna University College of Computing Studies, Caliraya, Resorts Club, Brgy. Lewin, Lumban, Laguna, March 23, 2023

"Mental Health Webinar (Prioritize Mental Health together: We Got This, Friend!)" Central Student Council, via Zoom Meeting, February 4, 2023

"Certified Quantum Computing Expert" Silicofeller Quantum Academy, Udemy, April 14, 2025

"Azure Cloud Security with Microsoft Azure for Beginners" Muharrem AYDIN, Udemy, April 14, 2025

"Generative AI: From Concept to Creation" Proton Expert Systems & Solutions, Udemy, April 14, 2025

"Data Science: Deep Learning and Neural Networks in Python" Lazy Programmer Inc., Lazy Programmer Team, Udemy, March 19, 2024

Award(s) and Commendation(s)

NOT APPLICABLE



Skill(s)/Talent(s)

Technical Skills

- Frontend Development - HTML, CSS, JavaScript and Frameworks
- Version Control - Git and GitHub

Soft Skills

- Systems level analysis
- Logical and critical reasoning
- Pragmatic, results oriented reasoning
- Strategic and tactical decision making
- Scientific method and hypothesis driven problem solving

Employment Details

Machine Operator

Production

CF Tech Foodcart and Advertising

Sitio Huwaran, National Highway, Brgy. Pagsawitan, Pagsanjan, Laguna

June 2025 – August 2025

Duties and Responsibilities:

- Operates CNC Laser cutting machine for Acrylic, Sintra Board and Tarpaulin Printer
- Assisting in material preparation, quality checking and safe machine operations

Barista | Pizza Chef

Front of House | Back of House

JuanMig's Resto

JP Rizal St, 2, Pagsanjan, Laguna

December 2020 – April 2023

I hereby certify that the above information is true and correct.

ANGELO R. GABOT
PRINTED NAME OVER SIGNATURE

1/20/26
DATE



LAGUNA UNIVERSITY

Academic Affairs Division

Laguna Sports Complex, Brgy. Bubukal, Santa Cruz, Laguna
Tel. Nos. (049) 576-4359

STUDENT INTERN CURRICULUM VITAE



GENESIS REBONG DELOS REYES

J.C. Rebong Street, Brgy. Nanhaya, Victoria, Laguna

delosreyesgenesis551@gmail.com

09686791889

Education

Baccalaureate: BS Computer Science with Specialization in Data Science
Laguna University
Laguna Sports Complex, Brgy. Bubukal, Sta. Cruz, Laguna
Inclusive dates (2021 – Present)

Secondary Education: General Academic Strand
San Francisco Integrated National High School
Brgy. San Francisco, Victoria, Laguna
Inclusive dates (2019 – 2021)

Liceo De Victoria
Brgy. Nanhaya, Victoria, Laguna
Inclusive dates (2015 – 2019)

Elementary Education: Liceo De Victoria
Brgy. Nanhaya, Victoria, Laguna
Inclusive dates (2010 – 2015)



Missionari Della Fede
Brgy. Nanhaya, Victoria, Laguna
Inclusive dates (2009 – 2010)

Conference(s), Training(s), Seminar(s) Attended

"CS Training Seminar 2024: Game Development"

Laguna University Computer Science Organization
Laguna University RLH Building
May 17, 2024

"CS Training Seminar 2024: MS Copilot"

Laguna University Computer Science Organization
Laguna University RLH Building
May 17, 2024

"CCS Summit 2024: The Limitless Power of AI into Humanity"

Laguna University Computer Science Organization
Rizal Re-Creation Center, Rizal, Laguna
March 18, 2024

"CCS Summit 2024: AI Chatbots and Emerging Frontiers in Future Projects"

Laguna University Computer Science Organization
Rizal Re-Creation Center, Rizal, Laguna
March 19, 2024

"Generative AI: From Concept to Creation"

Proton Expert Systems & Solutions
Udemy
April 15, 2025

"Azure Cloud Security with Microsoft Azure For Beginners"

Muharrem AYDIN, OAK Academy Team
Udemy
April 15, 2025

"Certified Quantum Computing Expert"

Silicofeller Quantum Academy
Udemy
April 15, 2025



Award(s) and Commendation(s)

- **Academic Honors (With Honors)** Liceo De Victoria – A.Y 2010 - 2015

Skill(s)/Talent(s)

TECHNICAL SKILLS

- Data Entry
- Documentation and Research
- Email Handling and File Management
- Computer Literacy (Google Docs, Spreadsheets, Presentations)
- Beginner Level Programming
- Familiarity with Data Base

SOFT SKILLS

- Attention to detail
- Willingness to learn
- Critical Thinking and Problem Solving
- Adaptable to Environment
- Time Management

Employment Details

N/A

I hereby certify that the above information is true and correct.

GENESIS R. DELOS REYES
PRINTED NAME OVER SIGNATURE

JANUARY 16, 2026
DATE

☐ I agree to the collection and processing of my data for the purpose of facilitating the application as student intern in accredited companies. I understand that my personal information is protected by RA 10173, Data Privacy Act of 2012, and that I am required to provide truthful information.



APPENDIX J

Manual

USER MANUAL



EduCompose

built for Laguna University

A writing-analytics
workspace

PREPARED BY:

Delos Reyes, Genesis
Catignas, Mishael
Antopina, Junie
Gabot, Angelo



2

LOGIN PAGE

EduCompose
Teacher's Companion for Google Classroom

- All-in-one tool easily integrates
- Detailed feedback generation
- Save time, improve learning

Welcome Back
Login to access your dashboard

Email:

Password:

☒ Remember me

[Don't have an account? Sign up now!](#)

You can log in the:

- Teacher
- Admin

Student log in

Welcome Back
Login with your student code to access your essays

Student Code:

Password:

☐ Remember me

Manage your student account through your teacher or administrator.

STUDENT ACCOUNT

White Better Sleep

How many hours of sleep do you get each night?

Let's Get Started!

How many hours of sleep do you get each night? (Select one)

☐ 6 hours

☐ 7 hours

☐ 8 hours

☐ 9 hours

☐ 10 hours

ADMIN ACCOUNT

Admin Dashboard

Overview

Students

100% (10/10)

Teachers

100% (10/10)

Assignments

100% (10/10)

Grades

100% (10/10)

Attendance

100% (10/10)

Discipline

100% (10/10)

Calendar

100% (10/10)

Settings

100% (10/10)

Help

100% (10/10)

TEACHER ACCOUNT

Overview

100% (10/10)

Students

100% (10/10)

Teachers

100% (10/10)

Assignments

100% (10/10)

Grades

100% (10/10)

Attendance

100% (10/10)

Discipline

100% (10/10)

Calendar

100% (10/10)

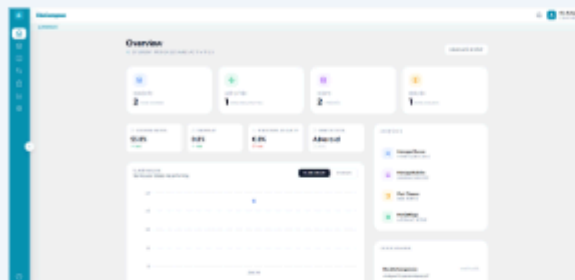
Settings

100% (10/10)

Help

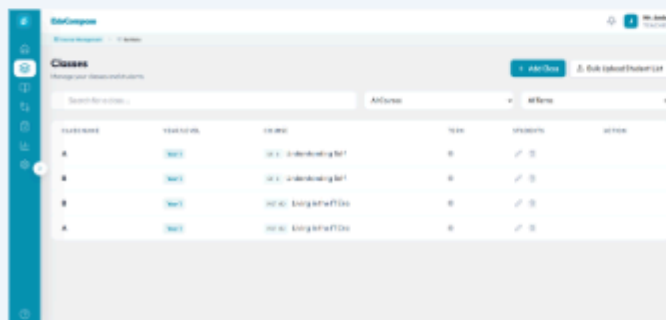
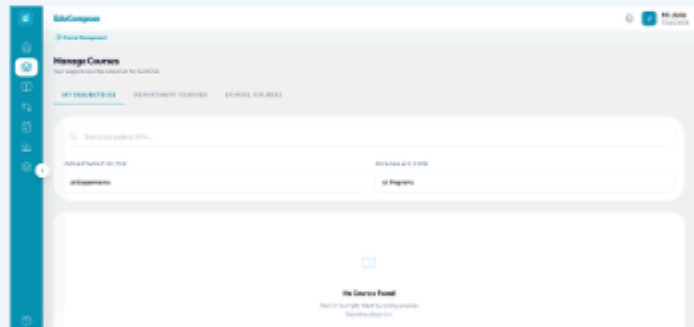
100% (10/10)

TEACHER ACCOUNT



The home dashboard serves as a central command center that provides an immediate visual summary of your teaching workload and student performance metrics. It allows you to track academic progress at a glance while providing quick-access shortcuts to essential tools for managing classes and grading assignments.

The Manage Courses page allows teachers to curate their personal teaching list by selecting and adding subjects from the department or school-wide curriculum.



The Teacher Classes page allows educators to organize their courses, add new classes, and manage student rosters for all their assigned subjects.



4

Searchable catalog of official subjects

You can add what courses you handle.

Activities page allows teachers to create new assignments, track total essay submissions, and manage all student tasks through a searchable card or table view

Assigned Classes table provides a high-level summary of your active courses, displaying the total number of enrolled students and essay submissions for each class

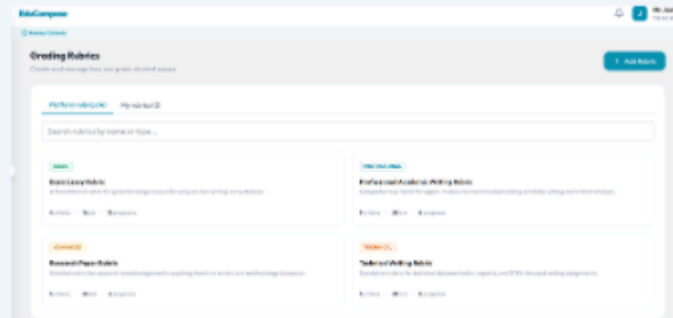
Similarity Detection page automatically flags assignments with duplicate or nearly identical content, helping teachers identify potential academic integrity issues across the class.

The screenshot displays the LMS interface with several key sections: 'Manage Courses' at the top with a search bar and a grid of course cards; 'Activities' below it with a search bar and a list of assignments; 'Assigned Classes' at the bottom left showing a table of active courses; and 'Similarity Detection' at the bottom right showing a search bar and a message about duplicate content. Blue arrows point from descriptive text to each of these sections.

CLASS	STUDENTS	SUBMISSIONS
2022-2023-2024	10	10

5

Platform Rubrics provide ready-to-use grading templates for different assignment types, allowing teachers to quickly apply professional evaluation standards to their classes



sample lang din ito PROFESSORIAL

Grading Intensity: Professional

Criteria: **4** Points **3** Points **2** Points **1** Points

CRITERIA	4 POINTS	3 POINTS	2 POINTS	1 POINTS
Evidence & Support MAX 4 PTS	EXCELLENT Demonstrates exceptional understanding and execution.	PROFICIENT Meets expectations with minor areas for improvement.	DEVELOPING Shows basic understanding but needs significant improvement.	BEGINNING Minimal evidence of understanding or skill.
Organization MAX 4 PTS	EXCELLENT Clear, logical structure with smooth transitions.	PROFICIENT Generally organized with some minor flow issues.	DEVELOPING Basic structure but lacks coherence.	BEGINNING Disorganized and difficult to follow.
Criteria 3 MAX 4 PTS	EXCELLENT [Description]	PROFICIENT [Description]	DEVELOPING [Description]	BEGINNING [Description]

SCORE GUIDE: **4** Excellent **3** Proficient **2** Developing **1** Beginning

Export Close

You can customize your own rubrics

+ Add Rubric

Add Rubric page offers three ways to create a grading standard: uploading an existing file, starting from a pre-made template, or building a custom rubric from scratch





6

Performance Overview page provides a comprehensive dashboard for teachers to monitor class-wide writing trends, average scores, and potential originality risks at a glance



Teacher Profile page allows educators to manage their professional information, school affiliation, and account security settings to ensure their portal details remain accurate

The Teacher Profile page includes a sidebar with navigation links: PROFILE, SECURITY, MY DATA, and ARCHIVE. The main content area is divided into two sections: 'Profile Information' and 'Account Settings'. The 'Profile Information' section contains fields for Name, Email, Phone Number, Address, and School. The 'Account Settings' section contains fields for Username, Password, and Security Questions. A 'Update Profile' button is located at the bottom right.

Change Password page allows you to update your account security by creating a new, strong password that meets the system's character and complexity requirements.

The Change Password page features three input fields: 'CURRENT PASSWORD', 'NEW PASSWORD', and 'CONFIRM PASSWORD'. Below the 'NEW PASSWORD' field, there is a note: 'Must be at least 8 characters, containing uppercase, lowercase, and numbers.' A 'Change Password' button is located at the bottom right.

The Archives page displays a list of student information with columns for Name, Email, and Phone Number. A 'Download Master Template' button is located at the bottom left. A 'Download All Data' button is located at the bottom right.

Provides a pre-formatted file that you can fill with your students' information

Archives page allows you to download master templates for student imports and access



7

STUDENT ACCOUNT

Welcome Back
Login with your student code to access your essays.

Student Code
123-4567

Password
[password]

☐ Remember me [Forgot password?](#)

Login to Portal

Manage your student account through your teacher or administrator.

Log In your student account.

Reset password trough gmail.

[BACK TO LOGIN](#)

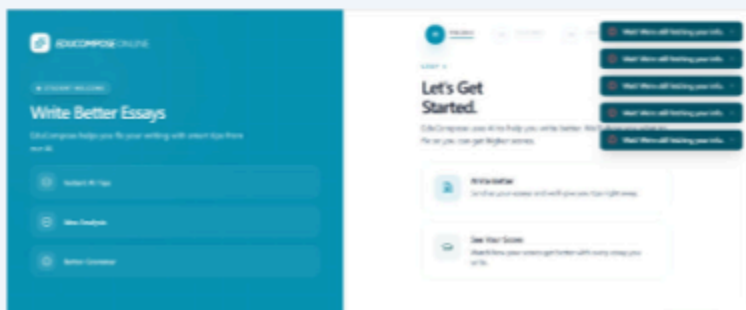
Reset Password

Enter your registered email address and we'll send you a link to reset your password.

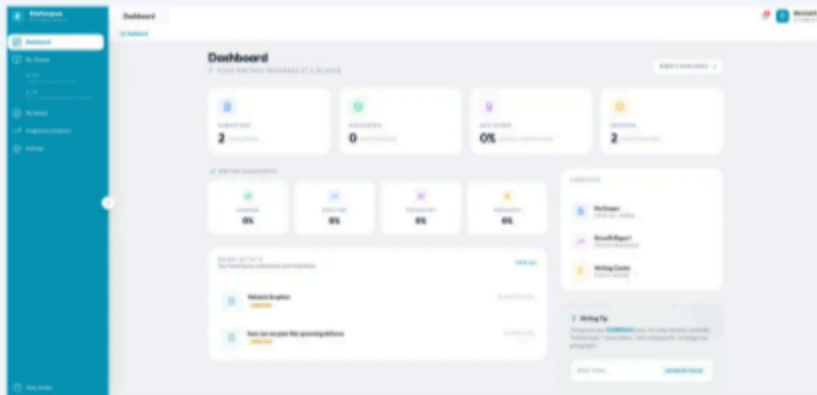
Email Address
name@example.com

Send Reset Link

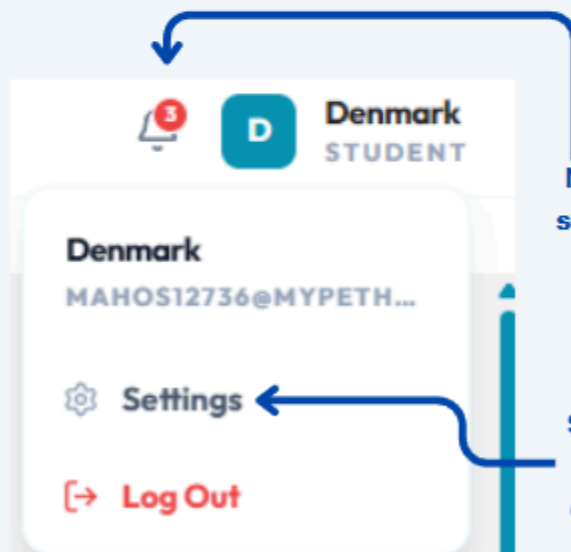
This page introduces the
EduCompose system
and guides users to start
setting up and using its
AI-powered essay
writing features.



8



This dashboard page shows the user's essay progress, scores, and recent activities, allowing them to monitor their writing performance and access important features.



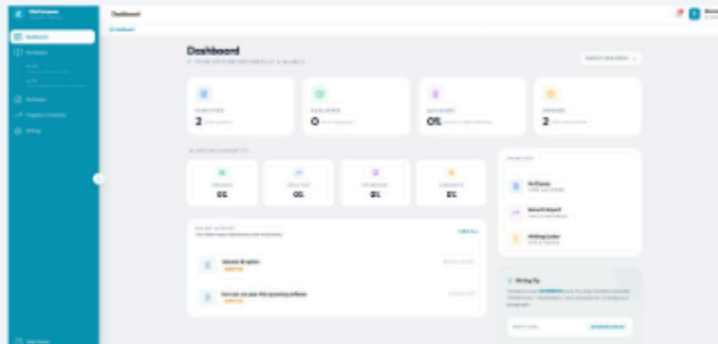
Notifications: Click the bell icon to see your latest updates and system alerts.

Settings: Use this option to update your personal information or change your account preferences.

Log Out: Click this to safely exit your account and end your current session.

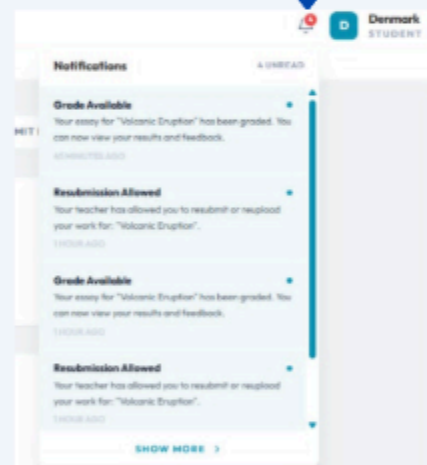


9



Student Dashboard provides a complete overview of your essay submissions, writing scores, and progress diagnostics to help you track your academic growth in one place.

The **Notifications** panel provides real-time updates on your essay status, alerting you when grades are available or when a teacher allows you to resubmit your work.



The **Class Detail** page displays information about a specific course, including the teacher's name, semester details, and a list of all your assigned

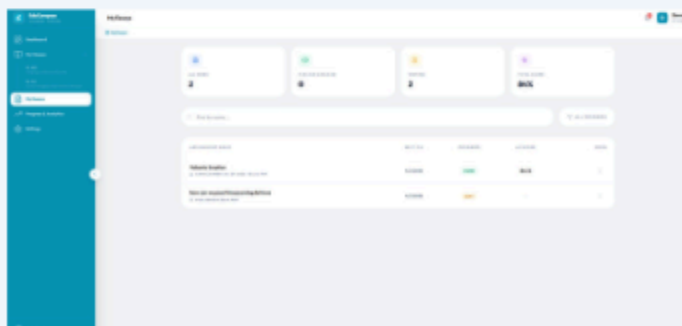
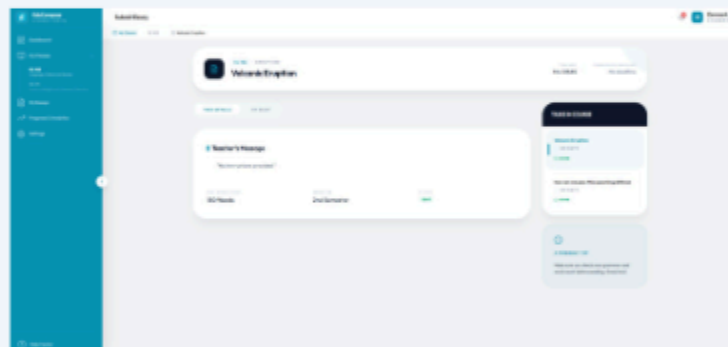


10

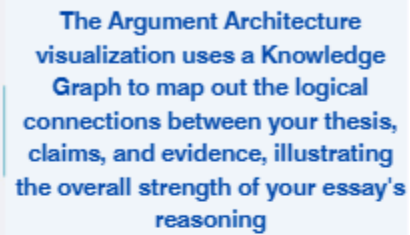


The Submission Results page confirms that your essay was sent successfully and provides an immediate AI score based on your writing's grammar, flow, and overall quality.

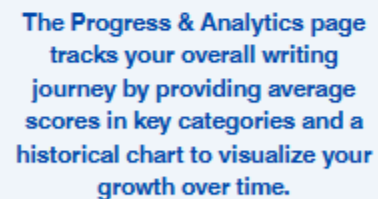
The Task Details tab outlines the specific requirements for an assignment, including instructions from your teacher, the minimum word count, and the current status of your submission.



The My Essays page provides a complete history of all your submitted work, allowing you to track the progress, submission dates, and AI scores for every assignment in one place.



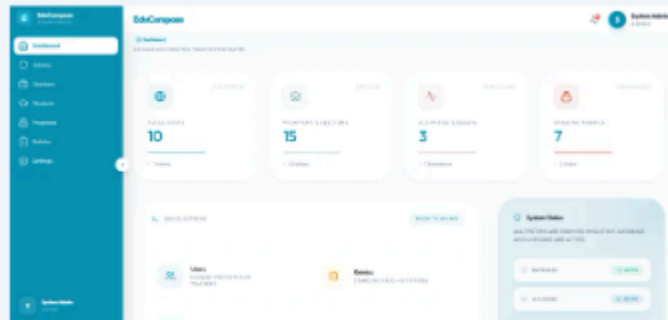
The screenshot displays the AWS IAM console interface. On the left, the navigation menu includes 'Groups', 'Users', 'Roles', 'Groups', 'Users', 'Roles', and 'Groups'. The main content area is titled 'Groups' and shows a list of groups. The 'AWS-ReadOnlyAccess' group is highlighted, showing a status of 'OK'. Below the list, the 'Permissions summary' section shows that the group has 0 permissions attached, 0 permissions in use, and 0 permissions not in use.



The screenshot shows the AWS IAM console 'My Profile' page. The left sidebar contains navigation links: Home, Users, Groups, Roles, Policies, Account Access, and Settings. The main content area is titled 'My Profile' and includes a 'Personal Details' section with fields for Name, Email, Username, Password, and Phone Number. Below this is a 'Security Information' section with fields for MFA Device and Security Questions. The right sidebar shows 'My Profile' summary information, including Name, Email, Username, and a link to 'Manage my profile'.



ADMIN ACCOUNT



Admin Dashboard provides a high-level overview of the entire system, allowing you to monitor total users, active programs, and the real-time status of the database and AI engine.

User Distribution and Resources panels provide a clear breakdown of active accounts by role and a detailed count of all academic data currently managed within the platform.



ID	NAME	STATUS	CREATED DATE	ACTIONS
1	Admin-Admin	ACTIVE	APR 12, 2024	

Page: 1

User Management page allows administrators to track all system accounts, manage user roles, and monitor account statuses through a searchable and paginated table.



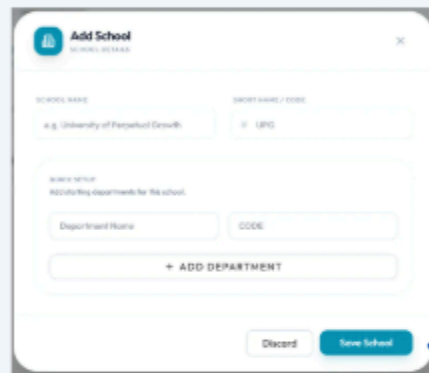
Edit Teacher modal allows administrators to update a teacher's personal details, school affiliation, and contact information to ensure all system records remain accurate

The screenshot shows the 'All Departments' view in the People app. At the top, there are filters for 'Phone, ID, or Email...', 'All Departments', 'Group View', and 'All Status'. Below this is a table with columns: 'ID', 'IDENTITY PICTURE', 'PLACED AT', 'STATUS', and 'ACTION'. The table lists four employees: 'Halla Hultala' (STATUS: + ACTIVE), 'Comptone Ponsile' (STATUS: + ACTIVE), 'Rene Mitchell' (STATUS: + ACTIVE), and 'Stephen Gray' (STATUS: + ACTIVE). Each row includes a small profile picture icon, the employee's name, a placeholder for an identity picture, and a status indicator with a green checkmark and the word 'ACTIVE'.

Student Actions menu provides quick shortcuts for administrators to edit profiles, view activity logs, and perform essential account maintenance like resending emails or account provisioning.

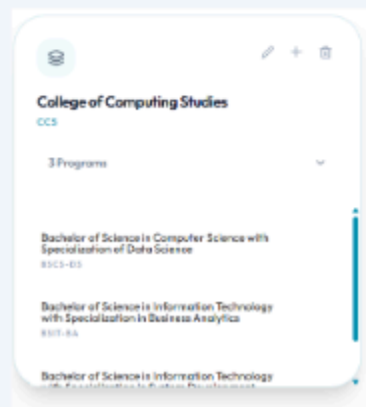


Institutional Hierarchy page allows administrators to manage the organizational structure of the institution by defining and organizing its various schools, departments, and academic programs.

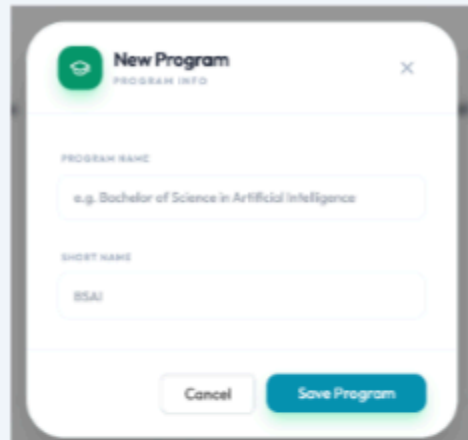


Add School modal allows administrators to register a new institution and immediately establish its initial academic departments through a convenient quick-setup feature.

New Department modal allows administrators to add a specific academic division to a school



Department Details view allows administrators to manage and see all specific degree programs and specializations hosted within a particular academic department



New Program

PROGRAM INFO

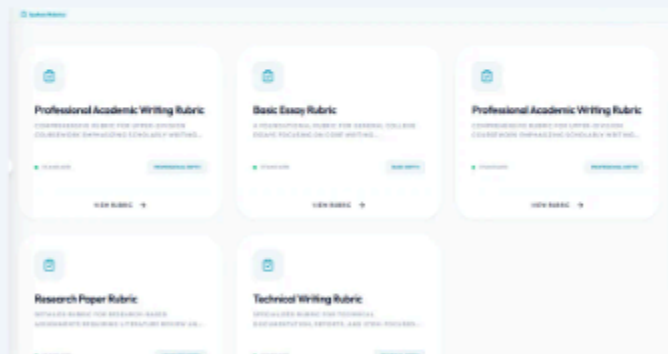
PROGRAM NAME
e.g. Bachelor of Science in Artificial Intelligence

SHORT NAME
BSAI

Cancel Save Program

You can also add a New Program.

The **System Rubrics** page allows administrators to manage a global library of standardized grading templates that provide consistent and professional evaluation criteria for all teachers on the platform.




System Settings

Academic Calendar

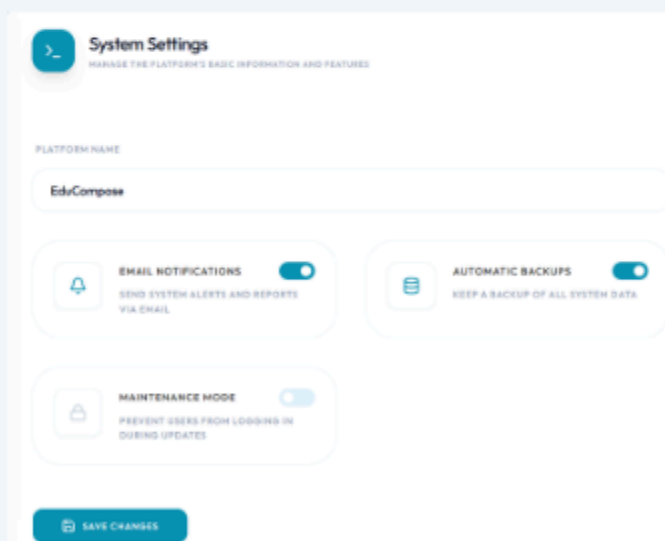
START YEAR 2023

END YEAR 2024

2023 2024 2025 2026 2027 2028 2029 2030 2031 2032 2033 2034 2035 2036 2037 2038 2039 2040 2041 2042 2043 2044 2045 2046 2047 2048 2049 2050 2051 2052 2053 2054 2055 2056 2057 2058 2059 2060 2061 2062 2063 2064 2065 2066 2067 2068 2069 2070 2071 2072 2073 2074 2075 2076 2077 2078 2079 2080 2081 2082 2083 2084 2085 2086 2087 2088 2089 2090 2091 2092 2093 2094 2095 2096 2097 2098 2099 2100 2101 2102 2103 2104 2105 2106 2107 2108 2109 2110 2111 2112 2113 2114 2115 2116 2117 2118 2119 2120 2121 2122 2123 2124 2125 2126 2127 2128 2129 2130 2131 2132 2133 2134 2135 2136 2137 2138 2139 2140 2141 2142 2143 2144 2145 2146 2147 2148 2149 2150 2151 2152 2153 2154 2155 2156 2157 2158 2159 2160 2161 2162 2163 2164 2165 2166 2167 2168 2169 2170 2171 2172 2173 2174 2175 2176 2177 2178 2179 2180 2181 2182 2183 2184 2185 2186 2187 2188 2189 2190 2191 2192 2193 2194 2195 2196 2197 2198 2199 2200 2201 2202 2203 2204 2205 2206 2207 2208 2209 2210 2211 2212 2213 2214 2215 2216 2217 2218 2219 2220 2221 2222 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The interface for Academic Sessions and Archives. It features three tabs: PRIMARY SESSION, SECONDARY SESSION, and EXTENDED SUMMER. Each tab has a table with columns for START MONTH and END MONTH. The PRIMARY SESSION table shows JANUARY and DECEMBER. The SECONDARY SESSION table shows JANUARY and MAY. The EXTENDED SUMMER table shows APRIL and JULY. Below the tables is a 'CALENDAR SYNC' section with a note: 'Sync academic calendar with our web faculty loads.' A 'SAVE CHANGES' button is at the bottom right. To the right of the session tables is a 'System Archives' section with a note: 'ACCESS HISTORICAL DATA AND ARCHIVED MEMBER RECORDS.' It contains two links: 'ACADEMIC ARCHIVE' (TEACHER LOADS & SCHEDULES) and 'USER ARCHIVE' (DEPARTMENTS & TEACHER MEMBERS).

Academic Sessions and Archives section allows administrators to define custom semester timelines and maintain a secure repository for historical teacher loads and inactive user records.

The System Settings interface. It has a title 'System Settings' and a subtitle 'MANAGE THE PLATFORM'S BASIC INFORMATION AND FEATURES'. There is a 'PLATFORM NAME' field with the value 'EduCompose'. Below this are three toggle switches: 'EMAIL NOTIFICATIONS' (SEND SYSTEM ALERTS AND REPORTS VIA EMAIL), 'AUTOMATIC BACKUPS' (KEEP A BACKUP OF ALL SYSTEM DATA), and 'MAINTENANCE MODE' (PREVENT USERS FROM LOGGING IN DURING UPDATES). A 'SAVE CHANGES' button is at the bottom.

Allows administrators to manage essential platform-wide features, including the platform name, automated backups, and the activation of maintenance mode during system updates.